BIOCHEMISTRY

Part 2

MCQs



2026

بسم الله الرحمن الرحيم

والصلاة والسلام على خير المرسلين محمد بن عبد الله وعلى آله وصحبه وسلم أما بعد، نقدم بين يديك ملف

(Biochemistry & Nutrition MCQ Part 2)

-	Title	Page
1	Assimilation of dietary fat	3
2	Ketone Bodies Metabolism	12
3	Lipogenesis	19
4	Biochemistry of amino acids and proteins (structure and function)	27
5	Decarboxylated amino acid (Biogenic amines)	43
6	Phenylalanine, Tyrosine Metabolism	51
7	Glycine metabolism	56
8	Nitrogen Metabolism and urea cycle	61
9	Oxygen Toxicity and Free Radical Injury	74
10	Integration of Metabolism	83
11	DNA Structure and chromatin organization	93
12	Purine and pyrimidine metabolism	100
13	DNA Replication	106
14	Ribonucleic acids (RNAs)	112
15	Protein Synthesis , genetic codons	126
16	DNA alteration (Mutations)	131

ابدأ مستعيناً بالله وكرر الحل ما استطعت واحرص على وقتك فهو رأس مالك وأغلى ما تملك

حسابات الفريق على منصات التواصل الاجتماعي

جميع الحقوق محفوظة لفريق مِعوان

ولا نحلل بيعها بعائد ربحي أو سرقتها بأي شكل من الأشكال

Lecture (10): Assimilation of dietary fat

Pre Exams , Formatives

1-	**Carnitine is involved in fatty acid ? (MID 31, FINAL 36, MID 37, Department Qs37)**						
A-	Synthesis	B-	Activation	C-	Oxidation	D-	Thiolysis

2-	**Palmatic acid oxidation produce (FINAL 34)**
A-	(106 ATP)

3-	**How many times b oxidation reapeted in palmatic acid (FINAL 34)**
A-	Repeated 7 times

4-	**Which step of B oxidation doesn-t require a coenzyme (FINAL 34)**
A-	Hydration of 2 enoyl co A to hydroxyacyl CoA

5-	**L-carnitine is used for (FINAL 34)**
A-	In B oxidation

6-	**Co-enzyme of B keto-thiolase is ? (MID 30)**						
A-	FAD	B-	NAD	C-	CoA SH	D-	ATP

7-	**In-oxidation of fatty acids, the second oxidation-reduction steps is catalyzed by ? (MID 31)**		
A-	Acyl-COA dehydrogenase	B-	B-hydroxyacylCOA dehydrogenase
C-	B-keto acyl reductase	D-	Enoylreductase
E-	Reductase		

8-	**Key regulatory enzyme in Beta oxidation (FINAL 36)**		
A-	Acetyl Co-A carboxylase	B-	Carnitine palmitoyltransferase I

9-	**Acyl-carnitine is formed by: (MID 37)**		
A-	Acyl-carnitine transferase I	B-	Acyl-carnitine transferase II
C-	Acyl-carnitine translocase	D-	Acyl-CoA synthetase

10-	**B-hydroxyacyl formed from: (FINAL 37)**		
A-	Enoyl acyl hydratase	B-	Acyl coA dehyrogenase

11-	**Malonyl coA inhibits: (FINAL 37 SNU)**				
A-	pyruvate dehydrogenase	B-	fatty acid synthase	C-	CPT1

12-	Carnitine palmitoyl transferase 1 is inhibited by... (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	Butyryl CoA	B-	Propionyl CoA	C-	Malonyl CoA	D-	Acetyl CoA

13-	BetA-oxidation of a fatty acid: (Formative 37, Quiz SNU-M1, SNU Bank)
A-	Involves two steps of reduction of fatty acid
B-	Produces ATP at substrate level
C-	Requires NAD+
D-	Has a preference for fatty acids shorter than 12 carbons

14-	Oxidation of fatty acids occurs (Formative 37, SNU Bank)		
A-	On the microsomes	B-	On inner mitochondrial membrane
C-	In the matrix of mitochondria	D-	In the cytosol

15-	During each cycle of B-oxidation (Final SNU-M1)
A-	One carbon atom is removed from the carboxyl end of the fatty acid
B-	One carbon atom is removed from the methyl end of the fatty acid
C-	Two carbon atoms are removed from the carboxyl end of the fatty acid
D-	Two carbon atoms are removed from the methyl end of the fatty acid

16-	B-Oxidation of fatty acids requires all the following coenzymes except (Final SNU-M1)						
A-	CoA	B-	FAD	C-	NAD	D-	NADP

17-	Which of the following statements describes best the function of carnitine? (FINAL 31)
A-	It transports medium-chain fatty acid into gut-epithelial cells
B-	It transports medium-chain fatty acids across the inner mitochondrial membrane
C-	It is a trans-acylation reaction catalyzed by a transferase
D-	It is a derivative of vitamin A

18-	Fatty acids enter the mitochondrial matrix for oxidation ? (FINAL 32)
A-	In the form of acyl-CoA s directly by active transport
B-	As free fatty acids
C-	Following conversion to carnitine
D-	Complexed with carnitin as acylcarnitines

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
C	A	A	A	A	C	B	B	A	A
11	12	13	14	15	16	17	18	19	20
C	C	C	C	C	D	C	D	-	-

Department Qs

1-	Lipolysis in adipose tissue: (SNU Bank, Formative 37, Quiz SNU-M1)		
A-	Is catalyzed by lipoprotein lipase	B-	Is enhanced in the fed state
C-	Produces fatty acid and glycerol	D-	Is enhanced by insulin secretion

2-	β-Oxidation of fatty acids requires all the following coenzymes except (SNU Bank)						
A-	CoA	B-	FAD	C-	NAD	D-	NADP

3-	During each cycle of β-oxidation (SNU Bank)
A-	One carbon atom is removed from the carboxyl end of the fatty acid
B-	One carbon atom is removed from the methyl end of the fatty acid
C-	Two carbon atoms are removed from the carboxyl end of the fatty acid
D-	Two carbon atoms are removed from the methyl end of the fatty acid.

4-	In β-oxidation of fatty acids which of the following are utilized as coenzymes? (SNU Bank)		
A-	NAD+ and NADP+	B-	FAD H2 and NADH + H+
C-	FAD and FMN	D-	FAD and NAD+

5-	The following hormones increase the mobilization of depot fat (lipolysis) EXCEPT: (SNU Bank, Lecture Qs33, Lecture Qs35, Lecture Qs37)		
A-	Epinephrine and nor epinephrine	B-	Insulin
C-	Growth hormone.	D-	Glucagon.

6-	The net energy production in the form of ATP that can produced by oxidation of one molecule of palmitate is: (SNU Bank)						
A-	12	B-	36	C-	106	D-	146

7-	Which of the following compounds is involved in the transport of long chain fatty acids into the mitochondrion: (SNU Bank, Lecture Qs33, Lecture Qs35, Lecture Qs37)						
A-	Oxaloacetate	B-	Citrate	C-	Carnitine	D-	Taurine

8-	During β-oxidation of fatty acids, the removal of two carbon units involves the following sequential reactions: (SNU Bank)
A-	Oxidation, dehydration, oxidation, cleavage
B-	Dehydrogenation, hydration, dehydrogenation, cleavage
C-	Reduction, hydration, dehydrogenation, cleavage
D-	Reduction, dehydration, reduction, cleavage

9-	Continuation of the fatty acid oxidation cycle in the metabolism of long chain fatty acids is dependent on the presence of all for the following except: (SNU Bank)		
A-	Acyl-CoA dehydrogenase	B-	Enoyl hydratase
C-	β-ketothiolase	D-	Thiokinase

10-	What is the role of L-carnitine in fatty acid metabolism? (SNU Bank)
A-	Facilitate the transport of fatty acid from the cytosol to mitochondria
B-	Serve as a cofactor for enzyme fatty aciimiting enzyme of lipolysis is d synthase
C-	Activator of acetyl CoA carboxylase
D-	Inhibitor

11-	The Major rate limiting enzyme in lipolysis is (SNU Bank)						
A-	HSL	B-	MAL	C-	DAL	D-	Hydrolase

12-	Which tissue forms the glycerol? (SNU Bank)						
A-	Areolar tissue	B-	Epithelium	C-	Adipose tissue	D-	Elastic tissue

13-	Propionyl CoA is formed on oxidation of: (Department Qs37)		
A-	Monounsaturated fatty acids	B-	Polyunsaturated fatty acids
C-	Fatty acids with odd number of carbon atoms	D-	None of these

14-	BetA-oxidation takes place in: (Department Qs37)		
A-	Mitochondria	B-	Extra mitochondria
C-	Lysozyme	D-	Endoplasmic reticulum

15-	Oxidation of palmitic acid to active acetate needs the repetition of BetA-oxidation cycles: (Department Qs37)						
A-	3 times	B-	6 times	C-	7 times	D-	8 times

16-	Glycerokinase is present in the following tissues except: (Department Qs37)						
A-	liver	B-	Kidney	C-	Intestine	D-	Adipose tissue

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	D	C	D	B	C	C	B	D	A
11	12	13	14	15	16	17	18	19	20
A	C	C	A	C	D	-	-	-	-

Lecture Qs

1-	**How many acetyl CoA groups are produced by the complete β oxidation of palmitic acid (C16)? (Lecture Qs33, Lecture Qs35, Lecture Qs37)**				
A-	16	B-	8	C-	7

2-	**How many oxidation cycles are necessary to completely oxidize palmitic acid (C16)? (Lecture Qs33, Lecture Qs35, Lecture Qs37)**				
A-	16	B-	8	C-	7

3-	**Which of the following statements is true? (Lecture Qs33)**
A-	High insulin/glucagon ratio activates lipolysis in muscle
B-	High insulin/glucagon ratio inhibits lipolysis in liver
C-	High insulin/glucagon ratio activates lipolysis in adipocytes
D-	Low insulin/glucagon ratio activates lipolysis in adipocytes

4-	**Activation of the fatty acid occur in: (Lecture Qs33)**						
A-	mitochondria	B-	nucleus	C-	Cytoplasm	D-	non of the above

5-	**Lipolysis produce: (Lecture Qs33)**		
A-	glycerol and cholesterol	B-	glycerol and fatty acid
C-	glycerol and ketone bodies	D-	all of the above

Answers - Lecture Qs

1	2	3	4	5	6	7	8	9	10
B	C	D	C	B	-	-	-	-	-

M3WAN Bank

1-	**The average adult diet contains approximately how much lipid per day, mainly in the form of triacylglycerols? (M3WAN - Core)**						
A-	10–20 grams	B-	50–75 grams	C-	100–150 grams	D-	300–400 grams

2-	**"Constant fat" (tissue fat), mainly phospholipid and cholesterol, differs from "depot fat" in that it: (M3WAN - Core)**
A-	Is affected by starvation and used for energy
B-	Is not affected by starvation and does not produce energy
C-	Is stored exclusively in adipocytes
D-	Is composed mainly of triglycerides

3-	**Lipolysis is defined as the hydrolysis of triacylglycerols to yield: (M3WAN - Core)**		
A-	Glycerol and ketone bodies	B-	Free glycerol and free fatty acids
C-	Acetyl-CoA and glycerol	D-	Diacylglycerol only

4-	**The ratE-limiting enzyme of lipolysis in adipose tissue is: (Important M3WAN - Core)**		
A-	Diacylglycerol lipase	B-	Monoacylglycerol lipase
C-	HormonE-sensitive triacylglycerol lipase	D-	Lipoprotein lipase

5-	**HormonE-sensitive lipase is activated by covalent modification, in which the active form is the: (Important M3WAN - Core)**		
A-	Dephosphorylated form	B-	Phosphorylated form
C-	Glycosylated form	D-	Ubiquitinated form

6-	**Which of the following hormones is antilipolytic, acting by reducing cAMP levels in adipocytes? (Important M3WAN - Core)**						
A-	Glucagon	B-	Epinephrine	C-	Insulin	D-	ACTH

7-	**Sequential hydrolysis of triacylglycerol proceeds through diacylglycerol and monoacylglycerol to yield glycerol, with each step releasing: (M3WAN - Core)**		
A-	A phosphate group	B-	One free fatty acid
C-	An acetyl-CoA molecule	D-	A ketone body

8-	**Glycerol released from adipose tissue lipolysis cannot be metabolized locally because adipocytes lack: (Important M3WAN - Core)**		
A-	Glycerol kinase	B-	Glycerol-3-phosphate dehydrogenase
C-	HormonE-sensitive lipase	D-	Thiokinase

9-	**Long-chain free fatty acids are transported in the blood mainly: (M3WAN - Core)**		
A-	Freely dissolved in plasma	B-	As chylomicron remnants
C-	Bound to albumin	D-	Bound to hemoglobin

10-	**The half-life of free fatty acids in the blood is approximately: (M3WAN - Hard)**						
A-	1–3 minutes	B-	1–3 hours	C-	24 hours	D-	3–5 days

11-	**Among the pathways of fatty acid oxidation, the most quantitatively important pathway is: (M3WAN - Core)**						
A-	α-oxidation	B-	ω-oxidation	C-	β-oxidation	D-	γ-oxidation

12-	**β-oxidation is defined as oxidation of the β-carbon of a fatty acid to form: (M3WAN - Core)**				
A-	β-hydroxy acid	B-	β-keto acid	C-	β-amino acid
D-	α-keto acid				

13-	**β-oxidation of fatty acids occurs mainly within the: (M3WAN - Core)**		
A-	Cytoplasm	B-	Mitochondrial matrix
C-	Nucleus	D-	Golgi apparatus

14-	**Although brain cells possess the enzymatic machinery for β-oxidation, the brain relies instead on ketone bodies for fuel because: (Important M3WAN - Core)**
A-	The brain lacks mitochondria
B-	The blooD-brain barrier prevents entry of fatty acids
C-	β-oxidation enzymes are absent in neurons
D-	Ketone bodies yield more ATP than fatty acids

15-	**Approximately what percentage of cardiac muscle fuel requirement is normally met by β-oxidation of fatty acids? (M3WAN - Hard)**						
A-	20%	B-	50%	C-	80%	D-	100%

16-	**Activation of a fatty acid prior to β-oxidation, forming acyl-CoA, is catalyzed by: (M3WAN - Core)**		
A-	Carnitine acyltransferase I	B-	Thiokinase (acyl-CoA synthetase)
C-	Acyl-CoA dehydrogenase	D-	Thiolase

17-	**Carnitine, required to transport long-chain acyl groups across the inner mitochondrial membrane, is synthesized in the liver and kidney from lysine and: (M3WAN - Core)**						
A-	Methionine	B-	Serine	C-	Glycine	D-	Arginine

18-	**The carnitine shuttle enzyme located on the outer surface of the inner mitochondrial membrane, considered the key regulatory enzyme of β-oxidation, is: (Important M3WAN - Core)**		
A-	Carnitine acyltransferase II	B-	Carnitine translocase
C-	Carnitine acyltransferase I	D-	Acyl-CoA synthetase

19-	**The first step of the β-oxidation spiral, introducing a double bond between the α and β carbons, is catalyzed by acyl-CoA dehydrogenase, whose coenzyme is: (M3WAN - Core)**						
A-	NAD+	B-	NADP+	C-	FAD	D-	TPP

20-	The second step of β-oxidation, hydration of the double bond, is catalyzed by enoyl-CoA hydratase, also known as: (M3WAN - Hard)						
A-	Crotonase	B-	Thiolase	C-	Thiokinase	D-	Lipase

21-	The third step of β-oxidation, oxidation of the β-hydroxyl group, is catalyzed by β-hydroxyacyl-CoA dehydrogenase using which coenzyme? (M3WAN - Core)						
A-	FAD	B-	NAD+	C-	TPP	D-	Biotin

22-	The final step of each β-oxidation cycle, cleaving the β-ketoacyl-CoA into acetyl-CoA and a shortened acyl-CoA, is catalyzed by: (M3WAN - Core)		
A-	Enoyl-CoA hydratase	B-	Acyl-CoA dehydrogenase
C-	Thiolase	D-	Carnitine translocase

23-	Complete β-oxidation of palmitic acid (16 carbons) requires how many cycles of the spiral, yielding 8 molecules of acetyl-CoA? (Important M3WAN - Core)						
A-	5 cycles	B-	6 cycles	C-	7 cycles	D-	8 cycles

24-	The net ATP yield from complete oxidation of one molecule of palmitate is approximately: (Important M3WAN - Core)						
A-	32 ATP	B-	80 ATP	C-	106 ATP	D-	129 ATP

25-	The general formula for calculating the energy yield of fatty acid oxidation, where N = number of carbons, is: (M3WAN - Hard)		
A-	[(N÷2−1)×4] + [(N÷2)×10] − 2	B-	[(N÷2)×4] + [(N÷2−1)×10] − 2
C-	[N×4] − [N×10]	D-	[(N−1)×10] + 2

26-	β-oxidation is regulated mainly at the level of carnitine acyltransferase I (CPT1); when cellular ATP levels are high, β-oxidation is: (Important M3WAN - Core)		
A-	Stimulated	B-	Inhibited
C-	Unaffected	D-	Converted to α-oxidation

27-	Besides energy production, β-oxidation is biologically important because the acetyl-CoA produced can be used for synthesis of cholesterol, acetylcholine, and: (M3WAN - Core)		
A-	Ketone bodies	B-	Glycogen directly
C-	Essential amino acids	D-	Purine bases

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	B	B	C	B	C	B	A	C	A
11	12	13	14	15	16	17	18	19	20
C	B	B	B	C	B	A	C	C	A
21	22	23	24	25	26	27	28	29	30
B	C	C	C	A	B	A	-	-	-

»واحذر داءَ الجَبابرةِ = (الكِبرَ)؛

فإنَّ الكِبرَ والحِرصَ والحَسدَ أوَّلُ ذنبٍ عُصِيَ اللهُ به،

فتَطاوُلُكَ على مُعلِّمِكَ كِبرِياءٌ، واستِنكافُكَ عَمَّن يُفيدُكَ ممَّن هو دُونَكَ كِبرِياءٌ، وتقصيرُكَ

عَنِ العَمَلِ بالعِلمِ حَمأةُ كِبرٍ وعُنوانُ حِرمانٍ.

العِلمُ حَربٌ للفَتى المُتعالِي

كالسَّيلِ حَربٌ للمكانِ العالِي

فالزَمْ -رحِمَكَ اللهُ- اللُّصوقَ إلى الأرضِ، والإزراءَ على نفسِكَ، وهَضمَها، ومُراغمَتَها عندَ

الاستِشرافِ لكِبرِياءٍ أو غَطرَسةٍ، أو حُبِّ ظُهورٍ، أو عُجبٍ...

ونحوِ ذلكَ مِن آفاتِ العِلمِ القاتِلةِ له، المُذهِبةِ لهَيبَتِه، المُطفِئةِ لنورِه، وكُلَّما ازدَدتَ علمًا أو رِفعةً

في ولايةٍ؛ فالزَمْ ذلك؛ تُحرِزْ سَعادةً عُظمَى، ومَقامًا يَغبِطُكَ عليهِ النَّاسُ.«

بكرٍ أبو زيدٍ -رحِمَهُ اللهُ.-

حِليَةُ طالبِ العِلمِ || ١٥

Lecture (11): Ketone Bodies Metabolism

Pre Exams , Formatives

1-	**Ketone bodies are synthesized in (FINAL 34, FINAL 32, SNU Bank)**						
A-	Adipose tissue	B-	Liver	C-	Muscles	D-	Brain

2-	**Which of the following is not included in keton bodies formation (FINAL 34)**		
A-	Thiolasethiokinase	B-	HGM COA synthetase
C-	HGM COA lyase	D-	HGM COA reductase

3-	**Before metabolism of acetoacetyl-CoA, it was? (2ND TRIAL 35)**		
A-	Acetone	B-	acetoacetate

4-	**Key regulatory enzyme in ketogenesis is? (2ND TRIAL 35)**
A-	HMG COA synthase

5-	**All the following statements correctly describe ketone bodies EXCEPT? (MID 30)**
A-	They may result from starvation
B-	They are present at high levels in uncontrolled diabetes
C-	They includE-OH-butyrate and acetone
D-	They are utilized by the liver during long term Starvation

6-	**the enzyme that is present in both ketogenesis and ketolysis is ..... (FINAL 36)**		
A-	Theokinase	B-	theophorase
C-	HMG coA synthase	D-	Beta hydroxybutyrate dehydrogenase

7-	**What is ketogenesis? (MID 37)**		
A-	Generation of acetyl-CoA from ketone bodies	B-	Generation of ketone bodies from acetyl-CoA
C-	Generation of acyl-CoA from ketone bodies	D-	Generation of ketone bodies from acyl-CoA

8-	**Why does liver not utilize ketone bodies (MID 37)**
A-	Absence of Thiophorase

9-	**Increases production of ketone bodies during: (FINAL 37)**		
A-	Protein consumption	B-	Prolonged fasting

10-	**Keton bodies form... (FINAL 37 SNU)**		
A-	Keto sugars	B-	Acetyl-CoA

11-	**Formation of ketone bodies: (Formative 37, Quiz SNU-M1, SNU Bank)**		
A-	Requires lactate dehydrogenase	B-	Starts from pyruvate
C-	Increases during fasting	D-	Occurs in the cytosol of the liver

12-	**All enzyme required for the synthesis of ketone bodies except (Formative 37, SNU Bank)**		
A-	HMG CoA synthetase	B-	Thiolase
C-	HMG CoA lyase	D-	HMG CoA reductase

13-	**What is the outcome of the accumulation of acetyl-CoA in the mitochondria of the liver? (FINAL 31, Department Qs37)**		
A-	It is used as an energy source	B-	It has broken down in to free fatty acids
C-	It gets converted to oxaloacetate	D-	It forms ketone bodies

14-	**All the following statements about ketone bodies are true EXCEPT? (FINAL 32)**		
A-	Their synthesis increases in diabetes mellitus	B-	They are synthesized in mitochondria
C-	They can deplete the alkali reserve	D-	They can be oxidized in the liver

15-	**All of the following tissue are capable of using ketone bodies EXCEPT ? (FINAL 32)**						
A-	skeletal muscles	B-	Renal cortex	C-	R-B-C	D-	Cardiac muscle

16-	**Keton bodies are by products of metabolism of ? (2ND TRIAL 32)**						
A-	Carbohydrates	B-	Protein	C-	Fat	D-	All of the above

Answers - Pre Exams , Formatives									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	**D**	**B**	**A**	**D**	**A**	**B**	**A**	**B**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	**D**	**D**	**D**	**C**	**C**	**-**	**-**	**-**	**-**

Department Qs

1-	**All of the following tissue are capable of using ketone bodies except (SNU Bank)**						
A-	Brain	B-	Renal cortex	C-	R-B-C-	D-	Cardiac muscle

2-	**Ketone bodies are: (SNU Bank)**
A-	Succinyl CoA, α-ketoglutarate and acetyl CoA.
B-	Oxaloacetate, acetaldehyde and malate.
C-	Acetoacetate, p-hydroxybutyrate and acetone.
D-	Non of the above.

3-	In prolonged starvation, the main energy source of brain is: (Department Qs37)						
A-	Glucose	B-	Ketone bodies	C-	Fructose	D-	Fatty acids

4-	Which of the following is not a ketone body: (Department Qs37)		
A-	Acetic acid	B-	Acetone
C-	betA-hydroxyButyric acid	D-	Aceto acetic acid

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	C	B	A	-	-	-	-	-	-

Lecture Qs

1-	ketogenesis occurs in (Lecture Qs37)						
A-	liver	B-	kidney	C-	Brain	D-	adipose tissues

2-	ALL of the following activates ketogenesis except..... (Lecture Qs37)		
A-	starvation	B-	Diabetes mellitus
C-	glucagon	D-	high carbohydrate diet

3-	The key enzyme of ketolysis is ..... (Lecture Qs37)		
A-	thiophorase	B-	HMG COA synthase
C-	HMG COA lyase	D-	fatty acid oxidase

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	D	A	-	-	-	-	-	-	-

M3WAN Bank

1-	The three compounds collectively known as ketone bodies are acetoacetic acid, acetone, and: (M3WAN - Core)		
A-	β-hydroxybutyric acid	B-	Pyruvic acid
C-	Lactic acid	D-	Malonic acid

2-	Ketogenesis occurs mainly in which organ, whose mitochondria contain the necessary enzymes? (Important M3WAN - Core)						
A-	Kidney	B-	Liver	C-	Skeletal muscle	D-	Heart

3-	**The net flow of ketone bodies from liver to extrahepatic tissues results from: (Important M3WAN - Core)**
A-	Active ketone body production in the liver coupled with low activity of their utilizing enzymes there
B-	Low production and high utilization in the liver
C-	Equal production and utilization capacity in liver and peripheral tissues
D-	Active transport pumps unique to hepatocytes

4-	**Acetoacetyl-CoA, the first intermediate of ketogenesis, can be formed by a reversal of the thiolase reaction, condensing two molecules of: (M3WAN - Core)**						
A-	Acetyl-CoA	B-	Malonyl-CoA	C-	Propionyl-CoA	D-	Succinyl-CoA

5-	**The major pathway for conversion of acetoacetyl-CoA to acetoacetate proceeds through condensation with acetyl-CoA to form an intermediate acted upon by: (Important M3WAN - Core)**		
A-	Thiophorase	B-	HMG-CoA synthase and HMG-CoA lyase
C-	Acetoacetate thiokinase	D-	Citrate synthase

6-	**Acetoacetate is reversibly converted to β-hydroxybutyrate by β-hydroxybutyrate dehydrogenase, using which coenzyme? (M3WAN - Core)**						
A-	FAD	B-	NADH + H+	C-	TPP	D-	Biotin

7-	**Acetone, one of the three ketone bodies, is formed by: (M3WAN - Core)**
A-	Enzymatic reduction of acetoacetate
B-	Spontaneous decarboxylation of acetoacetate
C-	Direct condensation of two acetyl-CoA molecules
D-	Oxidative decarboxylation by HMG-CoA lyase

8-	**The characteristic fruity (acetone) odor of the breath in uncontrolled diabetes reflects: (M3WAN - Core)**		
A-	Ketonemia	B-	Hyperglycemia alone
C-	Elevated blood urea	D-	Elevated bilirubin

9-	**Insulin decreases ketogenesis primarily by: (Important M3WAN - Core)**
A-	Directly inhibiting HMG-CoA synthase
B-	Inhibiting lipolysis, thereby decreasing plasma FFA delivered to the liver
C-	Stimulating glucagon release
D-	Increasing carnitine acyltransferase I activity

10-	**In the well-fed state, ketogenesis is inhibited in the liver mainly because: (Important M3WAN - Core)**
A-	Increased glycerol-3-phosphate favors esterification of fatty acids over β-oxidation
B-	Carnitine acyltransferase activity increases markedly
C-	Acetyl-CoA production is maximal
D-	Glucagon secretion rises

11-	**During starvation, increased glucagon secretion promotes ketogenesis mainly by: (Important M3WAN - Core)**
A-	Stimulating adipose tissue lipolysis and raising plasma FFA
B-	Increasing hepatic glycogen synthesis
C-	Inhibiting hormonE-sensitive lipase
D-	Increasing insulin secretion

12-	**The liver is unable to utilize the ketone bodies it produces because it lacks the enzymes: (Important M3WAN - Core)**
A-	HMG-CoA synthase and lyase
B-	Thiophorase and acetoacetate thiokinase
C-	Acyl-CoA dehydrogenase and thiolase
D-	β-hydroxybutyrate dehydrogenase and thiolase

13-	**Which of the following tissues can utilize ketone bodies as an important energy source, obtaining up to 75% of energy needs from acetoacetate during prolonged starvation? (Important M3WAN - Core)**						
A-	Liver	B-	Brain	C-	Erythrocytes	D-	Skin

14-	**The major pathway of ketolysis in extrahepatic tissues, transferring CoA from succinyl-CoA to acetoacetate, is catalyzed by: (Important M3WAN - Core)**
A-	Thiolase
B-	Thiophorase (succinyl-CoA-acetoacetate CoA transferase)
C-	HMG-CoA lyase
D-	Citrate lyase

15-	**The minor pathway of ketolysis activates acetoacetate directly using ATP and CoA, catalyzed by: (M3WAN - Hard)**		
A-	Acetoacetyl-CoA synthetase	B-	Thiophorase
C-	Thiolase	D-	HMG-CoA synthase

16-	**In the bioenergetics of ketolysis via the thiophorase pathway, there is an energetic "loss" equivalent to one GTP because: (M3WAN - Hard)**
A-	Succinyl-CoA, normally yielding GTP in the Krebs cycle, is diverted for CoA transfer
B-	ATP is consumed to activate acetoacetate
C-	NADH is not produced
D-	FADH2 formation is bypassed

17-	**The complete oxidation of one mole of acetoacetate via ketolysis and the Krebs cycle yields approximately how many moles of ATP? (Important M3WAN - Hard)**						
A-	12 mol	B-	17 mol	C-	23 mol	D-	26 mol

18-	**Because conversion of β-hydroxybutyrate to acetoacetate additionally generates one NADH (worth \~3 ATP), oxidation of β-hydroxybutyrate yields approximately: (M3WAN - Hard)**						
A-	20 mol ATP	B-	23 mol ATP	C-	26 mol ATP	D-	29 mol ATP

19-	**Ketosis is defined as an increase in blood ketone body levels above the normal range of approximately: (M3WAN - Core)**						
A-	0.3–2 mg/dL	B-	10–20 mg/dL	C-	50–100 mg/dL	D-	100–150 mg/dL

20-	**Ketoacidosis is best described as: (M3WAN - Core)**
A-	A type of respiratory alkalosis
B-	A type of metabolic acidosis resulting from ketosis
C-	A type of metabolic alkalosis
D-	Unrelated to aciD-base balance

21-	**The metabolic acidosis of severe ketosis develops because ketone bodies, being moderately strong acids, are neutralized mainly by: (Important M3WAN - Core)**		
A-	Hemoglobin buffering	B-	Bicarbonate, which becomes depleted
C-	Phosphate buffering exclusively	D-	Ammonia buffering exclusively

22-	**Severe ketoacidosis can cause hyperkalemia mainly because the accompanying fall in blood pH promotes: (M3WAN - Hard)**
A-	Movement of K+ from intracellular fluid into blood
B-	Renal K+ reabsorption
C-	Increased aldosterone secretion
D-	Movement of K+ from blood into cells

23-	Among the following, which is classified as a non-pathological cause of ketosis? (M3WAN - Core)		
A-	Diabetes mellitus	B-	Starvation
C-	Renal glucosuria	D-	Phlorhizin poisoning

24-	A diet very high in fat and very low in carbohydrate can precipitate ketosis; such a diet is typically composed of approximately what percentage of calories from fat? (M3WAN - Hard)						
A-	10%	B-	30%	C-	58%	D-	80%

25-	Diabetes mellitus is classified as a pathological cause of ketosis, and insulin is generally regarded as: (Important M3WAN - Core)		
A-	A ketogenic hormone	B-	An antiketotic hormone
C-	Having no relationship to ketogenesis	D-	A hormone that only affects ketolysis

26-	Comparing the ATP yield of palmitate metabolism, complete oxidation via the Krebs cycle yields approximately 129 mol ATP, whereas conversion to ketone bodies (acetoacetate) yields only approximately: (M3WAN - Hard)						
A-	23 mol ATP	B-	33 mol ATP	C-	80 mol ATP	D-	106 mol ATP

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
A	B	A	A	B	B	B	A	B	A
11	12	13	14	15	16	17	18	19	20
A	B	B	B	A	A	C	C	A	B
21	22	23	24	25	26	27	28	29	30
B	A	B	B	B	B	-	-	-	-

أرضُ الفطرةِ رحبةٌ قابلةٌ لما يُغرسُ فيها؛ فإن غرستَ شجرةَ الإيمانِ والتقوى أورثتَ حلاوةَ الأبد، وإن غرستَ شجرةَ الجهلِ والهوى فكلُّ الثمرِ مرٌّ!

• ابنُ القيّمِ . .

Lecture (12): Lipogenesis

Pre Exams , Formatives

1-	**Acetyl CoA carboxylase is activated by (FINAL 34, FINAL 36, FINAL 32, Department Qs33)**						
A-	Palmityl CoA	B-	Carbon dioxide	C-	Citrate	D-	Malonyl CoA

2-	**Chain elongation occure in: (FINAL 34, Formative 37, Quiz SNU-M1)**						
A-	Golgi apparatus	B-	Cytoplasm	C-	lysosomes	D-	Mitochondria

3-	**B ketoacyl residue attached to fatty acid synthase is reduced (FINAL 34)**						
A-	NADPH	B-	NADH	C-	FADH2	D-	Thiolase

4-	**During lipogenesis, the following sequential reactions occur: (FINAL 37 SNU, Formative 37, Quiz SNU-M1)**
A-	Oxidation, dehydration, oxidation, cleavage
B-	Dehydrogenation, hydration, dehydrogenation, cleavage
C-	Reduction, hydration, dehydrogenation, cleavage
D-	Reduction, dehydration, reduction, condensation

5-	**Higher rate of lipogenesis is associated with? (MID 30)**		
A-	High proportion of carbohydrate in diet	B-	Restricted caloric intake
C-	High fat diet	D-	Deficiency of insulin

6-	**A lipogenic hormone is? (MID 31)**				
A-	Glucagon	B-	Cortisol	C-	Insulin
D-	Epinephrine	E-	growth hormone		

7-	**key regulatory enzyme of lipogenesis is ..... (FINAL 36)**
A-	Acetyl CoA carboxylase

8-	**Chain elongation in..... (FINAL 37 SNU)**						
A-	Lysosomes	B-	Mitochondria	C-	Golgi	D-	Cytoplasm

9-	**Denovo synthesis of fatty acids occurs mainly in: (Formative 37, Quiz SNU-M1, SNU Bank)**		
A-	Cytoplasm	B-	Extra mitochondria
C-	Mitochondria	D-	Ribosomes

10-	NAD PH-H + used for? (FINAL 31)		
A-	Lipogenesis	B-	Lipolysis
C-	Both a , b	D-	fatty acid oxidation

11-	The key enzyme in fatty acid synthesis is? (FINAL 31)		
A-	Acetyl CoA carboxylase	B-	Beta hydroxyl acyl dehydratase
C-	Enoyl reductase	D-	Acetyl transacylase

12-	In microsomal synthesis of fatty acids, the source is? (FINAL 31)						
A-	Acetyl-COA	B-	Propionyl-COA	C-	Malonyl-COA	D-	Acetoacetyl-COA

13-	A precursor for fatty acid synthesis is? (FINAL 31)						
A-	Crotinyl-COA	B-	Acetoacetate	C-	Acetyl-COA	D-	Propionyl-COA

14-	The source of glycerol for fatty acid esterification in adipocytes is? (FINAL 31)
A-	Obtained primarily from phosphorylation of glycerol by glycerol kinase
B-	For the most part derived from glucose
C-	Formed by glycogenesis
D-	Inhibited by insulin stimulation

15-	For the de novo fatty acid synthesis. the co-enzyme required is ? (FINAL 32, Department Qs33)		
A-	NADH	B-	NADPH
C-	Thiamine pyrophosphate	D-	FADH2

16-	Limiting factor for fatty acid synthesis ? (2ND TRIAL 32)		
A-	Acetoacetate synthetase	B-	Acyl Transferase
C-	Acetyl CoA carboxylase	D-	Biotin

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
C	D	A	D	A	C	A	B	A	A
11	12	13	14	15	16	17	18	19	20
A	C	C	B	B	C	-	-	-	-

Department Qs

1-	During lipogenesis, the following sequential reactions (SNU Bank)
A-	Oxidation, dehydration, oxidation, cleavage

B-	Dehydrogenation, hydration, dehydrogenation, cleavage
C-	Reduction, hydration, dehydrogenation, cleavage
D-	Reduction, dehydration, reduction, condensation

2-	**Chain elongation occure in: (SNU Bank)**						
A-	lysosomes	B-	Microsomes	C-	Cytoplasm	D-	Golgi apparatus

3-	**Hepatic lipogenesis is stimulated by: (Department Qs33)**						
A-	CAMP	B-	Glucagon	C-	Epinephrine	D-	Insulin

4-	**Sources of NADPH for fatty acid synthesis include all the following reactions, except: (Department Qs33)**		
A-	GlucosE-6-phosphate dehydrogenase	B-	6-phospho gluconate dehydrogenase
C-	Cytoplasmic malate dehydrogenase	D-	Cytoplasmic isocitrate dehydrogenase

5-	**In the biosynthesis of triacylglycerol from glycerol phosphate, all the following are intermediates, except: (Department Qs33)**		
A-	Lysophosphatidic acid	B-	Phosphatidic acid
C-	1,2-diacyl glycerol	D-	Monoacyl glycerol

6-	**Chain elongation of fatty acids takes place in which of the following subcellular fraction: (Department Qs33)**						
A-	Cytoplasm	B-	Nucleus	C-	Microsomes	D-	Golgi complex

7-	**All of the following about the process of fatty acid biosynthesis are true, except: (Department Qs33)**		
A-	It occurs in the cytosol	B-	It requires NADPH
C-	It requires biotin	D-	It requires FADH2

8-	**The desaturation of essential fatty acids occurs in which of the following subcellular fractions: (Department Qs33)**						
A-	Nuclear	B-	Lysosomal	C-	Mitochondrial	D-	Microsomal

9-	**Fatty acid synthesis differs from beta oxidation in all, except: (Department Qs33)**		
A-	Uses NADPH as reducing power	B-	Require co-enzyme A
C-	Catalyzed by multienzyme complex	D-	Activated by insulin

10-	**Glucose will be converted into fatty acids if the diet has excess of (Department Qs33)**						
A-	Carbohydrates	B-	Proteins	C-	Fat	D-	Vitamins

11-	Which of the following enzyme is not used in the synthesis of triacylglycerol? (Department Qs37)		
A-	Glycerol-3-phosphate acyltransferase	B-	Acylglycerophophate acyltransferase
C-	Phosphatidic acid phosphohydrolase	D-	Glycogen phosphorylase

12-	Elongation of palmityl-COA to stearyl COA occurs mainly inside: (Department Qs37)						
A-	Microsomes	B-	Mitochondria	C-	Nucleus	D-	Cytosol

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	B	D	D	D	C	D	D	B	A
11	12	13	14	15	16	17	18	19	20
D	A	-	-	-	-	-	-	-	-

Lecture Qs

1-	Which of these is NOT a difference between fatty acid synthesis and beta oxidation? (Lecture Qs34)
A-	Synthesis requires an enzyme with a biotin cofactor, but oxidation does not.
B-	Synthesis utilizes NADP+/NADPH, but oxidation uses NAD+/NADH
C-	Synthesis takes place in the mitochondria, but oxidation takes place in cytosol.
D-	Synthesis is conducted on an enzyme complex, but oxidation enzymes are not associated.

2-	Which of the following is not a stage of fatty acid synthesis? (Lecture Qs34)		
A-	Condensation of precursors.	B-	Rearrangement.
C-	Reduction.	D-	Dehydration.

3-	Which of the following is the regulated step of fatty acid synthesis in eukaryotes? (Lecture Qs34)
A-	Carboxylation of acetyl CoA.
B-	Transportation of mitochondrial acetyl CoA into the cytosol.
C-	Assembly of the fatty acid chain.
D-	All of the above.

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	B	A	-	-	-	-	-	-	-

M3WAN Bank

1-	**Lipogenesis is best defined as: (M3WAN - Core)**
A-	The hydrolysis of triacylglycerols to free fatty acids and glycerol
B-	The biosynthesis of triacylglycerols
C-	The oxidative breakdown of fatty acids
D-	The conversion of ketone bodies to fatty acids

2-	**The steps of triacylglycerol biosynthesis include fatty acid biosynthesis, activation of fatty acids, biosynthesis of glycerol-3-phosphate, and: (M3WAN - Core)**
A-	β-oxidation of the newly formed fatty acids
B-	Combination of fatty acids and glycerol to form triacylglycerols
C-	Conversion of glycerol to pyruvate
D-	Formation of ketone bodies

3-	**The de novo (extramitochondrial) pathway of fatty acid synthesis mainly produces which fatty acid? (M3WAN - Core)**		
A-	Stearic acid (18C)	B-	Oleic acid (18:1)
C-	Palmitic acid (16C)	D-	Arachidonic acid (20:4)

4-	**De novo fatty acid synthesis occurs mainly in the cytosol of: (M3WAN - Core)**
A-	Liver and adipose tissue (and lactating mammary gland)
B-	Skeletal muscle only
C-	Kidney only
D-	Brain only

5-	**The two essential starting materials for de novo fatty acid synthesis are acetyl-CoA and: (Important M3WAN - Core)**						
A-	NADH	B-	NADPH	C-	FADH2	D-	GTP

6-	**Since mitochondria are impermeable to acetyl-CoA, it is transported to the cytosol for fatty acid synthesis mainly in the form of: (Important M3WAN - Core)**						
A-	Citrate	B-	Malate	C-	Pyruvate	D-	Oxaloacetate

7-	**In the cytoplasm, citrate is cleaved back into acetyl-CoA and oxaloacetate by: (M3WAN - Core)**		
A-	Citrate synthase	B-	Aconitase
C-	ATP-citrate lyase	D-	Isocitrate dehydrogenase

8-	The first and ratE-limiting step of fatty acid synthesis is catalyzed by: (Important M3WAN - Core)		
A-	Fatty acid synthase	B-	Acetyl-CoA carboxylase
C-	Thiolase	D-	ATP-citrate lyase

9-	Acetyl-CoA carboxylase catalyzes carboxylation of acetyl-CoA to form: (Important M3WAN - Core)		
A-	Malonyl-CoA	B-	Succinyl-CoA
C-	Propionyl-CoA	D-	Methylmalonyl-CoA

10-	The multienzyme complex that assembles palmitic acid from acetyl-CoA and malonyl-CoA units, consisting of seven covalently linked enzyme activities, is: (M3WAN - Core)		
A-	Pyruvate dehydrogenase complex	B-	α-Ketoglutarate dehydrogenase complex
C-	Fatty acid synthase complex	D-	Carnitine shuttle complex

11-	Acetyl-CoA carboxylase is allosterically activated by citrate, which signals: (Important M3WAN - Core)
A-	A well-fed state with plentiful acetyl-CoA supply
B-	Cellular energy depletion
C-	Active β-oxidation
D-	Ketosis

12-	Acetyl-CoA carboxylase is allosterically inhibited by: (Important M3WAN - Core)		
A-	Insulin directly	B-	Long-chain acyl-CoA
C-	GlucosE-6-phosphate	D-	Pyruvate

13-	Covalent regulation of acetyl-CoA carboxylase is such that the active form is the: (Important M3WAN - Core)		
A-	Phosphorylated form	B-	Dephosphorylated form
C-	Ubiquitinated form	D-	Glycosylated form

14-	Glucagon influences lipogenesis by: (Important M3WAN - Core)
A-	Stimulating fatty acid synthesis and inhibiting β-oxidation
B-	Inhibiting fatty acid synthesis while promoting lipid breakdown and β-oxidation
C-	Having no effect on lipid metabolism
D-	Directly activating fatty acid synthase

15-	Insulin promotes lipogenesis mainly by causing dephosphorylation and activation of: (Important M3WAN - Core)		
A-	HormonE-sensitive lipase	B-	Acetyl-CoA carboxylase
C-	Carnitine acyltransferase I	D-	Thiophorase

16-	**Malonyl-CoA, the product of the ratE-limiting step of lipogenesis, plays a key regulatory role by: (Important M3WAN - Hard)**
A-	Activating hormonE-sensitive lipase
B-	Inhibiting carnitine acyltransferase I (CPT1), thereby suppressing β-oxidation during active lipogenesis
C-	Directly activating β-oxidation
D-	Stimulating ketogenesis

17-	**After its synthesis, palmitate may undergo esterification, desaturation, or: (M3WAN - Core)**
A-	Elongation to form longer-chain fatty acids (e-g-, C18, C20, C22, C24)
B-	Immediate excretion in bile
C-	Conversion to ketone bodies exclusively
D-	Direct conversion to cholesterol

18-	**Mitochondrial chain elongation of fatty acids, which essentially reverses β-oxidation, becomes active mainly under conditions of: (M3WAN - Hard)**
A-	High NADH+H+/NAD+ ratio (as in anaerobic conditions)
B-	High oxygen availability
C-	Low acetyl-CoA availability
D-	High insulin activity only

19-	**Microsomal chain elongation of fatty acids, producing C22 and C24 fatty acids for incorporation into sphingolipids, is particularly active during: (M3WAN - Hard)**		
A-	Fasting	B-	Nerve myelination
C-	Ketosis	D-	Anaerobic exercise

20-	**Desaturases introduce double bonds into fatty acid chains, but mammalian cells cannot introduce a double bond at which position, making certain fatty acids dietary essentials? (Important M3WAN - Core)**						
A-	Δ9	B-	Δ12	C-	Δ5	D-	Δ2

21-	**Triacylglycerol synthesis occurs mainly in the microsomes of the liver, kidney, intestine, adipose tissue, and: (M3WAN - Core)**		
A-	Lactating mammary gland	B-	Erythrocytes
C-	Skeletal muscle exclusively	D-	Cornea

22-	**Activation of glycerol to glycerol-3-phosphate by glycerol kinase occurs efficiently in liver, kidney, and intestine, but this enzyme is absent or low in: (Important M3WAN - Core)**		
A-	Adipose tissue and muscle	B-	Lactating mammary gland
C-	Pancreas	D-	Liver

23-	**In muscle and adipose tissue, glycerol-3-phosphate for triacylglycerol synthesis is instead derived from the glycolytic intermediate dihydroxyacetone phosphate via: (M3WAN - Core)**		
A-	Glycerol-3-phosphate dehydrogenase	B-	Glycerol kinase
C-	Aldolase	D-	Phosphoglycerate kinase

24-	**Activation of fatty acids prior to their incorporation into triacylglycerols requires ATP and CoA-SH and is catalyzed by: (M3WAN - Core)**		
A-	Acyl-CoA synthetase	B-	Acetyl-CoA carboxylase
C-	Fatty acid synthase	D-	Thiolase

25-	**The reaction catalyzed by acyl-CoA synthetase during fatty acid activation releases which by-products, in addition to acyl-CoA? (M3WAN - Hard)**		
A-	AMP and pyrophosphate	B-	ADP and phosphate
C-	GDP and phosphate	D-	NAD+ and CO2

26-	**Regarding the three systems for fatty acid synthesis in mammals, which is correctly matched with its subcellular site? (M3WAN - Core)**		
A-	De novo synthesis — mitochondria	B-	Chain elongation — cytosol only
C-	De novo synthesis — cytosol	D-	Desaturation — nucleus

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	B	C	A	B	A	C	B	A	C
11	12	13	14	15	16	17	18	19	20
A	B	B	B	B	B	A	A	B	B
21	22	23	24	25	26	27	28	29	30
A	A	A	A	A	C	-	-	-	-

عَنِ الحَسَنِ البصريِّ -رحمَهُ اللهُ- قالَ:

"ابنَ آدمَ إنَّما أنتَ أيامٌ كُلَّما ذهبَ يومٌ ذهبَ بعضُكَ."

-روضةُ الزاهدين 📖 .

Lecture (13): Biochemistry of amino acids and proteins (structure and function)

Pre Exams , Formatives

1-	Tertiary structure of a protein describes? (MID 30, FINAL 35)		
A-	The order of amino acids	B-	Location of disulphide bonds
C-	Loop regions of proteins	D-	The protein folding and organization.

2-	The transfer of A-amino group to A-ketoglutarate is called (FINAL 34, 2ND TRIAL 35, FINAL 32)						
A-	transamination	B-	decarboxylation	C-	Deamination	D-	transmethylation

3-	Which one of the following is semi-essential amino acid for humans? (FINAL 35, Lecture Qs34)						
A-	Valine	B-	Arginine	C-	Lysine	D-	Tyrosine

4-	All of the following are sulfur containing amino acids EXCEPT (2ND TRIAL 35)						
A-	Cysteine	B-	Threonine	C-	Cystine	D-	Methionine

5-	transfer of amine group between amino acids is called (Formative 37, Quiz SNU-M1)						
A-	deamination	B-	transformation	C-	transmethylation	D-	transamination

6-	Pyruvic acid results from transamination of.....? (MID 30)						
A-	L-alanine	B-	L-serine	C-	glycine	D-	L-aspartic acid

7-	Which among the following is a nutritionally essential amino acid for man ? (MID 30)						
A-	Alanine	B-	Glycine	C-	Tyrosine	D-	Tryptophan

8-	Which of the listed proteins are simple ? (MID 31)								
A-	Albumins	B-	Hemoglobin	C-	Cazein	D-	Myoglobin	E-	Chlorophil

9-	Primary structure of proteins is formed thanks to one type of bonds. Point out it? (MID 31)				
A-	Peptide bond	B-	Disulfide bond	C-	Ether bond
D-	Hydrogen bond	E-	Metal bond		

10-	In scurvy, which amino acid that is normally part of collagen is not synthesized ? (MID 31)				
A-	Hydroxy Tryptophan	B-	Hydroxy Tyrosine	C-	Hydroxy Alanine
D-	Hydroxy Proline	E-	hydroxyl serine		

11-	One of the following amino acid is pure glycogenic? (MID 33, Lecture Qs34)						
A-	Leucine	B-	Lysine	C-	Tyrosine	D-	Alanine

12-	**Globin protein of hemoglobin is an example of .....? (MID 33, Lecture Qs34)**		
A-	Primary protein structure	B-	Secondary protein structure
C-	Tertiary protein structure	D-	Quaternary protein structure

13-	**Excess accumulation of β-amyloid b protein in central nervous system is associated with .....? (MID 33)**
A-	Huntington disease
B-	Parkinsonism
C-	Alzheimer-s disease
D-	Amyotrophic lateral sclerosis

14-	**the only ketogenic amino acid is (MID 34)**
A-	Leucine

15-	**amino acids are unique due to (MID 34)**						
A-	side chain	B-	Amino group	C-	Hydrogen atom	D-	Carboxylic group

16-	**Which of the following is optically inactive (MID 34)**
A-	glyceine

17-	**tripeptide is a peptide derived from (MID 34)**
A-	3 amino acids and 2 peptide bonds

18-	**Which protein structure is not liable to denaturation: (FINAL 36)**						
A-	primary	B-	secondary	C-	tertiary	D-	quaternary

19-	**glutamate is converted to proline by ..... (FINAL 36)**		
A-	Cyclization and oxidation	B-	Cyclization and reduction
C-	Oxidation	D-	Reduction

20-	**The primary reaction in tranamination?: (FINAL 37)**
A-	a transfer amino group between alpha keto acid and amino acid
B-	transfer carboxyl group between two amino acids
C-	Oxiditive of side chain in amino acid

21-	**Glutathione is? (Formative 35, FINAL 31, MID 32, Department Qs 35)**				
A-	Polypeptide	B-	contain 2 peptide bond	C-	non of the above

22-	**Which of the following compounds can not participate in transamination reactions? (Formative 35)**						
A-	Succinate	B-	Aspartate	C-	Pyruvate	D-	Alanine

23-	Which bond is present in the primary structure of protein? (Formative 35)						
A-	Hydrogen	B-	Ester bond	C-	Peptide bond	D-	Disulfide bond

24-	An essential amino acid in man is? (Formative 35)						
A-	Proline	B-	Tyrosine	C-	Threonine	D-	Aspartate

25-	which of the following bonds is present between amino acids (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	peptide bond	B-	ester link	C-	glycoside link	D-	glycolytic link

26-	unit of formation of proteins is... (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	amino acid	B-	glycerol	C-	fatty acids	D-	sugar

27-	transporter of amino group between amino acids is called (Quiz SNU-M1,SNU Bank)						
A-	amino acids	B-	aminotransferase	C-	alcohol	D-	fatty acids

28-	Removal of amino group from the amino acids to , - keto acids called? (FINAL 31)		
A-	Transamination	B-	Deamination
C-	Trans - deamination	D-	Trans methylation

29-	Tryptophan is? (FINAL 31)		
A-	Essential amino acid	B-	Both glucogenic and ketogenic
C-	Both a , b	D-	Semi essential

30-	Glutamic acid is? (FINAL 31)		
A-	Acidic amino acid	B-	Non essential amino acid
C-	Both a , b	D-	Semi essential

31-	Albumin is? (FINAL 31)		
A-	Insoluble in water	B-	Heat coagulable protein
C-	A plant protein	D-	A protein of low biological value

32-	Albumins and globulins are defined as? (FINAL 31)		
A-	Derived protein	B-	Conjugated protein
C-	Fibrous protein	D-	Globular protein

33-	A protein rich in proline and hydroxy proline is? (FINAL 31)						
A-	Globin	B-	Collagen	C-	Casein	D-	Histone

34-	Glycine Is characterized by? (FINAL 31)		
A-	Absence of an asymmetric carbon	B-	Absence of optical activity
C-	The shortest amino acid	D-	All of these

35-	which one of the following amino acids is an imino acid? (FINAL 31)						
A-	valine	B-	Leucine	C-	Threonine	D-	Proline

36-	Primary structure of a protein is formed by? (MID 32)						
A-	Hydrogen bonds	B-	Peptide bonds	C-	Disulphide bonds	D-	All of these

37-	A protein rich in cysteine is? (MID 32)						
A-	Collagen	B-	Keratin	C-	Haemoglobin	D-	Gelatin

38-	Collagen is very rich in? (MID 32)						
A-	Glycine	B-	Serine	C-	Aspartic acid	D-	Glutamic acid

39-	α-Ketoglutarate after accepting an amino group becomes ? (FINAL 32)
A-	Ketoglutarate
B-	glycine
C-	glutamic acid
D-	aspartic acid
E-	tyrosine

40-	Glycine and proline are the most abundant amino acids in the structure of ? (FINAL 32)						
A-	Hemoglobin	B-	Myoglobin	C-	Insulin	D-	Collagen

41-	Collagen is rich in ? (2ND TRIAL 32, Department Qs37)		
A-	Proline and glycine	B-	Glutamate and glycine
C-	Alanine and glycine	D-	Glutamate and proline

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
D	A	B	B	D	A	D	A	A	D
11	12	13	14	15	16	17	18	19	20
D	D	C	A	A	A	A	A	B	A
21	22	23	24	25	26	27	28	29	30
B	A	C	C	A	A	B	B	C	C

31	32	33	34	35	36	37	38	39	40
B	D	B	D	D	B	B	A	B	D
41	42	43	44	45	46	47	48	49	50
A	-	-	-	-	-	-	-	-	-

Department Qs

1-	hydroxy amino acid is (SNU Bank)						
A-	glycine	B-	serine	C-	histamine	D-	alanine

2-	which of the following bond is the strongest in protein structure (SNU Bank)		
A-	hydrogen bond	B-	electrostatic bond
C-	peptide bond	D-	ester bond

3-	basic amino acid is (SNU Bank)						
A-	tryptophan	B-	alanine	C-	glycine	D-	arginine

4-	phenylalanine is..... (SNU Bank)		
A-	non essential amino acids	B-	vitamin
C-	enzyme	D-	essential amino acid

5-	tyrosine is..... (SNU Bank)		
A-	aromatic amino acid	B-	basic amino acid
C-	acidic amino acid	D-	sulfur containing amino acid

6-	Which one Of The Following Is Aromatic Amino Acid. (Department Qs33)						
A-	Phenyle Alanine	B-	Glycine	C-	Lysine	D-	Methonine

7-	All proteins contain the (Department Qs33, Lecture Qs34)		
A-	Same 20 amino acids	B-	Different amino acids
C-	300 Amino acids occurring in nature	D-	Only a few amino acids

8-	Proteins contain (Department Qs33)		
A-	Only L - amino acids	B-	Only D-amino acids
C-	DL-Amino acids	D-	Both (A) and (B)

9-	Sulphur containing amino acid is (Department Qs33, Lecture Qs34)						
A-	Methionine	B-	Leucine	C-	Valine	D-	Asparagine

10-	An example of sulphur containing amino acid is (Department Qs33)		
A-	2-Amino-3-mercaptopropanoic acid	B-	2-Amino-3-methylbutanoic acid
C-	2-Amino-3-hydroxypropanoic acid	D-	Amino acetic acid

11-	All the following are sulphur containing amino acids found in proteins except (Department Qs33, Lecture Qs34)						
A-	Cysteine	B-	Cystine	C-	Methionine	D-	Threonine

12-	An aromatic amino acid is (Department Qs33, Lecture Qs34)						
A-	Lysine	B-	Tyrosine	C-	Taurine	D-	Arginine

13-	An essential amino acid in man is (Department Qs33, Lecture Qs34)						
A-	Aspartate	B-	Tyrosine	C-	Methionine	D-	Serine

14-	Non essential amino acids (Department Qs33, Lecture Qs34)
A-	Are not components of tissue proteins
B-	May be synthesized in the body from essential amino acids
C-	Have no role in the metabolism
D-	May be synthesized in the body in diseased states

15-	Which one of the following is semiessential amino acid for humans? (Department Qs33)		
A-	Valine	B-	Arginine + Histadine
C-	Lysine	D-	Tyrosine

16-	An example of polar amino acid is (Department Qs33)						
A-	Alanine	B-	Leucine	C-	Arginine	D-	Valine

17-	The amino acid with a nonpolar side chain is (Department Qs33)						
A-	Serine	B-	Valine	C-	Asparagine	D-	Threonine

18-	Which of the following is a tripeptide? (Department Qs33)						
A-	Anserine	B-	Oxytocin	C-	Glutathione	D-	Kallidin

19-	Glycine is a characterized by: (Department Qs 35)		
A-	Having (H) as a radical	B-	having (OH) as a radical
C-	Non of the above		

20-	which of the following describe serine A-a: (Department Qs 35)		
A-	Acidic amino acids	B-	have hydroxyethyl as aside chain

C-	Have hydroxymetly as aside chain

21-	**All the following amino acids are basic except: (Department Qs 35)**				
A-	Argnine	B-	Lysine	C-	valine

22-	**All the following amino acids are aromatic except: (Department Qs 35)**				
A-	Phenylalanine	B-	tryptophan	C-	glycine

23-	**which of the following amino acids is essential: (Department Qs 35)**		
A-	Tyrosine	B-	phenylalanine

24-	**which of the following amino acids is nonessential: (Department Qs 35)**				
A-	Serine	B-	threonine	C-	lysine

25-	**Arginine amino acid is: (Department Qs 35)**				
A-	Essential	B-	semiessential	C-	no of the above

26-	**All amino acids in human are: (Department Qs 35)**				
A-	B-amino acids	B-	L-amino acids	C-	D-amino acids

27-	**which amino acid is purely ketogenic: (Department Qs 35)**				
A-	Leucine	B-	Isoleucine	C-	alanine

28-	**All these A-a can give either glucose, ketone bodies except: (Department Qs 35)**				
A-	Isoleucine	B-	phenylalanine	C-	valine

29-	**which amino acid destabilizes a helix: (Department Qs 35)**				
A-	Valine	B-	proline	C-	leucine

30-	**a helix is: (Department Qs 35)**		
A-	Three dimential structure	B-	resist denaturation
C-	Is a type of 2 n dry protein structure		

31-	**tertiary structure is: (Department Qs 35)**
A-	Called native protein
B-	held by peptide bond
C-	Protein consist of 2 or more polypeptide chain linked together

32-	keratin is rich in: (Department Qs 35)				
A-	Cystein	B-	valine	C-	tryptophan

33-	All of the following proteins have high biological value except: (Department Qs 35)				
A-	Albumin	B-	globulin	C-	collagen

34-	The repeating units of proteins are (Department Qs 35)						
A-	glucose units	B-	amino acids	C-	fatty acids	D-	peptides

35-	which of the following is basic protein: (Department Qs 35)				
A-	Elastin	B-	Albumin	C-	histones

36-	The primary structure of protein represents (Department Qs 35)
A-	Linear sequence of amino acids joined by peptide bond
B-	3-dimensional structure of protein
C-	helical structure of protein
D-	sub unit structure of protein

37-	The simplest amino acid is (Department Qs 35)						
A-	Proline	B-	methionine	C-	glycine	D-	serine

38-	Aromatic amino acids include (Department Qs 35)		
A-	Phenylalanine, tyrosine and tryptophan	B-	Phenylalanine, serine and tryptophan
C-	Threonine, tyrosine and tryptophan	D-	Asparagine, tyrosine and tryptophan

39-	Indol ring is present in: (Department Qs37)						
A-	Tryptophan	B-	Valine	C-	Histidine	D-	Methionine

40-	With respect to proteins, the primary structure refers to the: (Department Qs37)
A-	helix or pleated sheet pattern of a polypeptide
B-	complex 3-dimensional folding pattern of a polypeptide
C-	interaction of more than one polypeptide chain
D-	ordered sequence of amino acids in a polypeptide

41-	One of the following is non essential amino acid: (Department Qs37)						
A-	Tyrosine	B-	Cystine	C-	Methionine	D-	Valine

42-	Glutathione is an example for: (Department Qs37)						
A-	Dipeptide	B-	Polypeptide	C-	Protein	D-	Tripeptide

43-	Aspartic acid is a (an): (Department Qs37)		
A-	Monoamino dicarboxylic acid	B-	Diamino monocarboxylic acid
C-	Aromatic amino acid	D-	Imino acid

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	C	D	D	A	A	A	A	A	A
11	12	13	14	15	16	17	18	19	20
D	B	C	B	B	C	B	C	A	C
21	22	23	24	25	26	27	28	29	30
C	C	B	A	B	B	A	C	B	C
31	32	33	34	35	36	37	38	39	40
A	A	C	B	C	A	C	A	A	D
41	42	43	44	45	46	47	48	49	50
A	D	A	-	-	-	-	-	-	-

Lecture Qs

1-	All the following are aromatic amino acids except (Lecture Qs33, Lecture Qs37)		
A-	TYROSINE	B-	TRYPTOPHAN
C-	PHENYLALANINE	D-	GLYCINE

2-	Proteins contain (Lecture Qs34)				
A-	Only L amino acids	B-	Only D amino acids	C-	D L Amino acids

3-	An example of sulphur containing amino acid is (Lecture Qs34)		
A-	2 Amino 3 mercaptopropionic acid	B-	2 Amino 3 methyl butanoic acid
C-	2 Amino 3 hydroxy propanoic acid	D-	Amino acetic acid

4-	An example of polar amino acid is (Lecture Qs34)				
A-	Alanine	B-	Leucine	C-	Arginine

5-	Excess accumulation of B-amyloid b protein in central nervous system is associated with (Lecture Qs34)				
A-	Huntington disease	B-	Parkinsonism	C-	Alzheimer-s disease

6-	Organic compound that contains carbon, hydrogen, oxygen, nitrogen, and in some cases sulfur: (Lecture Qs35)

A-	Protein	B-	Lipid	C-	Carbohydrate	D-	Nucleic acid

7-	**A protein may have up to: (Lecture Qs35)**		
A-	Three levels of structure	B-	Four levels of structure
C-	Five levels of structure	D-	two levels of structure

8-	**The tertiary structure of a protein describes: (Lecture Qs35)**		
A-	The sequence of amino acids	B-	Location of disulfide bonds
C-	Amino-terminal-end amino acid	D-	The nature of protein folding

9-	**The primary structure of a protein describes: (Lecture Qs35)**		
A-	The sequence of amino acids	B-	Location of disulfide bonds
C-	Amino-terminal-end amino acid	D-	The nature of protein folding

10-	**when the hydrogen bonds formed inside the same polypeptide chain resulting in the formation of: (Lecture Qs35)**		
A-	helical structure	B-	sheets
C-	globular structure	D-	fibrous structure

11-	**when the hydrogen bonds formed between different polypeptide chains resulting in the formation of: (Lecture Qs35)**		
A-	helical structure	B-	sheets
C-	globular structure	D-	fibrous structure

Answers - Lecture Qs									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	**A**	**A**	**C**	**C**	**A**	**B**	**D**	**A**	**A**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

M3WAN Bank

1-	**Peptide chains containing fewer than 50 amino acids are termed peptides or polypeptides, while the term protein is reserved for chains exceeding: (M3WAN - Core)**						
A-	20 amino acids	B-	50 amino acids	C-	100 amino acids	D-	10 amino acids

2-	**All of the following are biological functions of proteins described in the textbook EXCEPT: (M3WAN - Core)**
A-	Maintaining osmotic pressure and water balance
B-	Acting as coagulation factors in blood clotting
C-	Storing genetic information as triplet codons
D-	Controlling gene expression during replication and translation

3-	**Regarding the stereochemistry of amino acids found in mammalian proteins, they are all: (M3WAN - Core)**		
A-	AlphA-amino acids in the L'configuration	B-	BetA-amino acids in the D'configuration
C-	AlphA-amino acids in the D'configuration	D-	BetA-amino acids in the L'configuration

4-	**An amino acid possessing one amino group and two carboxylic groups is chemically classified as: (M3WAN - Core)**						
A-	Acidic	B-	Basic	C-	Neutral	D-	Aromatic

5-	**All of the following are classified as basic amino acids EXCEPT: (M3WAN - Core)**						
A-	Lysine	B-	Ornithine	C-	Histidine	D-	Glutamic acid

6-	**Tyrosine, which contains a phenyl ring in its side chain, is structurally classified as: (M3WAN - Core)**						
A-	Aliphatic	B-	Basic	C-	Aromatic	D-	Heterocyclic

7-	**Which amino acid contains a heterocyclic ring in its side chain? (M3WAN - Core)**						
A-	Glycine	B-	Tyrosine	C-	Lysine	D-	Tryptophan

8-	**Glycine, which has no ring structure in its side chain, is structurally classified as: (M3WAN - Core)**						
A-	Aliphatic	B-	Aromatic	C-	Heterocyclic	D-	Basic

9-	**Which of the following amino acids contains a hydroxyl group in its side chain? (M3WAN - Core)**						
A-	Leucine	B-	Threonine	C-	Alanine	D-	Isoleucine

10-	Hydroxyproline and hydroxylysine are grouped together with serine and threonine because all of them possess: (Important M3WAN - Core)		
A-	A hydroxyl group on their side chain	B-	A phenyl ring
C-	Two amino groups	D-	A heterocyclic ring

11-	Which of the following is a nutritionally essential amino acid that must be supplied in the diet? (M3WAN - Core)						
A-	Alanine	B-	Tyrosine	C-	Glycine	D-	Histidine

12-	Which amino acid is classified metabolically as both ketogenic and glucogenic? (Important M3WAN - Hard)						
A-	Tryptophan	B-	Lysine	C-	Isoleucine	D-	Leucine

13-	Leucine is classified metabolically as: (M3WAN - Core)		
A-	Glucogenic only	B-	Ketogenic and glucogenic
C-	Neither ketogenic nor glucogenic	D-	Purely ketogenic

14-	A peptide formed of three amino acids linked by two peptide bonds is called : (M3WAN - Core)				
A-	Dipeptide	B-	Tripeptide	C-	Oligopeptide
D-	Polypeptide				

15-	A chain of 10 to 50 amino acids linked by peptide bonds is classified as: (M3WAN - Hard)				
A-	Polypeptide	B-	Tripeptide	C-	Oligopeptide
D-	Protein				

16-	Glutathione is a biologically active tripeptide composed of glutamate, cysteine and: (M3WAN - Core)						
A-	Glycine	B-	Alanine	C-	Serine	D-	Lysine

17-	Glutathione inactivates the insulin hormone through which mechanism? (M3WAN - Hard)		
A-	Hydrolyzing its peptide bonds	B-	Breaking its disulfide bonds
C-	Phosphorylating its tyrosine residues	D-	Competing for its membrane receptor

18-	Glutathione protects the cell membrane from damage and prevents hemolysis mainly because it acts as: (M3WAN - Core)
A-	A structural protein
B-	An antioxidant that prevents lipid peroxidation
C-	A transport protein for lipids
D-	A hormone receptor

19-	**The pH at which a protein carries equal numbers of positive and negative charges, with no net electric charge, is called its: (Important M3WAN - Core)**		
A-	Melting point	B-	Isoelectric point
C-	Buffering point	D-	Denaturation point

20-	**At its isoelectric point, a protein typically shows: (M3WAN - Core)**		
A-	Maximum solubility	B-	Minimum solubility and tends to precipitate
C-	Maximum net charge	D-	No change in solubility

21-	**Separation of amino acids by the precipitation method depends mainly on: (M3WAN - Hard)**
A-	Maximizing the net charge on the amino acid
B-	Increasing solubility at extreme pH
C-	Denaturing the amino acid with heat
D-	Neutralizing the amino acid at its isoelectric point

22-	**In an acidic medium, proteins become: (M3WAN - Core)**
A-	Positively charged and migrate to the cathode
B-	Negatively charged and migrate to the anode
C-	Neutral and remain stationary
D-	Negatively charged and migrate to the cathode

23-	**The primary structure of a protein is formed and maintained mainly by: (Important M3WAN - Core)**		
A-	Peptide bonds	B-	Hydrogen bonds
C-	Disulfide bonds	D-	Hydrophobic bonds

24-	**Which level of protein structure describes the overall three-dimensional folding of a single polypeptide chain? (Important M3WAN - Core)**						
A-	Primary	B-	Secondary	C-	Tertiary	D-	Quaternary

25-	**The association of the globin chains of hemoglobin into one functional unit is an example of which level of protein structure? (M3WAN - Core)**						
A-	Primary	B-	Secondary	C-	Tertiary	D-	Quaternary

26-	**Loss of the native form of a protein by denaturation disrupts which structural levels? (Important M3WAN - Hard)**
A-	Primary only
B-	Secondary, tertiary and quaternary, but not primary
C-	Primary and secondary only

D-	All four levels equally

27-	**During protein denaturation, which of the following bonds are lost? (M3WAN - Hard)**
A-	Peptide and disulfide bonds
B-	Only peptide bonds
C-	Only disulfide bonds
D-	Hydrogen, hydrophobic and electrostatic bonds

28-	**Which of the following is a physical agent capable of causing protein denaturation? (M3WAN - Core)**		
A-	Repeated thawing and freezing	B-	Concentrated acid
C-	Heavy metal salts	D-	Organic solvents

29-	**Which of the following is classified as a chemical agent of protein denaturation? (M3WAN - Core)**		
A-	Strong alkalis	B-	High temperature
C-	X-ray radiation	D-	Digestive enzymes

30-	**Denaturation of a protein typically produces which of the following physical changes? (M3WAN - Hard)**		
A-	Increased viscosity	B-	Increased solubility
C-	Increased diffusibility	D-	Decreased viscosity

31-	**A protein of high biological value is characterized by all of the following EXCEPT: (M3WAN - Core)**
A-	Mainly animal origin
B-	Easy digestibility
C-	Well-balanced amounts of essential amino acids
D-	Deficiency in one or more essential amino acids

32-	**Proteins with an axial ratio greater than 10, whose polypeptide chains lie parallel to one another, are classified as: (Important M3WAN - Core)**		
A-	Conjugated proteins	B-	Fibrous proteins
C-	Globular proteins	D-	Derived proteins

33-	**Which of the following is an example of a globular protein? (M3WAN - Core)**						
A-	Keratin	B-	Collagen	C-	Insulin	D-	Myosin

34-	**Complete hydrolysis of a simple protein yields: (M3WAN - Core)**		
A-	Amino acids and carbohydrates only	B-	Amino acids only
C-	Amino acids and lipids only	D-	Amino acids and nucleic acids only

35-	Keratin, the scleroprotein found in hair, nail and skin, is characteristically rich in which amino acid? (M3WAN - Hard)						
A-	Glycine	B-	Proline	C-	Cysteine	D-	Lysine

36-	The molecular structure of collagen consists of: (M3WAN - Hard)
A-	A single globular polypeptide chain
B-	Three polypeptide chains arranged in a triple helix
C-	Four polypeptide chains forming a tetramer
D-	A branched polysaccharide-protein complex

37-	Gelatin differs from collagen mainly in that gelatin: (M3WAN - Core)
A-	Is insoluble and non-digestible
B-	Is richer in cysteine
C-	Is a conjugated protein
D-	Is a boiling product of collagen that is easily digestible

38-	Elastin, a scleroprotein, is mainly found in which of the following tissues? (M3WAN - Hard)		
A-	Hair, nail and skin	B-	Bone and cartilage
C-	Lung, uterus, tendons and large arteries	D-	Spleen, liver and lymph nodes

39-	Casein, the milk protein, is classified as a conjugated protein of which type? (M3WAN - Core)						
A-	Phosphoprotein	B-	Lipoprotein	C-	Glycoprotein	D-	Metalloprotein

40-	Ferritin and transferrin are classified as metalloproteins because they are conjugated with: (M3WAN - Core)						
A-	Copper	B-	Lipid	C-	Iron	D-	Carbohydrate

41-	Hemoglobin and cytochrome, which are conjugated with a pigment, belong to which class of conjugated proteins? (M3WAN - Hard)						
A-	Glycoproteins	B-	Chromoproteins	C-	Lipoproteins	D-	Nucleoproteins

42-	**Which of the following is a recognized function of glycoproteins according to the textbook? (Important M3WAN - Core)**
A-	Acting as mucins that lubricate and protect epithelial linings of the gastrointestinal, urogenital and respiratory tracts
B-	Carrying oxygen from the lungs to the tissues
C-	Forming the triple helix of connective tissue
D-	Acting as the main uncoupler of oxidative phosphorylation

43-	**Secondary derived proteins, such as proteoses, peptones and polypeptides, are produced through: (M3WAN - Core)**		
A-	Conjugation with a non-protein moiety	B-	Denaturation of the native protein
C-	Oxidation of the peptide bond	D-	Hydrolysis of protein

Answers -									
1	2	3	4	5	6	7	8	9	10
B	C	A	A	D	C	D	A	B	A
11	12	13	14	15	16	17	18	19	20
D	B	D	B	A	A	B	B	B	B
21	22	23	24	25	26	27	28	29	30
D	A	A	C	D	B	D	A	A	A
31	32	33	34	35	36	37	38	39	40
D	B	C	B	C	B	D	C	A	C
41	42	43	44	45	46	47	48	49	50
B	A	D	-	-	-	-	-	-	-

"الإخلاصُ للهِ = أن يكون اللهُ هو مقصودُ المرءِ ومُرادِه، فحينئذٍ تتفجَّرُ ينابيعُ الحكمة من قلبِه على لسانه."

النبوات لابن تيمية | | صـ ١٤٧

Lecture (14): Decarboxylated amino acid (Biogenic amines)

Pre Exams , Formatives

1-	Which of the following amino acids produce a vasoconstrictor on decarboxylation? (MID 30)						
A-	Histidine	B-	Tyrosine	C-	Threonine	D-	Arginine

2-	histamine substance is formed of (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	hydroxy proline	B-	Proline	C-	histidine	D-	glycine

3-	Which of the following gives rise to y-amino butyrate? (FINAL 31, FINAL 32)		
A-	Glutamate synthase	B-	Glutamate decarboxylation
C-	Glutamine synthase	D-	Pyruvate decarboxylation

4-	Major donor of methyl group in the body is? (FINAL 31)		
A-	S-adenosyl – methionine	B-	Ethanolamine
C-	Non of the above	D-	cysteine

5-	Decarboxylation of Histidine gives? (FINAL 31)						
A-	Histamine	B-	Adrenaline	C-	Non of the above	D-	hormones

6-	Decarboxylation of glutamic acid produce? (FINAL 31)		
A-	Gamma amino butyric acid	B-	Aceto acetic acid
C-	Both a , b	D-	Glutamine

7-	End metabolic product used in diagnosis of phenochromocytoma is ? (FINAL 32)						
A-	van mandlic acid	B-	catochlamines	C-	both of them	D-	non of the above

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
B	C	B	A	A	A	B	-	-	-

Department Qs

1-	Serine when decarboxylated will produce: (Department Qs33)		
A-	Ethanolamine	B-	Beta mercaptoethanolamine
C-	Histamine	D-	Putrescine

2-	Acetylcholine is derived from which amino acid: (Department Qs33)						
A-	Tyrosine	B-	Tryptophan	C-	Glutamic acid	D-	Serine

3-	The methyl donor in methylation reactions is: (Department Qs33)		
A-	Methyl cobalamin	B-	Methyl malonyl COA
C-	S- adenosyl methionine	D-	S- adenosyl homocysteine

4-	Which amino acid will give rise to an inhibitory neurotransmitter: (Department Qs33)						
A-	Histidine	B-	Glutamic acid	C-	Ornithine	D-	Tyrosine

5-	Gamma amino butyric acid is derived from which amino acid: (Department Qs33)						
A-	Tyrosine	B-	Tryptophan	C-	Glutamic acid	D-	Serine

6-	Histidine when decarboxylated will produce: (Department Qs33)		
A-	Ethanolamine	B-	Beta mercaptoethanolamine
C-	Histamine	D-	Putrescine

7-	The convertion of DOPA into DOPAMINE requires: (Department Qs33)		
A-	Tetrahydrobiopterin	B-	Tetrahydrofolate
C-	Methylcobalamine	D-	Pyridoxal phosphate

8-	Which of the following amino acids can serve as a precursor of serotonine: (Department Qs33)						
A-	Tyrosine	B-	Threonine	C-	Tryptophan	D-	serine

9-	The decarboxylation of which amino acid produces a vasodilating compound: (Department Qs33, Department Qs37)						
A-	Arginine	B-	Aspartic acid	C-	Histidine	D-	Glutamine

10-	Which of the following statement is not correct for the Gamma amino butyric acid (Department Qs33)
A-	It is an inhibitory neurotransmitter.
B-	It is synthesized in the brain from glutamate.
C-	The enzyme glutamate decarboxylase is directly responsible for its synthesis.
D-	It is an essential amino acid.

11-	All the following statements about serotonin are true except: (Department Qs37)		
A-	It causes vasodilatation	B-	It causes bronchoconstriction
C-	It is metabolized by monoamine oxidase	D-	Its metabolite is 5-hydroxyindole acetic acid

12-	Glutamic acid is a precursor of: (Department Qs37)						
A-	Folic acid	B-	GABA	C-	Glutathione	D-	All of these

Answers - Department Qs									
11	12	13	14	15	16	17	18	19	20
A	D	C	B	C	C	D	C	C	D
21	22	23	24	25	26	27	28	29	30
A	D	-	-	-	-	-	-	-	-

Lecture Qs

1-	Serine decarboxylase need: (Lecture Qs33)						
A-	PLP	B-	TPP	C-	Lipoic acid	D-	COASH

2-	Histamine produced from histadine by: (Lecture Qs33)						
A-	Transamination	B-	Decarboxylation	C-	Deamination	D-	Transmethylation

3-	A vasodilating compound is produced by decarboxylation of the amino acid: (Lecture Qs34)						
A-	Arginine	B-	Aspartic acid	C-	Glutamine	D-	Histidine

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	B	D	-	-	-	-	-	-	-

M3WAN Bank

1-	Biogenic amines are defined as products of: (M3WAN - Core)
A-	Transamination of amino acids
B-	Simple decarboxylation of amino acids or their derivatives
C-	Oxidative deamination of amino acids
D-	Hydroxylation of amino acids

2-	The decarboxylase enzymes responsible for forming biogenic amines require which coenzyme? (M3WAN - Core)		
A-	Pyridoxal phosphate (PLP)	B-	Thiamin pyrophosphate
C-	Biotin	D-	FAD

3-	Biogenic amines are classified according to structure into mono-amines, di-amines, and: (M3WAN - Core)						
A-	Poly-amines	B-	Tri-amines only	C-	Neutral amines	D-	Cyclic amines

4-	Dopamine, a hypothalamic neurotransmitter, is formed by decarboxylation of: (M3WAN - Core)		
A-	Tyrosine directly	B-	3,4-dihydroxyphenylalanine (DOPA)
C-	Phenylalanine	D-	Tryptophan

5-	Dopamine serves as the direct precursor for which biogenic amine in adrenergic nerve endings and the adrenal medulla? (M3WAN - Core)		
A-	Serotonin	B-	Norepinephrine (noradrenaline)
C-	Histamine	D-	GABA

6-	Conversion of norepinephrine to epinephrine requires methylation, with the methyl group donated by: (Important M3WAN - Core)		
A-	S-adenosylmethionine (SAM)	B-	Tetrahydrofolate
C-	Biotin	D-	Vitamin B12

7-	Ethanolamine, an important component of cephalin and plasmalogen, is synthesized from: (M3WAN - Core)						
A-	Tyrosine	B-	Serine	C-	Histidine	D-	Glutamate

8-	Choline, formed from ethanolamine by transmethylation, is required for synthesis of phospholipids and, via choline acetyltransferase, for synthesis of: (Important M3WAN - Core)						
A-	Histamine	B-	Acetylcholine	C-	Serotonin	D-	Dopamine

9-	S-adenosylmethioninE-derived "SAM amine" is used mainly for the biosynthesis of: (M3WAN - Hard)		
A-	Polyamines such as spermidine and spermine	B-	Histamine
C-	Serotonin	D-	GABA

10-	Histamine, formed by decarboxylation of histidine, is synthesized in gastric mucosa, basophils, mast cells, and: (M3WAN - Core)		
A-	Hepatocytes exclusively	B-	Histaminergic receptors in the brain
C-	Renal tubular cells	D-	Pancreatic islet cells

11-	Histamine increases gastric HCl and pepsin secretion by acting through H1 and H2 receptors; cimetidine, an H2-receptor blocker, is used clinically to treat: (Important M3WAN - Core)		
A-	Gastric hyperacidity and peptic ulcer	B-	Allergic rhinitis exclusively

C-	Hypertension	D-	Bronchial asthma

12-	**In addition to its role in gastric acid secretion, histamine also functions as: (M3WAN - Core)**
A-	Vasoconstrictor reducing vascular permeability
B-	Vasodilator increasing vascular permeability, contributing to allergic inflammation
C-	Inhibitory neurotransmitter only
D-	Direct precursor of serotonin

13-	**Serotonin, a biogenic amine formed by decarboxylation of tryptophan, is synthesized mainly in the intestinal mucosal cells known as: (Important M3WAN - Core)**						
A-	Argentaffin cells	B-	Kupffer cells	C-	Melanocytes	D-	Chief cells

14-	**Which of the following is a recognized action of serotonin? (M3WAN - Core)**		
A-	Vasodilation and inhibition of apoptosis	B-	Vasoconstriction and induction of apoptosis
C-	Inhibition of smooth muscle contraction	D-	Direct stimulation of gastric acid secretion

15-	**Serotonin acts as an excitatory neurotransmitter in the brain and also causes smooth muscle contraction, manifesting clinically as: (M3WAN - Core)**		
A-	Bronchoconstriction and diarrhea	B-	Bronchodilation and constipation
C-	Vasodilation and hypotension only	D-	Increased gastric acid secretion only

16-	**γ-Aminobutyric acid (GABA), an inhibitory neurotransmitter, is formed by decarboxylation of: (Important M3WAN - Core)**						
A-	Aspartic acid	B-	Glutamic acid	C-	Glycine	D-	Histidine

17-	**GABA-A receptors mediate presynaptic inhibition mainly by increasing permeability to: (Important M3WAN - Core)**						
A-	Sodium	B-	Calcium	C-	Chloride	D-	Potassium

18-	**GABA-B receptors mediate postsynaptic inhibition mainly by increasing permeability to: (Important M3WAN - Core)**						
A-	Chloride	B-	Potassium	C-	Sodium	D-	Calcium

19-	**Deficiency of pyridoxal phosphate (vitamin B6), by decreasing brain GABA concentration, is a recognized cause of: (Important M3WAN - Core)**		
A-	Convulsions, especially in infants	B-	Peptic ulcer
C-	Alkaptonuria	D-	Ketosis

20-	Di-amines are defined as decarboxylation products of diamino-monocarboxylic (basic) amino acids; putrescine, an example, serves as a precursor for: (M3WAN - Core)		
A-	Serotonin and dopamine	B-	Spermidine and spermine
C-	Histamine and GABA	D-	Choline and ethanolamine

21-	Catabolism of biogenic monoamines is catalyzed mainly by: (M3WAN - Core)		
A-	Diamine oxidase (DAO)	B-	Monoamine oxidase (MAO)
C-	Decarboxylase	D-	Transaminase

22-	Catabolism of di-amines and polyamines is catalyzed mainly by: (M3WAN - Core)		
A-	Monoamine oxidase (MAO)	B-	Diamine oxidase (DAO)
C-	Tyrosinase	D-	Xanthine oxidase

23-	Both monoamine oxidase and diamine oxidase require which two cofactors for activity? (Important M3WAN - Hard)		
A-	PLP and copper	B-	FAD and iron
C-	NAD+ and zinc	D-	Biotin and manganese

24-	Which amino acid serves as the precursor for the mono-amine dopamine, ultimately via decarboxylation of DOPA? (M3WAN - Core)		
A-	Phenylalanine/tyrosine	B-	Histidine
C-	Tryptophan	D-	Glutamate

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	A	A	B	B	A	B	B	A	B
11	12	13	14	15	16	17	18	19	20
A	B	A	B	A	B	C	B	A	B
21	22	23	24	25	26	27	28	29	30
B	B	A	A	-	-	-	-	-	-

قال الإمام ابن تيمية – رحمه الله:–

عفّوا تعفّ نساؤكم وأبناؤكم، وبِرّوا آباءكم تبرّكم

أبناؤكم؛ فإنّ الجزاء من جنس العمل.

مجموع الفتاوى (319/15)

Lecture (15): Phenylalanine , Tyrosine Metabolism

Pre Exams , Formatives

1-	**we can obtain of melanin pigment (FINAL 34, FINAL 32)**		
A-	tyrosine	B-	sulfur containing amino acid
C-	both of them	D-	None of the above

2-	**Alkaptonuria occurs due to deficiency of the enzyme ? (MID 30)**		
A-	Maleylacetoacetate isomerase	B-	Homogentisate oxidase
C-	p-Hydroxyphenylpyruvate hydroxylase	D-	Fumarylacetoacetate hydrolase

3-	**Which of the following amino acids are ketogenic ? (MID 30)**		
A-	Tyrosine and Alanine	B-	Phenylalanine and Glycine
C-	Leucine and Tryptophan	D-	Tryptophan and Serine

4-	**Formation of melanin from tyrosine requires the action of ? (MID 31)**				
A-	Dopadecarboxylation	B-	Diamine oxidase	C-	Peroxidase
D-	Tyrosinase	E-	Hydroxy tyrosine		

5-	**which of amino acid used in formation of dopamine (FINAL 36)**						
A-	Tyrosine	B-	Tryptophan	C-	Histidine	D-	Methionine

6-	**Albinism occur because deficiency of: (FINAL 37)**				
A-	Tyrosine hydroxidase	B-	Tyrosinase	C-	homogentisate oxidase

7-	**glycine can be formed of... (Formative 37, Quiz SNU-M1,SNU Bank)**						
A-	phenylalanine	B-	tyrosine	C-	tryptophan	D-	serine

8-	**tyrosine is... (Formative 37, Quiz SNU-M1)**		
A-	aromatic amino acid	B-	sulfur containing amino acid
C-	acidic amino acid	D-	basic amino acid

9-	**phenylalanine is... (Formative 37, Quiz SNU-M1)**		
A-	vitamin	B-	non essential amino acids
C-	essential amino acid	D-	Enzyme

10-	**Tyrosine gives rise to a family of catecholamines that does not include? (FINAL 31, FINAL 32)**						
A-	Dopamine	B-	Norepinephrine	C-	Epinephrine	D-	Cortisol

11-	Deficiency of phenyl alanine hydroxylase enzyme cause? (FINAL 31)						
A-	Alkaptonuria	B-	Phenyl ketonuria	C-	Both a , b	D-	Favism

12-	Definicy of homogentisic oxygenase enzyme leads to ....? (FINAL 32)						
A-	leucoderma	B-	alkaptenuria	C-	both of them	D-	non of the above

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
A	B	C	D	A	B	D	A	C	D
11	12	13	14	15	16	17	18	19	20
B	B	-	-	-	-	-	-	-	-

Department Qs

1-	Phenyle Alanine Is Converted To Tyrosine By. (Department Qs33)		
A-	Hekokinase Enzyme	B-	Phenyle Alanine Hydroxlase Enzyme
C-	Tyrosinase Enzyme	D-	Glucokinase Enzyme

2-	Melanine Pigment is Formed Of (Department Qs33)						
A-	Glycine	B-	Leucine	C-	Tyrosine	D-	Histidine

3-	Thyroid Hormones Are Formed of ... (Department Qs33)						
A-	Tyrosine	B-	isoleusine	C-	Hydroxyproline	D-	Lysine

4-	Epinephrine is derived from which amino acid: (Department Qs33)						
A-	Tyrosine	B-	Tryptophan	C-	Glutamic acid	D-	Serine

5-	Which amino acid is quickly converted to tyrosine? (Department Qs37)						
A-	Glycine	B-	Arginine	C-	Leucine	D-	Phenylalanine

6-	Alkaptonuria, an inherited metabolic disorder of L-tyrosine metabolism is due to lack of : (Department Qs37)		
A-	Homogentisate oxidase	B-	Tyrosine Transaminase
C-	Para hydroxyl phenyl pyruvate hydroxylase	D-	Tyrosine oxidae

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	C	A	A	D	A	-	-	-	-

Lecture Qs

1-	All the following are products of phenylalanine except (Lecture Qs33, Lecture Qs37)		
A-	Thyroxine	B-	Melanin
C-	Sertonin	D-	Adrenaline , noradrenaline

2-	Urine in alkaptinuria is changed after voiding into.... Color (Lecture Qs33, Lecture Qs37)						
A-	Black	B-	Yellow	C-	Red	D-	Deep blue

3-	Phenylketonuria is a disease characterized by absence of Enzyme (Lecture Qs33, Lecture Qs37)		
A-	Phenylalanine hydroxylase	B-	Phenylpyruvate
C-	Tyrosinase	D-	Transaminase

Answers - Lecture Qs

1	2	3	4	5	6	7	8	9	10
C	A	A	-	-	-	-	-	-	-

M3WAN Bank

1-	Which of the following amino acids does NOT contain an aromatic ring? (M3WAN - Core)						
A-	Tyrosine	B-	Tryptophan	C-	Phenylalanine	D-	Glycine

2-	Phenylalanine is classified as an aromatic, essential amino acid that is metabolically: (M3WAN - Core)		
A-	Purely glucogenic	B-	Purely ketogenic
C-	Both glucogenic and ketogenic	D-	Metabolically inert

3-	Conversion of phenylalanine to tyrosine is catalyzed by phenylalanine hydroxylase, a reaction that is: (Important M3WAN - Core)		
A-	Reversible, requiring biotin	B-	Irreversible, requiring tetrahydrobiopterin
C-	Reversible, requiring PLP	D-	Irreversible, requiring vitamin B12

4-	Since degradation of phenylalanine mostly occurs through tyrosine, tyrosine itself is classified as: (M3WAN - Core)
A-	A non-aromatic, essential amino acid
B-	A non-essential amino acid, glucogenic and partly ketogenic
C-	Purely ketogenic and essential
D-	Unrelated metabolically to phenylalanine

5-	Tyrosine catabolism in the liver produces fumarate (glucogenic) and acetoacetate (ketogenic), with which intermediate formed along the pathway? (Important M3WAN - Core)		
A-	Homogentisic acid	B-	Phenylpyruvate
C-	Melanin	D-	Dopamine

6-	Homogentisic acid is normally oxidized further by the enzyme: (Important M3WAN - Core)		
A-	Tyrosinase	B-	Homogentisate oxidase
C-	Phenylalanine hydroxylase	D-	Tyrosine hydroxylase

7-	Tyrosine serves as the biosynthetic precursor for which three classes of biologically important compounds? (M3WAN - Core)
A-	Melanin, thyroid hormones, and catecholamines
B-	Melanin, ketone bodies, and bile acids
C-	Serotonin, histamine, and GABA
D-	Purines, pyrimidines, and porphyrins

8-	Melanin, the pigment of skin, hair, and eyes, is synthesized in melanocytes from tyrosine through the action of: (Important M3WAN - Core)		
A-	Tyrosine hydroxylase	B-	Tyrosinase
C-	Phenylalanine hydroxylase	D-	Homogentisate oxidase

9-	Tyrosinase catalyzes hydroxylation of tyrosine to form: (M3WAN - Core)		
A-	Homogentisic acid	B-	3,4-dihydroxyphenylalanine (DOPA)
C-	Dopamine	D-	Thyroxine

10-	Generalized absence of melanin synthesis due to tyrosinase deficiency is the inborn error known as: (Important M3WAN - Core)						
A-	Alkaptonuria	B-	Phenylketonuria	C-	Albinism	D-	Leucoderma

11-	Localized absence or degeneration of melanocytes, producing white patches of skin without a generalized enzyme defect, is termed: (M3WAN - Hard)						
A-	Albinism	B-	Leucoderma	C-	Alkaptonuria	D-	Ochronosis

12-	Synthesis of catecholamines from tyrosine occurs mainly in the adrenal medulla and central nervous system, beginning with hydroxylation of tyrosine to DOPA by the ratE-limiting enzyme: (Important M3WAN - Core)		
A-	Tyrosinase	B-	Tyrosine hydroxylase
C-	Homogentisate oxidase	D-	Phenylalanine hydroxylase

13-	Tyrosine hydroxylase, like phenylalanine hydroxylase, requires which coenzyme? (M3WAN - Core)		
A-	Tetrahydrobiopterin	B-	Biotin
C-	Thiamin pyrophosphate	D-	Pyridoxal phosphate

14-	In individuals with albinism (tyrosinase deficiency), catecholamine synthesis is characteristically: (Important M3WAN - Hard)
A-	Absent, because tyrosinase is required
B-	Normal, because catecholamine synthesis uses tyrosine hydroxylase, a different enzyme
C-	Increased above normal
D-	Diverted entirely to melanin production

15-	Synthesis of the thyroid hormones triiodothyronine (T3) and thyroxine (T4) requires iodination of tyrosine residues within the protein: (M3WAN - Core)						
A-	Albumin	B-	Thyroglobulin	C-	Transferrin	D-	Ceruloplasmin

16-	Thyroxine (T4) is more precisely described chemically as: (M3WAN - Hard)		
A-	Monoiodotyrosine	B-	Diiodotyrosine
C-	Triiodothyronine	D-	Tetraiodothyronine

17-	Phenylketonuria (PKU), the most common metabolic disorder of amino acid metabolism, is most commonly caused by deficiency of: (Important M3WAN - Core)		
A-	Tyrosinase	B-	Phenylalanine hydroxylase
C-	Homogentisate oxidase	D-	Tyrosine hydroxylase

18-	PKU may also result from a defect in the enzyme responsible for regenerating the coenzyme of phenylalanine hydroxylase, namely: (M3WAN - Hard)		
A-	Dihydrobiopterin reductase	B-	Dihydrofolate reductase
C-	Methylene tetrahydrofolate reductase	D-	Xanthine oxidase

19-	PKU is inherited in which pattern? (M3WAN - Core)		
A-	X-linked recessive	B-	Autosomal dominant
C-	Autosomal recessive	D-	X-linked dominant

20-	In PKU, accumulated phenylalanine is diverted into alternative pathways, producing excessive amounts of phenylpyruvate, phenylacetate, and: (Important M3WAN - Core)						
A-	Phenyllactate	B-	Homogentisate	C-	Dopamine	D-	Melanin

21-	**The hypopigmentation (fair hair, light skin, blue eyes) characteristic of untreated PKU is explained by: (Important M3WAN - Core)**
A-	Direct destruction of melanocytes by phenylalanine
B-	Competitive inhibition of tyrosinase by accumulated phenylalanine
C-	Deficiency of dietary tyrosine only
D-	Increased homogentisate oxidase activity

22-	**The characteristic "mousy" body odor in PKU results from elevated plasma and urinary levels of phenylalanine and its alternative metabolites, including: (M3WAN - Core)**
A-	Phenylpyruvate, phenylacetate, and phenyllactate
B-	Homogentisic acid
C-	Melanin precursors
D-	Catecholamines

23-	**A nonspecific screening test for phenylpyruvate in urine, producing a green color, is the: (M3WAN - Core)**		
A-	Ferric chloride test	B-	Benedict-s test
C-	Biuret test	D-	Ninhydrin test

24-	**Definitive diagnosis of PKU is established by measuring plasma phenylalanine concentration and the ratio of: (Important M3WAN - Core)**		
A-	Phenylalanine to tyrosine, which is elevated	B-	Tyrosine to tryptophan
C-	Phenylalanine to homogentisate	D-	Melanin to DOPA

25-	**Alkaptonuria, an autosomal recessive disorder of tyrosine metabolism caused by homogentisate oxidase deficiency, is characterized by urine that: (Important M3WAN - Core)**
A-	Remains colorless on standing
B-	Turns black or brown ("cokE-colored") on standing due to oxidation of homogentisate
C-	Turns bright yellow immediately
D-	Contains excess phenylpyruvate

26-	**In alkaptonuria, deposition of oxidized homogentisate pigment in connective tissue, bones, nose, and ear produces a condition known as: (Important M3WAN - Core)**						
A-	Ochronosis	B-	Leucoderma	C-	Xanthoma	D-	Hemochromatosis

27-	**Alkaptonuria is also characteristically associated with which joint complication, due to deposition of alkapton pigment? (M3WAN - Core)**		
A-	Gouty arthritis	B-	Arthritis due to pigment deposition in joints
C-	Osteoporosis	D-	Rheumatic fever

28-	**Albinism is characteristically associated with all of the following clinical features EXCEPT: (Important M3WAN - Hard)**		
A-	Photophobia due to lack of ocular pigment	B-	Increased susceptibility to skin cancer
C-	Sensitivity of skin to sunlight	D-	Black discoloration of urine on standing

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
D	C	B	B	A	B	A	B	B	C
11	12	13	14	15	16	17	18	19	20
B	B	A	B	B	D	B	A	C	A
21	22	23	24	25	26	27	28	29	30
B	A	A	A	B	A	B	D	-	-

تلمّحوا عُلوَّ قدر يعقوب ببلائه وعزّ يوسف في صبره، وليكن حظكم من هذه القصة: ﴿إنه من يتّقِ ويصبر فإن الله لا يُضيع أجر المُحسنين﴾.

وليتفكر العاصي في لذّاتٍ فنيت وتبعاتٍ بقيت، وليتدبّر الصابر لذة مديحة ثبتت ومرارة مصابرة خلَت، والأمر بآخره وللعواقب يعمل المُتيقّظ.

📖 ابن الجوزي | التبصرة.

Lecture (16): Glycine metabolism

Pre Exams , Formatives

1-	The source of carbon and nitrogen atoms of the porphyrin from (FINAL 36)				
A-	Leucine and aspartate	B-	Glycine and fumarate	C-	Glycine and succinyl coA

2-	glycine is used in (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	fomation of fatty acids	B-	formation of acetone
C-	formation of hemoglobin	D-	formation of ketone bodies

3-	Glycine enters in formation of? (FINAL 31)						
A-	Glutathione	B-	Bile salts	C-	Both a , b	D-	Estrogen

4-	Precursor of glycine is ? (FINAL 32)						
A-	Proline	B-	Glutamine	C-	Serine	D-	Glutamate

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
C	C	C	C	-	-	-	-	-	-

Department Qs

1-	Glutathione formed of the following amino acids except:- (Department Qs33)						
A-	Glycine	B-	Glutamic acid	C-	Serine	D-	Cysteine

2-	Transmethylation of guanidoacetate occur in :- (Department Qs33)						
A-	Liver	B-	Kidney	C-	Intestine	D-	Spleen

3-	Glycine is: (Department Qs33)		
A-	ketogenic	B-	Essential
C-	Ketogenic and glucogenic	D-	Non of the above

4-	In mammalian tissues serine can be a biosynthetic precursor of (Department Qs33)						
A-	Methionine	B-	Glycine	C-	Tryptophan	D-	Phenylalanine

5-	Glycine participates in the synthesis of: (Department Qs 35)				
A-	Heme and purine	B-	Histamine and creatine	C-	Carnosine and anserine
D-	Pyrimidine and spermine	E-	GABA and bile salts		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	A	D	B	A	-	-	-	-	-

Lecture Qs

1-	Which out of the following amino acids in not required for creatine synthesis? (Lecture Qs33)						
A-	Methionine	B-	Serine	C-	Glycine	D-	Arginine

2-	Which out of the following amino acids is not needed for the synthesis of Glutathione? (Lecture Qs33, Lecture Qs34)						
A-	Serine	B-	Cysteine	C-	Glutamic acid	D-	Glycine

3-	Glycine is required for the synthesis of all of the following except: (Lecture Qs33)		
A-	Creatine	B-	Glutathione
C-	Heme	D-	y-aminobutyric acid (GABA)

4-	In mammalian tissues serine can be a biosynthetic precursor of (Lecture Qs34)				
A-	Methionine	B-	Glycine	C-	Tryptophan

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
B	A	D	B	-	-	-	-	-	-

M3WAN Bank

1-	Glycine is classified nutritionally and metabolically as: (M3WAN - Core)		
A-	An essential, ketogenic amino acid	B-	A non-essential, glucogenic amino acid
C-	An essential, glucogenic amino acid	D-	A non-essential, ketogenic amino acid

2-	Besides de novo synthesis, glycine can be synthesized in the body by a reversible reaction from: (M3WAN - Core)						
A-	Serine	B-	Tyrosine	C-	Lysine	D-	Tryptophan

3-	In addition to serine, glycine can also be synthesized from glyoxylate and: (M3WAN - Hard)		
A-	Threonine and choline	B-	Histidine and arginine
C-	Phenylalanine and tyrosine	D-	Methionine only

4-	Catabolic fates of glycine include production of CO2 and NH3, formation of glyoxylic acid and pyruvate, and: (M3WAN - Core)
A-	Detoxication reactions by conjugation
B-	Direct conversion to urea
C-	Direct conversion to glucose without intermediates
D-	Formation of ketone bodies exclusively

5-	Collagen, the most abundant protein in the body, is particularly rich in glycine, which constitutes approximately what fraction of its amino acid content? (Important M3WAN - Core)						
A-	5%	B-	15%	C-	30%	D-	50%

6-	Glycine contributes which atoms to the purine ring during purine biosynthesis? (Important M3WAN - Core)						
A-	N1, C2, N3	B-	N7, C4, C5	C-	C6, N9, C8	D-	N3, C6, N1

7-	Glutathione (GSH), synthesized from glycine, cysteine, and glutamate, functions in amino acid transport, as a coenzyme for heme synthesis, and as: (Important M3WAN - Core)
A-	Hydrogen donor in oxidation-reduction reactions
B-	Direct precursor of purines
C-	Direct precursor of creatine
D-	Structural component of collagen

8-	Reduced glutathione protects red blood cell membranes from oxidative damage, thereby preventing hemolysis caused by oxidants such as: (Important M3WAN - Core)		
A-	Hydrogen peroxide	B-	Bilirubin
C-	Urea	D-	Homogentisic acid

9-	In the liver, glutathione inactivates insulin through reductive cleavage of the two disulfide linkages of insulin, a reaction catalyzed by: (M3WAN - Hard)		
A-	Insulin-glutathione transhydrogenase	B-	Glutathione peroxidase
C-	Glutathione reductase	D-	Glutathione synthetase

10-	Glycine conjugates with cholic acid to form the bile salt: (M3WAN - Core)		
A-	Taurocholic acid	B-	Glycocholic acid
C-	Chenodeoxycholic acid	D-	Deoxycholic acid

11-	Heme biosynthesis begins with condensation of glycine with which compound to form δ-aminolevulinic acid? (Important M3WAN - Core)						
A-	Succinyl-CoA	B-	Acetyl-CoA	C-	Malonyl-CoA	D-	Propionyl-CoA

12-	Creatine biosynthesis requires glycine together with arginine and methionine, the latter functioning in its active form as: (Important M3WAN - Core)		
A-	Homocysteine	B-	S-adenosylmethionine (SAM)
C-	Methionine sulfoxide	D-	Cystathionine

13-	The normal serum creatinine concentration is approximately: (M3WAN - Hard)						
A-	0.6–1.2 mg/dL	B-	2–5 mg/dL	C-	5–10 mg/dL	D-	10–20 mg/dL

14-	Normal urinary creatinine excretion is approximately: (M3WAN - Hard)						
A-	0–200 mg/day	B-	1–2 g/day	C-	5–10 g/day	D-	20–30 g/day

15-	Primary hyperoxaluria results from a genetic defect involving glycine transaminase together with impaired: (Important M3WAN - Core)
A-	Oxidative decarboxylation of glyoxylate to formate
B-	Reduction of oxaloacetate
C-	Conjugation of glycine with cholic acid
D-	Transamination of alanine

16-	The clinical consequence of primary hyperoxaluria (Type I) is increased formation of urinary oxalate calculi, potentially leading to: (Important M3WAN - Core)
A-	Renal (kidney) stone formation composed of calcium oxalate
B-	Uric acid stone formation only
C-	Cystine stone formation
D-	Struvite stone formation

17-	"Oxalosis," a complication of primary hyperoxaluria, refers to: (M3WAN - Hard)
A-	Deposition of oxalate crystals in various tissues
B-	Excess urinary excretion of oxalate with no tissue deposition
C-	Oxidative destruction of glycine
D-	Formation of oxalate exclusively in the liver

18-	Glycine is required for the synthesis of all of the following EXCEPT: (Important M3WAN - Hard)						
A-	Creatine	B-	Glutathione	C-	Heme	D-	Cholesterol

19-	Which of the following is NOT a recognized source for de novo synthesis of glycine in the body? (M3WAN - Hard)						
A-	Serine	B-	Threonine	C-	Choline	D-	Isoleucine

20-	The two chenodeoxycholic acid derivatives formed by glycine conjugation and taurine conjugation are, respectively, glycochenodeoxycholic acid and: (M3WAN - Core)		
A-	Taurochenodeoxycholic acid	B-	Glycocholic acid
C-	Cholic acid	D-	Deoxycholic acid

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	A	A	A	C	B	A	A	A	B
11	12	13	14	15	16	17	18	19	20
A	B	A	B	A	A	A	D	D	A

إخواني: مَن فعل ما يُحب لقِيَ ما يكره، ومَن صبَر على ما يكره نال ما يُحب، لا تقطع مشاورة العقل قبل مُشاورة الهوى، فإن المُستبدّ برأيه واقفٌ على مداحض الزّلل، لم تزل أكف العقل ضابطة أعنة النفوس غير أن العزم ينقلب، ركوب الأخطار يسوق الأقدار . .

من قرأ والناس نيام تكلّم والناس سُكُوت .

ابن الجوزي | التبصرة .

Lecture (17): Nitrogen Metabolism and urea cycle

Pre Exams , Formatives

1-	carbamoyl-phosphate is formed from (FINAL 34, FINAL 32)		
A-	ammonia and CO2	B-	Ornithine
C-	Citrulline	D-	Arginine

2-	The first amino group entering into urea cycle is (FINAL 34, Quiz 37, Quiz 36, FINAL 31)		
A-	Carbomyl phosphate	B-	Ornithine
C-	Cituilline	D-	Argininosuccinate

3-	Arginase deficiency leads to (FINAL 34)				
A-	Hyperammonemia	B-	excess urea excretion	C-	excess arginine excretion

4-	The end product of ammonia in brain (FINAL 34)						
A-	Arginine	B-	Urea	C-	Uric acid	D-	Glutamine

5-	The end product of protein nitrogen metabolism in man (FINAL 34)
A-	Urea

6-	What is first amino acid in the urea cycle (FINAL 34)
A-	Citroulline

7-	Transamination of aspartate (FINAL 34)
A-	Oxaloacetate

8-	Urea production occurs almost exclusively in Kidneys.. (Quiz 37, Quiz 36, FINAL 31)				
A-	Urine	B-	Blood	C-	Liver

9-	The first enzyme in the pathway carbamoyl phosphate synthase I, is allosterically activated by.. (Quiz 37, Quiz 36)		
A-	Acetyl coA	B-	N-acetylglutamate
C-	Carbamoyl phosphate	D-	Glutamate

10-	The nitrogen atoms of urea produced in the urea cycle are derived from ? (MID 30)		
A-	Nitrate	B-	ammonia and aspartic acid
C-	Nitrite	D-	Ammonia

11-	The reaction of glutamate and NH4+ to yield glutamine is catalyzed by ? (MID 31)				
A-	Uridylyl transferase	B-	Adenylyl transferase	C-	Glutamate synthase
D-	Glutamine synthase	E-	Urease		

12-	The main site of urea synthesis in mammals is? (MID 31, FINAL 32, Department Qs33)						
A-	Liver	B-	Skin	C-	Intestine	D-	Kidney

13-	Urea formatin from ammonia occurs at (FINAL 36)		
A-	Cytoplasm of muscle	B-	Mitochondria of brain
C-	both cytoplasm and mitochondria of liver cells		

14-	Site of Urea cycle is ..... (FINAL 36)
A-	mitochondrial and cytosol of liver

15-	Deficiency of arginase causes: (FINAL 36)		
A-	Hyperammonemia	B-	Arginine deficiency
C-	Excess of ornithine	D-	Excess of urea

16-	Site of Carbamoyl phosphate is: (FINAL 37)		
A-	Golgi body	B-	Cytoplasm
C-	RER	D-	mitochondrial matrix

17-	A second amino group is transferred to cituilline from.. (Quiz 36)						
A-	Aspartate	B-	Guanine	C-	Alanine	D-	Glutamate

18-	urea cycle occurs in...of liver (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	mitochondria and cytoplasm	B-	basement membrane
C-	smooth endoplasmic reticulum	D-	rough endoplasmic reticulum

19-	in urea cycle especially in liver and kidney we can..... (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	not need energy	B-	produce energy
C-	no formation of amino acid	D-	form amino acid arginine

20-	urea cycle occurs in.. (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	Muscle	B-	Brain	C-	lung	D-	liver

21-	Amonia is (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	has no effect to body	B-	toxic substance to body
C-	non toxic substance to body	D-	beneficial substance to body

22-	Major pathway of nitrogen excretion in man is as? (FINAL 31)						
A-	Creatine	B-	Urea	C-	Both a , b	D-	Ammonia

23-	Principal pathway of disposal of ammonia is? (FINAL 31)		
A-	Urea formation	B-	Glutamine synthesis
C-	Non of the above	D-	a and b

24-	Citrullinaemia is a type of inherited hyperammonaemin resulting from inherited deficiency of? (FINAL 31)		
A-	Citrulline synthease	B-	Arginosuccinate synthase
C-	Ornithine transcarbamoylase	D-	Carbamoyl phosphate synthas

25-	Normal serum creatinine level is? (FINAL 31)						
A-	0.2-0.4 mg/dL	B-	0.3-0.6 mg/dL	C-	0.6-1.2 mg/dL	D-	1.4-2.8 mg/dL

26-	Creatinine is synthesized from the following amino acids, EXCEPT? (FINAL 31)						
A-	Arginine	B-	Aspartic acid	C-	Glycine	D-	Methionine

27-	All these reaction take place inside the mitochondria EXCEPT? (FINAL 31)						
A-	Glycolysis	B-	KreB-s cycle	C-	Urea cycle	D-	Electron transfer

28-	carbamoyl phosphate synthetase I? (FINAL 32)		
A-	presents in cytoplasm of liver cells	B-	presents in mitochondria of liver cells
C-	presents in cytoplasm of kidney cells	D-	presents in mitochondria of kidney cells

29-	Key enzyme in ureA-synthesis is ? (2ND TRIAL 32, Department Qs37)		
A-	Urease	B-	Ornithine
C-	Arginine	D-	Carbamyl synthetase

30-	Major source of ammonia in the kidney is ? (2ND TRIAL 32, Department Qs37)						
A-	Glutamine	B-	Aspartate	C-	Glutamate	D-	Urea

31-	Which is the byproduct of the urea cycle ? (2ND TRIAL 32, Department Qs37)						
A-	Succinate	B-	Aspartate	C-	Ornithine	D-	Fumarate

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
A	A	A	D	A	A	A	C	B	B
11	12	13	14	15	16	17	18	19	20
D	A	C	A	A	D	A	A	D	D
21	22	23	24	25	26	27	28	29	30
B	B	A	B	C	B	A	B	D	A
31	32	33	34	35	36	37	38	39	40
D	-	-	-	-	-	-	-	-	-

Department Qs

1-	**transfer of amine group between amino acids is called 1 (SNU Bank)**						
A-	transformation	B-	deamination	C-	transamination	D-	transmethylation

2-	**ALT and AST enzymes are called (SNU Bank)**						
A-	vitamins	B-	carbohydrates	C-	lipid	D-	aminotransferase

3-	**transfer of amino group from one amino acid to anther.... (SNU Bank)**		
A-	can make sugar acid	B-	can make alcohol
C-	can make vitamin	D-	can make alpha keto acid and new amino acid

4-	**ALT and AST can surely diagnose (SNU Bank)**		
A-	brain diseases	B-	respiratory diseases
C-	liver diseases	D-	stomach diseases

5-	**increase blood serum level of amonia is called (SNU Bank)**						
A-	hyperglycaemia	B-	hypoglycemia	C-	hyperammonemia	D-	hyperlipidemia

6-	**A compound serving a link between citric acid cycle and urea cycle is: (Department Qs33)**						
A-	Malate	B-	Citrate	C-	Succinate	D-	Fumarate

7-	**The enzymes of urea synthesis are found in: (Department Qs33)**		
A-	Mitochondria only	B-	Cytosol only
C-	Both mitochondria and cytosol	D-	Nucleus

8-	**The number of ATP required for urea synthesis is: (Department Qs33)**						
A-	0	B-	1	C-	2	D-	3

9-	**Most of the ammonia released from L-amino acids reflects the coupled action of transaminase and: (Department Qs33)**		
A-	L-glutamate dehydrogenase	B-	L-amino acid oxidase
C-	Histidase	D-	Serine dehydratase

10-	**In urea synthesis, the amino acid functioning solely as an enzyme activator: (Department Qs33)**		
A-	N-acetyl glutamate	B-	Ornithine
C-	Citrulline	D-	Arginine

11-	**The enzyme carbamoyl phosphate synthetase requires: (Department Qs33)**						
A-	Mg++	B-	Ca++	C-	Na+	D-	K+

12-	**Control of urea cycle involves the enzyme: (Department Qs33)**		
A-	Carbamoyl phosphate synthetase	B-	Ornithine transcarbamoylase
C-	Argininosuccinase	D-	Arginase

13-	**Transfer of the carbamoyl moiety of carbamoyl phosphate to ornithine is catalysed by a liver mitochondrial enzyme: (Department Qs33)**		
A-	Carbamoyl phosphate synthetase	B-	Ornithine transcarbamoylase
C-	N-acetyl glutamate synthetase	D-	N-acetyl glutamate hydrolase

14-	**Carbamoyl phosphate synthatase 1: (Department Qs 35)**		
A-	Is cystolic enzyme	B-	Requires one ATP
C-	Deficiency causes hyperammonemia type I	D-	Is activated by glutamine
E-	Catalyses a reversible reaction		

15-	**Most urinary ammonium ions are derived by the enzymatic reaction of: (Department Qs 35)**				
A-	Glutamine	B-	Glutamic acid	C-	Histamine
D-	6-amino purine derivatives	E-	Urea		

16-	**Carbamoyl phosphate: (Department Qs 35)**
A-	Is formed from ATP and carbon dioxide
B-	Is hydrolysed to ammonia and carbon dioxide
C-	Is increased due to deficiency of ornithine transcarbamylase
D-	Is produced from degradation of arginine
E-	Diffuses out of the mitochondria during urea synthesis

17-	**In the urea cycle: (Department Qs 35)**
A-	The whole process is cystosolic
B-	One ATP molecule is required
C-	Carbamoyl phosphate synthetase I catalyses a reversible reaction
D-	Fumerate and urea are released
E-	Glutamine is an intermediate

18-	**Which of the following compounds can not participate in transamination reactions: (Department Qs 35)**				
A-	Oxaloacetate	B-	Pyruvate	C-	Succinate
D-	Aspartate	E-	Alanine		

19-	**Which metabolite of TCA cycle is used in detoxification of ammonia in brain? (Department Qs37)**		
A-	Alpha ketoglutarate	B-	Ornithine
C-	Oxaloacetate	D-	Glycine

20-	**The nitrogen of the body is supplies by: (Department Qs37)**						
A-	Triacylglycerol	B-	Lipid	C-	Glucose	D-	Protein

21-	**A coenzyme required in transamination reactions is: (Department Qs37)**		
A-	Coenzyme A	B-	Coenzyme Q
C-	Biotin	D-	Pyridoxal phosphate

22-	**Transamination of aspartic acid will produce: (Department Qs37)**		
A-	oxaloacetic acid	B-	AlphA-ketoglutarate
C-	Pyruvate	D-	All of these

Answers - Department Qs									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	**D**	**D**	**C**	**C**	**D**	**C**	**D**	**A**	**A**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	**A**	**B**	**C**	**A**	**C**	**D**	**C**	**A**	**D**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	**A**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

Lecture Qs

1-	Major activator of carbomyl phosphate synthetase 1 is: (Lecture Qs33, Lecture Qs37)		
A-	Glutamine	B-	GABA
C-	Nacetyl-glutamate	D-	Glutamate

2-	What is the primary function of the Urea Cycle in nitrogen metabolism? (Lecture Qs37)		
A-	ATP synthesis	B-	Nitrate reduction
C-	Ammonia detoxification	D-	Gluconeogenesis

3-	What is the fate of arginine produced in the Urea Cycle? (Lecture Qs37)		
A-	Hydrolysis to urea and ornithine	B-	Incorporation into proteins
C-	Conversion to citrulline	D-	Excretion as ammonia

4-	What is the precursor for the synthesis of urea in the Urea Cycle? (Lecture Qs37)		
A-	Glutamate	B-	Aspartate
C-	Carbamoyl phosphate	D-	Citrulline

5-	Urea production occurs almost exclusively in (Lecture Qs37)						
A-	Kidneys	B-	Liver	C-	Blood	D-	Urine

6-	Nitrogen atoms of urea produced in the urea cycle are derived from (Lecture Qs37)		
A-	Ammonia and aspartic acid	B-	Nitrate
C-	Nitrite	D-	Ammonia

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	C	A	C	B	A	-	-	-	-

M3WAN Bank

1-	Unlike carbohydrates and fats, amino acids in excess of biosynthetic needs are characteristically: (M3WAN - Core)
A-	Stored in adipose tissue
B-	Stored as a dedicated amino acid pool in muscle indefinitely
C-	Rapidly degraded
D-	Converted entirely to glycogen for storage

2-	**The first phase of amino acid catabolism principally involves: (M3WAN - Core)**
A-	Removal of the α-amino group by transamination and oxidative deamination
B-	Direct hydrolysis of the peptide bond only
C-	β-oxidation of the carbon skeleton
D-	Direct excretion of the intact amino acid

3-	**Transamination is defined as the transfer of an amino aciD-s α-amino group to which acceptor molecule, forming glutamate? (M3WAN - Core)**						
A-	Oxaloacetate	B-	Pyruvate	C-	α-Ketoglutarate	D-	Succinate

4-	**The enzymes catalyzing transamination reactions, found in both cytosol and mitochondria of liver, kidney, intestine, and muscle, are collectively called: (M3WAN - Core)**		
A-	Decarboxylases	B-	Aminotransferases (transaminases)
C-	Oxidases	D-	Hydroxylases

5-	**With the exception of two amino acids, all standard amino acids can undergo transamination. The two non-transaminable exceptions are: (Important M3WAN - Hard)**		
A-	Lysine and threonine	B-	Alanine and glutamate
C-	Serine and glycine	D-	Arginine and histidine

6-	**Aminotransferases are typically named according to: (M3WAN - Core)**
A-	The specific amino group acceptor, since donors vary widely
B-	The specific amino group donor, since the acceptor is almost always α-ketoglutarate
C-	Their tissue of origin exclusively
D-	Their molecular weight

7-	**The two most clinically important aminotransferase reactions are catalyzed by: (Important M3WAN - Core)**
A-	Alanine aminotransferase (ALT) and aspartate aminotransferase (AST)
B-	Glutamate dehydrogenase and glutaminase
C-	Arginase and argininosuccinase
D-	Tyrosinase and homogentisate oxidase

8-	**All aminotransferases require which coenzyme, a derivative of vitamin B6? (M3WAN - Core)**		
A-	Pyridoxal phosphate	B-	Thiamin pyrophosphate
C-	Flavin adenine dinucleotide	D-	Nicotinamide adenine dinucleotide

9-	During the aminotransferase reaction mechanism, the pyridoxal form of the coenzyme accepts an amino group to form: (M3WAN - Hard)		
A-	Pyridoxamine phosphate	B-	Pyridoxic acid
C-	Pyridoxine	D-	Thiamin pyrophosphate

10-	ALT (GPT), found mainly in the cytosol of liver cells, is of greatest diagnostic value in: (Important M3WAN - Core)		
A-	Acute viral hepatitis	B-	Myocardial infarction
C-	Skeletal muscle disease only	D-	Acute pancreatitis

11-	AST (GOT), present in both cytosol and mitochondria of cardiac, hepatic, muscle, pancreatic, and renal cells, is elevated in all of the following EXCEPT: (Important M3WAN - Hard)
A-	Myocardial infarction
B-	Liver disease
C-	Myositis
D-	Uncomplicated viral upper respiratory infection

12-	Among all amino acids, the one that undergoes rapid oxidative deamination, catalyzed by glutamate dehydrogenase, is: (Important M3WAN - Core)						
A-	Alanine	B-	Aspartate	C-	Glutamate	D-	Glycine

13-	The sequential action of transamination followed by glutamate dehydrogenasE-catalyzed oxidative deamination allows amino groups from most amino acids to ultimately be released as: (M3WAN - Core)						
A-	Urea directly	B-	Free ammonia	C-	Nitrate	D-	Nitrogen gas

14-	The transport mechanism used by most peripheral tissues to carry ammonia safely to the liver involves combination of ammonia with glutamate, catalyzed by: (Important M3WAN - Core)		
A-	Glutaminase	B-	Glutamine synthetase
C-	Glutamate dehydrogenase	D-	Aspartate aminotransferase

15-	Glutamine, the nontoxic transport form of ammonia, is cleaved in the liver to regenerate glutamate and free ammonia by: (Important M3WAN - Core)		
A-	Glutamine synthetase	B-	Glutaminase
C-	Arginase	D-	Argininosuccinase

16-	The second major mechanism for ammonia transport, used primarily by muscle, involves transamination of pyruvate to form: (Important M3WAN - Core)						
A-	Glutamine	B-	Alanine	C-	Aspartate	D-	Glutamate

17-	The normal concentration of ammonia in blood is approximately: (M3WAN - Hard)		
A-	10–80 µg/dL	B-	1–5 mg/dL
C-	10–20 mg/dL	D-	100–200 µg/dL

18-	In liver cirrhosis, ammonia bypasses hepatic detoxification through portosystemic collaterals and reaches the brain, where its detoxification to glutamine depletes: (Important M3WAN - Core)		
A-	Glucose	B-	α-Ketoglutarate of the Krebs cycle
C-	Pyruvate	D-	Oxaloacetate

19-	The resulting depletion of α-ketoglutarate in hepatic encephalopathy leads to: (Important M3WAN - Core)		
A-	Enhanced Krebs cycle activity	B-	Energy failure in brain cells
C-	Increased hepatic urea synthesis	D-	Increased gluconeogenesis in the brain

20-	Among the sources of blood ammonia, a significant contribution in states of constipation or gastrointestinal hemorrhage comes from: (M3WAN - Core)
A-	Intestinal bacterial putrefaction of dietary protein
B-	Renal glutaminase activity
C-	Hepatic glutamine synthetase
D-	Muscle protein turnover only

21-	Intestinal antiseptics such as neomycin are clinically beneficial in liver disease because they: (Important M3WAN - Core)
A-	Reduce intestinal bacterial production of ammonia
B-	Directly increase hepatic urea cycle activity
C-	Directly chelate blood ammonia
D-	Inhibit glutaminase in the kidney

22-	The urea cycle, a fivE-reaction cyclic pathway synthesizing urea from ammonia and CO2, occurs in the mitochondria and cytosol of the: (M3WAN - Core)						
A-	Kidney	B-	Liver	C-	Muscle	D-	Intestine

23-	Of the two nitrogen atoms in the urea molecule, one is derived from free ammonia and the other from: (Important M3WAN - Core)						
A-	Aspartate	B-	Glutamate	C-	Glutamine	D-	Alanine

24-	The first, ratE-limiting step of the urea cycle, forming carbamoyl phosphate from ammonia, CO2, and phosphate, is catalyzed by: (Important M3WAN - Core)		
A-	Ornithine transcarbamoylase	B-	Carbamoyl phosphate synthetase I
C-	Argininosuccinate synthetase	D-	Arginase

25-	Carbamoyl phosphate synthetase I requires Mg2+ or Mn2+ and is allosterically activated by: (Important M3WAN - Core)		
A-	N-acetylglutamate	B-	Citrate
C-	Acetyl-CoA	D-	ATP alone

26-	Transfer of the carbamoyl group from carbamoyl phosphate to ornithine, forming citrulline, is catalyzed by: (Important M3WAN - Core)		
A-	Argininosuccinate synthetase	B-	Ornithine transcarbamoylase
C-	Carbamoyl phosphate synthetase I	D-	Argininosuccinase

27-	Ornithine transcarbamoylase activity is clinically useful as a test of: (M3WAN - Hard)		
A-	Renal function	B-	Liver function
C-	Cardiac function	D-	Pancreatic function

28-	Citrulline combines with aspartate to form argininosuccinate, in a reaction requiring ATP and Mg2+, catalyzed by: (Important M3WAN - Core)		
A-	Argininosuccinate synthetase	B-	Argininosuccinase
C-	Arginase	D-	Ornithine transcarbamoylase

29-	Cleavage of argininosuccinate by argininosuccinase produces arginine and: (Important M3WAN - Core)		
A-	Oxaloacetate, which enters gluconeogenesis	B-	Fumarate, which joins the Krebs cycle
C-	Malate directly	D-	Citrate

30-	The final step of the urea cycle, cleaving arginine into urea and regenerating ornithine, is catalyzed by: (Important M3WAN - Core)		
A-	Arginase	B-	Argininosuccinase
C-	Carbamoyl phosphate synthetase I	D-	Ornithine transcarbamoylase

31-	Synthesis of urea cycle enzymes is increased during starvation and a very-high-protein diet, but decreased in: (M3WAN - Hard)
A-	Well-fed states with predominantly carbohydrate and fat diets, or protein-free diets
B-	Chronic high-protein diets
C-	Prolonged fasting

D-	States of severe trauma

32-	**Acquired hyperammonemia, most commonly resulting from liver cirrhosis, may be caused by all of the following EXCEPT: (Important M3WAN - Hard)**		
A-	Alcoholism	B-	Hepatitis
C-	Biliary obstruction	D-	A congenital urea cycle enzyme mutation

33-	**Inherited hyperammonemias, caused by genetic defects in urea cycle enzymes, characteristically manifest: (Important M3WAN - Core)**
A-	Only in adulthood after high protein intake
B-	During the first week after birth, and lead to mental retardation if untreated
C-	Exclusively as a benign, self-limiting condition
D-	With no neurological consequences

34-	**Normal blood urea concentration is approximately: (M3WAN - Core)**						
A-	5–10 mg/dL	B-	20–40 mg/dL	C-	60–80 mg/dL	D-	100–120 mg/dL

35-	**Elevated blood urea concentration, termed uremia, characteristically occurs in: (M3WAN - Core)**		
A-	Advanced renal failure	B-	Advanced liver failure
C-	Acute pancreatitis only	D-	Uncomplicated dehydration only

36-	**In advanced liver disease, blood urea characteristically: (Important M3WAN - Core)**
A-	Increases due to enhanced urea cycle activity
B-	Decreases, with increasing blood ammonia concentration
C-	Remains completely unchanged
D-	Is undetectable

37-	**Uremia resulting from decreased blood volume, as in severe dehydration, is classified as: (M3WAN - Core)**		
A-	Renal uremia	B-	Post-renal uremia
C-	PrE-renal uremia	D-	Hepatic uremia

38-	**Uremia resulting from obstruction of urine flow, as by stones, strictures, or tumors, is classified as: (M3WAN - Core)**		
A-	PrE-renal uremia	B-	Renal uremia
C-	Post-renal uremia	D-	Functional uremia

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
C	A	C	B	A	B	A	A	A	A
11	12	13	14	15	16	17	18	19	20
D	C	B	B	B	B	A	B	B	A
21	22	23	24	25	26	27	28	29	30
A	B	A	B	A	B	B	A	B	A
31	32	33	34	35	36	37	38	39	40
A	D	B	B	A	B	C	C	-	-

Lecture (18): Oxygen Toxicity and Free Radical Injury

Pre Exams , Formatives

1-	**which of the following is useful in the treatment of atherosclerosis ? (MID 33, FINAL 34, FINAL 32)**		
A-	polyunsaturated fatty acids	B-	monounsaturated fatty acids
C-	Palmitoleic acid	D-	Oleic acid

2-	**Which of the following is the precursor for nitric oxide (FINAL 34, FINAL 31, FINAL 32)**						
A-	Arginine	B-	Leucine	C-	Isoleucine	D-	Lysine

3-	**vitamin E is considerd (FINAL 35)**		
A-	enzymatic antioxidant	B-	non enzymatic antioxidant
C-	cytochrome p 450	D-	nutrient antioxidant

4-	**super oxide dismutase is considerd (FINAL 35, FINAL 37)**		
A-	enzymatic antioxidant	B-	non enzymatic antioxidant
C-	nutrient antioxidant	D-	free radical

5-	**vitamin E has an important function as it is (Quiz 37, Quiz 36, FINAL 31)**						
A-	an antioxidant	B-	enzyme	C-	cofactor	D-	non of the above

6-	**Catalase enzyme convers? (MID 30)**		
A-	Superoxide anion into H2O2	B-	H2O2 into H2O and O2
C-	H2O2 into H2O	D-	None of the above

7-	**In course of metabolic process active forms of oxygen including superoxide anion radical are formed in the human body. By means of what enzyme is this anion inactivated? (MID 31)**				
A-	Super oxide dismutase	B-	Glutathione reductase	C-	Peroxidase
D-	Catalase	E-	Glutathione peroxidase		

8-	**The Beneficial Effect Of Free Radical Is Called? (MID 31)**				
A-	Lipid Peroxidation	B-	Respirator Brust	C-	DNA Mutation
D-	Non Of The Above	E-	Oxidative stress		

9-	**Nutrient Antioxidant Is ? (MID 31)**				
A-	Vitamin c	B-	UNC Acid	C-	Glutathione Peroxidase
D-	Non Of The Above	E-	Glucose		

10-	Zinc and selenium are ..... (FINAL 36)						
A-	Lipid	B-	Vitamins	C-	Protein	D-	Antioxidant

11-	free radicals affect (Formative 37, Quiz SNU-M1, Formative SNU-M1)		
A-	protein.	B-	DNA structure only
C-	carbohydrates, lipid, DNA structures	D-	protein structure only

12-	free radicals have..... (Formative 37, Quiz SNU-M1, Formative SNU-M1, SNU Bank)		
A-	harmful effects only	B-	no effects at all
C-	harmful and beneficial effects	D-	beneficial effects only

13-	Protect RBCs from haemolysis? (FINAL 31)						
A-	Purines	B-	Glutathione	C-	Non of the above	D-	protein

14-	congenital deficiency of glucosE-6-phosphate dehydrogenase enzyme cause? (FINAL 31)						
A-	Favism	B-	Dwarfism	C-	Non of the above	D-	anemia

15-	Vitamin E is the most important natural .....? (2ND TRIAL 32)						
A-	Anti-allergic	B-	Anti-hemorrhagic	C-	Anti-oxidant	D-	Anti-pellagra.

16-	effects of free radicals occur in..... (Formative SNU 1)		
A-	proteins only	B-	lipid only
C-	carbohydrates only	D-	all of the above

17-	free radicals have ...on cells and tissues (Formative SNU 1)		
A-	harmful and beneficial effects	B-	harmful effects only
C-	No effects	D-	beneficial effects only

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
A	A	D	A	A	B	A	B	A	D
11	12	13	14	15	16	17	18	19	20
C	C	B	A	C	D	A	-	-	-

Department Qs

1-	free radicals affect (SNU Bank)		
A-	protein structure only	B-	carbohydrates structure only
C-	DNA structure only	D-	protein, carbohydrates,lipid,DNA structures

2-	**nutrient antioxidant is (SNU Bank)**		
A-	Glutathione	B-	super oxide dismutase
C-	vitamin C	D-	catalase

3-	**respiratory brust is ..... (SNU Bank)**		
A-	harmful effect to the cell	B-	benificial effect to the cell
C-	no effect to the cell	D-	no answer is correct

4-	**which of the following is not antioxidant (SNU Bank)**						
A-	vitamin c	B-	glycine	C-	vitamin E	D-	vitamin A

5-	**Free Radical Is..... (Department Qs33)**
A-	An Atom Free
B-	Amolocule That Contains One Or More Unparied Electron in Its Outer Orbital
C-	Inactive Atorn
D-	Non Of The Above

6-	**Role Of Free Radical In Body Is.... (Department Qs33)**		
A-	Harmfull	B-	Benificial Respiraturs Barste
C-	Both	D-	, b
E-	Non Of The Above		

7-	**Harmfull Effects Of Free Radical Affect (Department Qs33, Lecture Qs34)**						
A-	Lipid	B-	Dna	C-	Protein	D-	All Of The Above

8-	**The Beneficial Effect Of Free Radical Is Called: (Department Qs37)**		
A-	Lipid Peroxidation	B-	Respiratory Burst
C-	DNA mutation	D-	None Of The Above

9-	**Metabolic Antioxidant Is: (Department Qs37)**						
A-	Albumin	B-	Vitamin A	C-	Selenium	D-	Catalase Enzyme

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
D	C	B	B	B	C	D	B	A	-

Amira Nazeer

1-	Role of free radicals in the body is (Dr. Amira Nazeer)						
A-	harmful only	B-	beneficial only	C-	both a and b	D-	none of the above

2-	glutathione peroxidase is; (Dr. Amira Nazeer)		
A-	chain breaking antioxidant	B-	preventive antioxidant
C-	reparative antioxidant	D-	none of the above

3-	vitamine C is (Dr. Amira Nazeer)		
A-	nutrient antioxidant	B-	enzymatic antioxidant
C-	metabolic antioxidant	D-	none of the above

4-	All of the following are antioxidants except; (Dr. Amira Nazeer)						
A-	vitamin c	B-	vitamin E	C-	selenium	D-	glycine

5-	Free radical is (Dr. Amira Nazeer)
A-	An atom free
B-	A molecule that contains one or more unpaired electrons in outer
C-	Inactive atom
D-	None of above

6-	Harmful effects of free radicals affect (Dr. Amira Nazeer)						
A-	Lipids	B-	Proteins	C-	Carbohydrates	D-	all of above

7-	which of the following is not a free radical? (Dr. Amira Nazeer)		
A-	Hydroxyl radical	B-	Ozone
C-	Nitric oxide	D-	Hydrogen peroxide

8-	enzyme which catalyze the reaction H2O2 give H2O + O2 (Dr. Amira Nazeer)		
A-	catalase	B-	glutathione reductase
C-	glutathione perioxidase	D-	Glutathione transferase

9-	Which of the following is Preventive antioxidant: (Dr. Amira Nazeer)		
A-	Superoxide dismutase	B-	vitamin E
C-	uric acid	D-	catalase

10-	which enzyme converts superoxide (O2-) to hydrogen peroxide , O2 (Dr. Amira Nazeer)		
A-	catalase	B-	Glutathione peroxidase
C-	Superoxide dismutase	D-	none of the above

11-	All of the following is metabolic antioxidant except: (Dr. Amira Nazeer)		
A-	glutathione	B-	Ceruloplasmin
C-	Glutathione reductase	D-	Albumin

12-	Propagation phase occur under; (Dr. Amira Nazeer)		
A-	aerobic conditions	B-	anaerobic
C-	both a and b	D-	None of the above

13-	what are the sources of free radicals; (Dr. Amira Nazeer)
A-	Result of drug metabolism
B-	damage caused by ionizing radiations on tissues (X-rays)
C-	Photoexcitation of organic molecules
D-	all of the above

14-	Which of the following is not an oxygen derived free radical (ROS)? (Dr. Amira Nazeer)		
A-	Nitric oxide	B-	Hydrogen peroxide
C-	Superoxide anion	D-	Hydroxyl radical

15-	Which of the following lipid is most affected by free radicals? (Dr. Amira Nazeer)		
A-	Long chain fatty acids	B-	Polyunsaturated fatty acids
C-	Saturated fatty acids	D-	Monounsaturated fatty acids

Answers - Amira Nazeer									
1	2	3	4	5	6	7	8	9	10
C	B	A	D	B	D	D	A	D	C
11	12	13	14	15	16	17	18	19	20
C	A	D	A	B	-	-	-	-	-

Lecture Qs

1-	role of free radicals in the body is: (Lecture Qs33, Lecture Qs34, Lecture Qs35)						
A-	Harmful only	B-	Benefecial only	C-	Both 1,2	D-	Non of the above

2-	**glutathione peroxidase is : (Lecture Qs33, Lecture Qs34, Lecture Qs35)**		
A-	Chain braking antioxidant.	B-	Preventive antioxidants.
C-	Reparative antioxidant.	D-	Non of the above.

3-	**Vitamin C Is : (Lecture Qs33, Lecture Qs34, Lecture Qs35)**		
A-	Nutrient Antioxidant	B-	Enzymatic Antioxidant
C-	Metabolic Antioxidant	D-	Non Of The Above

4-	**all of th following are antioxidants except : (Lecture Qs33, Lecture Qs34, Lecture Qs35)**						
A-	VITAMIN C	B-	VITAMIN E	C-	SELENIUM	D-	GLYCINE

5-	**Free Radical Is..... (Lecture Qs34)**
A-	An Atom Free
B-	A molecule That Contains One Or More Unpaired Electron In Its Outer Orbital
C-	Inactive Atom
D-	Non Of The Above

Answers - Lecture Qs									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	**B**	**A**	**D**	**B**	**-**	**-**	**-**	**-**	**-**

M3WAN Bank

1-	**A free radical is best defined as a molecule or molecular fragment that: (M3WAN - Core)**
A-	Contains one or more unpaired electrons in its outer orbital
B-	Has an excess negative charge
C-	Contains only paired electrons
D-	Is always a stable, inert species

2-	**Reactive oxygen species (ROS) are generated as: (M3WAN - Core)**
A-	Products of complete reduction of oxygen to water
B-	Products of the partial reduction of molecular oxygen
C-	Products of glycolysis exclusively
D-	Byproducts of DNA replication only

3-	**Which of the following is correctly classified, despite extreme reactivity, as NOT technically a free radical? (Important M3WAN - Core)**		
A-	Superoxide anion	B-	Hydroxyl radical

C-	Hydrogen peroxide	D-	Nitric oxide

4-	**Environmental sources contributing to ROS generation include cytochrome P450-related drug metabolism, ionizing radiation, and: (M3WAN - Core)**

A-	Cigarette smoke and alcohol	B-	Dietary fiber intake
C-	Vitamin C supplementation	D-	Regular aerobic exercise alone

5-	**Characteristic properties of reactive oxygen species include extreme reactivity, short life span, tissue damage, and: (M3WAN - Core)**
A-	Generation of new ROS through chain reactions
B-	Complete chemical stability
C-	Exclusive restriction to the nucleus
D-	Inability to cross cell membranes

6-	**Lipid peroxidation proceeds through three sequential stages: initiation, propagation, and: (M3WAN - Core)**						
A-	Termination	B-	Reduction	C-	Hydrolysis	D-	Amplification

7-	**The initiation phase of lipid peroxidation involves removal of a hydrogen atom from a polyunsaturated fatty acid, caused by: (Important M3WAN - Core)**		
A-	Superoxide anion	B-	Hydroxyl radical
C-	Singlet oxygen only	D-	Nitric oxide

8-	**During the propagation phase of lipid peroxidation, the fatty acid radical (L•) reacts with oxygen to form: (Important M3WAN - Core)**		
A-	A peroxy radical (LOO•)	B-	Hydrogen peroxide directly
C-	A stable alcohol	D-	Superoxide anion

9-	**Free radical damage to proteins characteristically involves oxidation of sulfhydryl groups and modification of amino acids such as methionine, cysteine, histidine, and: (M3WAN - Core)**						
A-	Tryptophan	B-	Glycine	C-	Alanine	D-	Serine

10-	**Free radical-induced damage to carbohydrates, particularly relevant in diabetes mellitus, is characterized by: (M3WAN - Core)**
A-	Increased susceptibility of glycated proteins to free radical attack
B-	Complete resistance of glycated proteins to oxidative damage
C-	Reduced glucose oxidation
D-	Formation of amylose exclusively

11-	**Free radical damage to nucleic acids characteristically produces: (M3WAN - Core)**
A-	DNA strand breaks and fragmentation of bases and deoxyribose
B-	Increased base pairing stability
C-	Formation of additional hydrogen bonds
D-	No detectable structural change

12-	**Chronic inflammatory diseases such as rheumatoid arthritis are self-perpetuated by free radicals released mainly by: (M3WAN - Core)**						
A-	Neutrophils	B-	Hepatocytes	C-	Erythrocytes	D-	Adipocytes

13-	**Oxidized low-density lipoprotein (LDL), formed by free radical action, is a key contributor to: (Important M3WAN - Core)**		
A-	Atherosclerosis and coronary heart disease	B-	Peptic ulcer disease
C-	Osteoporosis	D-	Hemolytic anemia

14-	**Free radicals contribute to carcinogenesis mainly through: (M3WAN - Core)**
A-	DNA damage causing mutagenicity and cytotoxicity
B-	Direct inhibition of all cellular enzymes
C-	Stabilization of the genome
D-	Enhanced DNA repair fidelity

15-	**Prolonged exposure of the lungs to 100% oxygen is associated with endothelial destruction and pulmonary edema, contributing to: (Important M3WAN - Core)**
A-	Adult respiratory distress syndrome (ARDS)
B-	Chronic bronchitis exclusively
C-	Pulmonary fibrosis unrelated to oxidative stress
D-	Asthma

16-	**Free radical-mediated destruction of pancreatic islet cells is implicated in the pathogenesis of: (M3WAN - Core)**		
A-	Insulin-dependent diabetes mellitus (IDDM)	B-	Diabetes insipidus
C-	Gout	D-	Hemochromatosis

17-	**Oxidative stress is a recognized contributing factor in the pathogenesis of all of the following EXCEPT: (Important M3WAN - Hard)**		
A-	Cataract formation	B-	Male infertility (reduced sperm motility)
C-	Parkinsonism and Alzheimer-s dementia	D-	Vitamin K-dependent clotting factor synthesis

18-	**Antioxidants that block the initial production of free radicals, such as catalase and glutathione peroxidase, are classified as: (Important M3WAN - Core)**		
A-	Preventive antioxidants	B-	Chain-breaking antioxidants
C-	Pro-oxidants	D-	Structural antioxidants

19-	**Antioxidants that inhibit the propagative phase of lipid peroxidation, such as vitamin E and uric acid, are classified as: (Important M3WAN - Core)**		
A-	Preventive antioxidants	B-	Chain-breaking antioxidants
C-	Enzymatic antioxidants exclusively	D-	Nutrient cofactors only

20-	**Among antioxidants classified by location, α-tocopherol is characteristically described as: (M3WAN - Core)**		
A-	Plasma antioxidant	B-	Cell membrane antioxidant
C-	Intracellular cytosolic antioxidant only	D-	Nuclear antioxidant

21-	**Which of the following is classified as a plasma antioxidant rather than an intracellular enzymatic antioxidant? (M3WAN - Core)**		
A-	Superoxide dismutase	B-	Catalase
C-	Ceruloplasmin	D-	Glutathione peroxidase

22-	**Superoxide dismutase (SOD), catalase, and glutathione peroxidase are classified together as: (M3WAN - Core)**		
A-	Enzymatic antioxidants	B-	Non-enzymatic antioxidants
C-	Nutrient antioxidants exclusively	D-	Chain-breaking antioxidants exclusively

23-	**Nutrient (dietary) non-enzymatic antioxidants include carotenoids, α-tocopherol, selenium, and: (M3WAN - Core)**						
A-	Vitamin C	B-	Vitamin K	C-	Vitamin B12	D-	Biotin

24-	**Superoxide dismutase catalyzes conversion of superoxide anion into: (Important M3WAN - Core)**		
A-	Hydrogen peroxide and oxygen	B-	Water directly
C-	Hydroxyl radical	D-	Nitric oxide

25-	**Superoxide dismutase is considered the first line of cellular defense against the injurious effects of: (Important M3WAN - Core)**		
A-	Superoxide	B-	Hydrogen peroxide
C-	Hydroxyl radical	D-	Singlet oxygen

26-	Catalase functions to metabolize: (M3WAN - Core)		
A-	Superoxide anion	B-	Hydrogen peroxide, into water and oxygen
C-	Lipid hydroperoxides exclusively	D-	Nitric oxide

27-	Glutathione peroxidase detoxifies hydrogen peroxide to water while converting reduced glutathione (GSH) to: (Important M3WAN - Core)		
A-	Oxidized glutathione (GSSG)	B-	GlutathionE-S-transferase
C-	Cysteine	D-	Homocysteine

28-	Regeneration of reduced glutathione from its oxidized form is catalyzed by glutathione reductase, which requires: (Important M3WAN - Core)						
A-	NADPH	B-	NADH	C-	FADH2	D-	ATP only

29-	The major cellular source of the NADPH required for glutathione regeneration is: (Important M3WAN - Core)		
A-	The HMP (hexose monophosphate) shunt	B-	Glycolysis
C-	The Krebs cycle	D-	β-oxidation

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
A	B	C	A	A	A	B	A	A	A
11	12	13	14	15	16	17	18	19	20
A	A	A	A	A	A	D	A	B	B
21	22	23	24	25	26	27	28	29	30
C	A	A	A	A	B	A	A	A	-

«العِلمُ النّبوي يُمَيِّز الحق من الباطل»

قال ابن القيم رحمه الله: فليس العلم كثرة النقل والبحث والكلام، ولكن نُورٌ يُمَيِّزُ بِهِ صحيح الأقوالِ مِنْ سقيمها، وحقها مِنْ باطِلِها، وما هو مِنْ مشكاة النُّبُوَّةِ مِمَّا هو مِنْ آراء الرجال

•اجتماع الجيوش الإسلامية صـ (٧٧)

Lecture (19): Integration of Metabolism

Pre Exams , Formatives

1-	During starvation, the first reserve nutrient to be depleted is? (MID 30)						
A-	Glycogen	B-	Proteins	C-	Triglycerides	D-	Cholesterol

2-	Which of the following regulates lipolysis in adipocytes (FINAL 34, FINAL 31)
A-	Activation of fatty acid synthesis mediated by CAMP
B-	Glycerol phosphorylation to prevent futile esterification of fatty acids
C-	Activation of triglyceride lipase as a result of hormone stimulated increases in CAMP levels
D-	Activation of CAMP production by Insulin

3-	The following Hormones increase the mobilization of depot fat (lipoysis) except (FINAL 34)		
A-	Epinephrine and norepinephrine	B-	Insulin
C-	Growth Hormone	D-	Glucagon

4-	Action of insulin on lipid metabolism is ? (MID 30)
A-	it increases lipolysis and increases triglyceride synthesis
B-	it decreases lipolysis and increases triglyceride synthesis
C-	decreases lipolysis and decreases triglyceride synthesis
D-	It increases synthesis of triglyceride and increased ketogenesis

5-	Which of the following statements is true about brain metabolism in starvation? (MID 31)
A-	The brain can use glucogenic amino acids for energy
B-	The brain can only use glucose
C-	Up to a quarter of energy requirement of the brain can come from fatty acids
D-	Up to three-quarters (75%) of energy requirement of the brain can come from ketone bodies
E-	All of the above

6-	in starvation, hepatic enzymes activate (FINAL 36)						
A-	Gluconeogenesis	B-	Lipogenesis	C-	Glycolysis	D-	glycogenesis

7-	best brain fuel energy is during feed state (FINAL 37 SNU, SNU Bank)						
A-	protein	B-	amino acids	C-	glucose	D-	fatty acids

8-	in absence of glucose brain will take energy from..... During starvation (FINAL 37 SNU, SNU Bank)						
A-	alcohol	B-	ketone bodies	C-	glycogen	D-	fatty acids

9-	in feed state body has its energy from (Formative 37, Quiz SNU-M1)		
A-	Carbohydrate	B-	lipid metabolism
C-	protein metabolism	D-	nucleoprotein metabolism

10-	in feed state which organ will start metabolism (Formative 37, Quiz SNU-M1, SNU Bank)						
A-	muscle	B-	Liver	C-	lung	D-	kidney

11-	Which of the following is not true about the fasting metabolic state? (FINAL 31)
A-	Enhanced lipolysis occurs in adipose tissue
B-	Ketogenesis is enhanced
C-	Gluconeogenesis occurs
D-	Glucose is formed from muscle glycogen via glucosE-6-phosphatase

12-	Starvation induces an increase in activity of the hepatic enzymes involved in? (FINAL 31)		
A-	Hexose monophosphate shunt	B-	Lipogenesis
C-	Glycolysis	D-	Gluconeogenesis

13-	Following a normal overnight fast, the body will depend only on the following as a source of energy? (FINAL 31)		
A-	Keton bodies	B-	Glycolysis
C-	Gluconeogenesis	D-	Glycogenolysis and gluconeogenesis

14-	What is the molecular basis of statement "Fat in the flame of carbohydrates"? (FINAL 31)
A-	Fat are hydrolyzed in the presence of carbohydrates
B-	Fatty acid and glucose are simultaneously oxidized
C-	Acetyl coA is the common product of fatty acid and glucose oxidation
D-	Acetyl coA Is oxidized completely in the presence of oxaloacetate in TCA cycle

15-	In prolonged starvation, the main energy source of brain is ? (2ND TRIAL 32)						
A-	Glucose	B-	Ketone bodies	C-	Fructose	D-	Fatty acids

16-	Insulin increases the following pathways in liver EXCEPT ? (2ND TRIAL 32, Department Qs37)		
A-	Fatty acid synthesis	B-	Glycogen synthesis
C-	Protein synthesis	D-	Glucose synthesis

17-	Acetyl CoA can be converted into all of the following EXCEPT? (2ND TRIAL 32, Department Qs37)						
A-	Glucose	B-	Fatty acid	C-	Cholesterol	D-	Ketone bodies

18-	**Which of the following statements is true? (Formative 34)**
A-	High insulin/glucagon ratio activates lipolysis in muscle
B-	High insulin/glucagon ratio activates lipolysis in adipocytes
C-	High insulin/glucagon ratio inhibits lipolysis in adipocytes
D-	Low insulin/glucagon ratio inactivates lipolysis in adipocytes

Answers - Pre Exams , Formatives									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	**C**	**B**	**B**	**D**	**A**	**C**	**B**	**A**	**B**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	**D**	**D**	**D**	**B**	**D**	**A**	**C**	**-**	**-**

Department Qs

1-	**in feed state body has its energy from (SNU Bank)**		
A-	protein metabolism	B-	lipid metabolism
C-	carbohydrate metabolism	D-	nucleoprotein metabolism

2-	**which of the following organ is the glucostate of the body (SNU Bank)**						
A-	liver	B-	muscle	C-	lung	D-	heart

3-	**Glucagone has receptor in all the following except (Department Qs33)**				
A-	Liver	B-	Muscle	C-	Both
D-	, B	E-	None of above		

4-	**Hypoglycemic hormone is (Department Qs33)**						
A-	Insulin	B-	Glucagone	C-	Adenaline	D-	All of above

5-	**The following mechanisms play a recognized part in fat metabolism, which is not true: (Department Qs33)**
A-	Conversion of adipose tissue triglyceride to free fatty acids and glycerol during fasting.
B-	Conversion of free fatty acids to glycogen in muscles during fasting.
C-	Utilization of keton bodies by brain tissue as an energy source during prolonged fasting.
D-	Conversion in adipose tissue of excess glucose to triglycerides

6-	**During prolonged starvation, which one of the following will take place: (Department Qs33)**
A-	The brain will utilize 10% of keton bodies formed
B-	One of the major sources of glucose will be alanine

C-	Muscles will synthesize most of glucose
D-	Pyruvate carboxylase will be inactivated

7-	Cyclic AMP Is formed from ATP by the enzyme (Department Qs33)		
A-	Insulin	B-	Epinephrine + glucagon
C-	Testosterone	D-	Progesterone

8-	Which of the following metabolite integrates glucose and fatty acid metabolism? (Department Qs33)						
A-	Acetyl COA	B-	Pyruvate	C-	Citrate	D-	Lactate

9-	Transport of glucose across the cell membrane is stimulated by insulin in: (Department Qs 35)						
A-	Brain	B-	RBCs	C-	Liver	D-	Skeletal muscles

10-	Diabetes mellitus is due to the deficiency of the action of which hormones: (Department Qs 35)				
A-	Glucagon	B-	Glucocorticoids	C-	Nor-epinephrine
D-	Insulin	E-	Thyroxin		

11-	Citrate exerts effects on the activities of certain enzymes involved in glycolysis and in fatty acid synthesis: (Department Qs 35)
A-	These effects are examples of competitive inhibition
B-	The effect on acetyl-CoA carboxylase is an example of negative feed back control.
C-	Effects on phosphofructokinase and on acetyl-CoA carboxylase are examples of allosteric effects
D-	Both citrate and palmitoyl-CoA have similar effects on acetyl-CoA-carboxylase.

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	A	B	A	B	B	B	A	D	D
11	12	13	14	15	16	17	18	19	20
C	-	-	-	-	-	-	-	-	-

Amira Nazeer

1-	Hypoglycemic hormone (Dr. Amira Nazeer)						
A-	Insulin	B-	Glucagon	C-	Adrenaline	D-	None of above

Answers - Amira Nazeer									
1	2	3	4	5	6	7	8	9	10
A	-	-	-	-	-	-	-	-	-

M3WAN Bank

1-	**Integration of metabolism refers to the coordination between: (M3WAN - Core)**		
A-	Carbohydrate, lipid, and protein metabolism	B-	DNA and RNA synthesis only
C-	Vitamin and mineral absorption only	D-	Enzyme kinetics exclusively

2-	**The two hormones primarily responsible for maintaining blood glucose homeostasis are: (M3WAN - Core)**		
A-	Cortisol and thyroxine	B-	Insulin and glucagon
C-	Epinephrine and norepinephrine	D-	Growth hormone and prolactin

3-	**Under conditions of positive caloric balance, a significant proportion of excess food energy is stored as: (M3WAN - Core)**						
A-	Glycogen or fat	B-	Ketone bodies	C-	Urea	D-	Free amino acids

4-	**Under negative caloric balance, fatty acids are preferentially oxidized over glucose in order to: (Important M3WAN - Core)**
A-	Spare glucose for tissues such as brain and red blood cells that require it under all conditions
B-	Increase hepatic glycogen stores
C-	Promote protein synthesis
D-	Inhibit ketogenesis

5-	**Glucogenic amino acids serve as precursors for glucose synthesis via pyruvate or intermediates of the: (M3WAN - Core)**		
A-	Pentose phosphate pathway	B-	Tricarboxylic acid (Krebs) cycle
C-	Urea cycle	D-	β-oxidation pathway

6-	**Ketogenic amino acids serve mainly as precursors for: (M3WAN - Core)**						
A-	Acetyl-CoA	B-	Oxaloacetate	C-	Glucose directly	D-	Urea

7-	**Acetyl-CoA is described as the "common metabolite" of energy metabolism because it: (Important M3WAN - Core)**
A-	Is produced from carbohydrate, lipid, and protein fuel sources and enters the citric acid cycle
B-	Is produced only from glucose
C-	Cannot be oxidized further
D-	Is unique to ketogenic pathways only

8-	The hexose monophosphate (HMP) shunt contributes to metabolic integration mainly by supplying NADPH for biosynthesis and: (M3WAN - Core)		
A-	Ribose sugar for nucleotide synthesis	B-	ATP for muscle contraction
C-	Ammonia for the urea cycle	D-	Ketone bodies for brain fuel

9-	In the well-fed state, the liver is described as the body-s "metabolic hub" because it: (M3WAN - Core)
A-	Removes about two-thirds of absorbed glucose and supplies fuel to other organs
B-	Is the sole site of glucose utilization in the body
C-	Cannot metabolize lipids
D-	Depends entirely on peripheral tissues for glucose supply

10-	In adipose tissue during the well-fed state, increased glycolysis serves mainly to provide: (Important M3WAN - Core)
A-	Glycerol phosphate for triacylglycerol synthesis
B-	ATP for muscle contraction
C-	NADH for the electron transport chain exclusively
D-	Ketone bodies

11-	Since adipocytes lack glycerol kinase, the glycerol-3-phosphate needed for triacylglycerol synthesis in adipose tissue must be derived from: (Important M3WAN - Core)		
A-	Glucose metabolism	B-	Direct phosphorylation of free glycerol
C-	Fatty acid oxidation	D-	Amino acid catabolism

12-	In the well-fed state, insulin inhibits triacylglycerol degradation in adipose tissue mainly by: (Important M3WAN - Core)
A-	Keeping hormonE-sensitive lipase in its dephosphorylated (inactive) form
B-	Directly activating hormonE-sensitive lipase
C-	Increasing glucagon secretion
D-	Inhibiting glucose transport

13-	In skeletal muscle during the well-fed state, an increased insulin-to-glucagon ratio together with available glucosE-6-phosphate stimulates: (M3WAN - Core)						
A-	Glycogenesis	B-	Gluconeogenesis	C-	Ketogenesis	D-	Lipolysis

14-	Which amino acids are taken up preferentially by skeletal muscle in the well-fed state for both protein synthesis and energy? (Important M3WAN - Core)
A-	The brancheD-chain amino acids (leucine, isoleucine, valine)
B-	The aromatic amino acids

C-	The sulfur-containing amino acids
D-	The acidic amino acids

15-	**Compared with skeletal muscle, heart muscle is distinguished by all of the following EXCEPT: (Important M3WAN - Hard)**
A-	Continuous rather than intermittent activity
B-	Completely aerobic metabolism
C-	Large glycogen and lipid energy reserves
D-	Use of glucose, free fatty acids, and ketone bodies as fuels

16-	**In the well-fed (absorptive) state, the brain relies almost exclusively on which fuel, given its lack of glycogen stores? (Important M3WAN - Core)**						
A-	Glucose	B-	Free fatty acids	C-	Ketone bodies	D-	Amino acids

17-	**Cerebral function becomes impaired when blood glucose falls to approximately: (M3WAN - Hard)**						
A-	60 mg/dL	B-	30 mg/dL	C-	90 mg/dL	D-	120 mg/dL

18-	**Free fatty acids contribute minimally to brain energy supply because they: (M3WAN - Core)**
A-	Cannot cross the blooD-brain barrier
B-	Are toxic to neurons
C-	Are rapidly converted to ketone bodies in the brain
D-	Require insulin for neuronal uptake

19-	**The hormonal pattern characteristic of starvation is: (M3WAN - Core)**		
A-	Decreased insulin and increased glucagon	B-	Increased insulin and decreased glucagon
C-	Both insulin and glucagon decreased	D-	Both insulin and glucagon increased equally

20-	**Liver glycogen stores become nearly depleted after approximately how many hours of fasting? (M3WAN - Core)**						
A-	2–4 hours	B-	10–18 hours	C-	48 hours	D-	72 hours

21-	**Hepatic gluconeogenesis begins approximately how many hours after the last meal and becomes fully active as glycogen is depleted? (Important M3WAN - Core)**						
A-	4–6 hours	B-	30 minutes	C-	24 hours	D-	1 hour

22-	**The main substrates for gluconeogenesis during fasting are amino acids, glycerol, and: (M3WAN - Core)**						
A-	Lactate	B-	Acetoacetate	C-	Cholesterol	D-	Uric acid

23-	**During starvation, hepatic ketogenesis becomes favored when acetyl-CoA production from fatty acid oxidation exceeds: (Important M3WAN - Core)**
A-	The oxidative capacity of the tricarboxylic acid cycle
B-	The capacity of glycolysis
C-	The capacity of the urea cycle
D-	Renal excretory capacity

24-	**Ketone bodies characteristically appear in the blood and urine by approximately which point in a fast? (M3WAN - Hard)**		
A-	Within the first hour	B-	By the second day
C-	Not until after 3 weeks	D-	Only during refeeding

25-	**In adipose tissue during starvation, activation of hormonE-sensitive lipase and increased lipolysis is driven mainly by high levels of: (Important M3WAN - Core)**		
A-	Epinephrine	B-	Insulin
C-	Aldosterone	D-	ThyroiD-stimulating hormone

26-	**The glycerol released from adipose tissue lipolysis during starvation is used by the liver mainly for: (M3WAN - Core)**		
A-	Gluconeogenesis	B-	Ketogenesis directly
C-	Glycogen synthesis	D-	Protein synthesis

27-	**During the first few days of starvation, rapid breakdown of muscle protein releases amino acids used by the liver for gluconeogenesis; the most important glucogenic amino acids released from muscle are: (Important M3WAN - Core)**		
A-	Alanine and glutamine	B-	Leucine and lysine
C-	Tryptophan and phenylalanine	D-	Methionine and cysteine

28-	**After approximately three weeks of starvation, muscle proteolysis decreases mainly because: (Important M3WAN - Core)**
A-	The brain has adapted to rely on ketone bodies, reducing its need for glucose derived from protein catabolism
B-	Muscle glycogen stores have been replenished
C-	Insulin secretion sharply increases
D-	Gluconeogenesis is no longer active

29-	Beyond approximately three weeks of starvation, the brain can derive up to what percentage of its energy needs from ketone bodies? (Important M3WAN - Core)						
A-	25%	B-	50%	C-	75%	D-	100%

30-	After about three weeks of fasting, skeletal muscle characteristically shifts to relying mainly on: (M3WAN - Hard)
A-	Fatty acids alone, decreasing ketone body utilization
B-	Glucose alone
C-	Ketone bodies alone
D-	Amino acids alone

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
A	B	A	A	B	A	A	A	A	A
11	12	13	14	15	16	17	18	19	20
A	A	A	A	C	A	B	A	A	B
21	22	23	24	25	26	27	28	29	30
A	A	A	B	A	A	A	A	C	A

عن أبي مسعود الأنصاري رضي الله عنه قال: جَاءَ رَجُلٌ إلى النبيّ صَلَّى اللَّهُ عليه وَسَلَّمَ، فَقَالَ: إنّي أُبدِعَ بي فَاحْمِلْنِي، فَقَالَ: ما عِندِي، فَقَالَ رَجُلٌ: يا رَسولَ اللهِ، أَنَا أَدُلُّهُ علَى مَن يَحْمِلُهُ، فَقَالَ رَسولُ اللهِ صَلَّى اللَّهُ عليه وَسَلَّمَ: مَن دَلَّ علَى خَيْرٍ فَلَهُ مِثْلُ أَجْرِ فَاعِلِهِ.

Lecture (20): DNA Structure and chromatin organization

Pre Exams & Formatives

1-	Most predominant protein associated with chromatin is (FINAL 34, FINAL 32)						
A-	Ubiquitin	B-	Nucleoplasmin	C-	DNA ligase	D-	Histone

2-	A nucleotide contain (FINAL 34, FINAL 32)		
A-	Base+pentose+ phosphate	B-	Base+ phosphate
C-	Base+ pentose	D-	None of the above

3-	5--3 - DNA strand is called ? (MID 30)						
A-	Temple strand	B-	Antisence strand	C-	Sence strand	D-	Anticoding strand

4-	A nucleoside consists of ? (MID 30)
A-	Nitrogenous base
B-	Purine or pyrimidine base + sugar
C-	Purine or pyrimidine base + phosphorous
D-	Purine or pyrimidine base + sugar + phosphorous

5-	Number of guanine and cytosine residues is equal in ? (MID 30)						
A-	mRNA	B-	tRNA	C-	rRNA	D-	DNA

6-	All the following statements are true with regard to Watson-Crick model of DNA, EXCEPT? (FINAL 31)		
A-	Double stranded helix	B-	Adenine bonds with Thymine
C-	DNA strands run in opposite directions	D-	Guanine bonds with Uracil

7-	Telomere? (FINAL 31)		
A-	Present in the middle of DNA	B-	Present only at 5 end
C-	Formed of repeated sequence	D-	not present in DNA

8-	nucleotide is formed of? (FINAL 31)		
A-	base, sugar and phosphate group	B-	base and sugar
C-	base, and phosphate group	D-	base, sugar and sulfate group

9-	Histones are small proteins formed from? (FINAL 31, Department Qs37)		
A-	Arginine only	B-	arginine and glycine
C-	arginine and histidine	D-	arginine and tyrosine

10-	The packaging unit of the chromatin is called? (FINAL 31, Department Qs37)						
A-	chromosome	B-	nucleosome	C-	Ribosome	D-	Microsome

11-	Genetic information flows from? (FINAL 31)		
A-	DNA to DNA	B-	DNA to RNA
C-	RNA to cellular proteins	D-	DNA to cellular proteins

12-	In Neucleic acid, nucleotides are linked to each other by ? (FINAL 32)		
A-	3,5 phosphodiester bond	B-	Ester bond
C-	Glycosidic bond	D-	None of the above

13-	DNA is characterized by presence of ? (FINAL 32)						
A-	Ribose	B-	Deoxyribose	C-	Fucose	D-	Uracil

14-	An Example of Nucleoside is ? (FINAL 32)						
A-	C AMP	B-	ATP	C-	AMP	D-	Adenosine

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
D	A	C	B	D	D	C	A	C	B
11	12	13	14	15	16	17	18	19	20
B	A	B	D	-	-	-	-	-	-

Department Qs

1-	purine base is (SNU Bank)						
A-	adinine	B-	uracil	C-	cytosine	D-	thymine

2-	nucleotide is... (SNU Bank)
A-	nitrogenous base , deoxy sugar and phosphate group
B-	nitrogenous base
C-	deoxy sugar
D-	vitamins

3-	nucleoside is..... (SNU Bank)		
A-	deoxy sugar	B-	nitrogenous base
C-	phosphate group	D-	nitrogenous base and deoxy sugar

4-	pyrimidine base is (SNU Bank)						
A-	adinine	B-	guanine	C-	protein	D-	thymine

5-	Purine Ring is (Department Qs33)						
A-	Uracil	B-	Thymin	C-	Adnine Guanen	D-	Cytosine

6-	**Pyrimidine Ring Is... (Department Qs33)**						
A-	Guanine	B-	Adnine	C-	Tymine	D-	Adenosine

7-	**Nucleoside Is (Department Qs33)**		
A-	Nitrogenous Base	B-	Sugar
C-	Nitrogenous Base And Ribose	D-	Phosphric Acid

8-	**Nucleotide Is (Department Qs33)**		
A-	Nucleoside	B-	Nucleoside And Phosphric Acd
C-	Nitrogenous Base	D-	Phosphoric Acid

9-	**which of the following can be enter in nucleic Acids (Department Qs 35)**				
A-	Keratin	B-	protamines	C-	non of the above.

10-	**Nucleotides consists of all the following except: (Department Qs 35)**						
A-	Nucleotide	B-	phosphate	C-	thymine	D-	sphingosine

11-	**All of the following pyrmidine nitrogenous bases except: (Department Qs 35)**		
A-	Uracil	B-	Adenine

12-	**Ribose enters in the following except: (Department Qs 35)**				
A-	Adenosine	B-	ATP	C-	guanine

13-	**in secondary DNA structure: (Department Qs 35)**		
A-	The two strands are complementary	B-	The two strands are parallel
C-	Non of the above.		

14-	**In DNA structure: (Department Qs 35)**
A-	The phosphoric acid project inward
B-	The hydrophilic deaxyribose stacked in the core
C-	Non of the above.

15-	**Tolmer: (Department Qs 35)**		
A-	Coding for specific gene.	B-	Has similar length along whole life.
C-	Its length is marker of cellular senscene.		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
A	A	D	D	C	C	C	B	B	D
11	12	13	14	15	16	17	18	19	20
B	C	A	C	C	-	-	-	-	-

Lecture Qs

1-	Which of the following bases is not present in DNA? (Lecture Qs33)						
A-	Adenine	B-	Guanine	C-	Thymine	D-	Uracil

2-	The structure shown is that of a (Lecture Qs33)						
A-	DNA nucleoside	B-	RNA nucleoside	C-	DNA nucleotide	D-	RNA nucleotide

3-	Which of the following statements is correct? (Lecture Qs33)		
A-	Adenine is a pyrimidine	B-	DNA is made of amino acids
C-	Nucleosides do not contain phosphorous	D-	RNA contains thymine

4-	Which of the following best describes cytosine? (Lecture Qs33)		
A-	pyrimidine, present in RNA and DNA	B-	pyrimidine, present only in DNA
C-	purine, present only in RNA	D-	purine, present in RNA and DNA

5-	Which part of the nucleotide is responsible for the formation of bonds in the DNA double helix? (Lecture Qs33)		
A-	Base	B-	Sugar
C-	Phosphate group	D-	Hydroxyl group of sugar

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
D	B	C	A	A	-	-	-	-	-

M3WAN Bank

1-	DNA is fundamentally important to the cell mainly because it: (M3WAN - Core)
A-	Serves as the genetic material controlling protein production
B-	Functions exclusively as a structural membrane component
C-	Is the primary energy storage molecule of the cell
D-	Cannot be transmitted between generations

2-	**DNA exists physiologically as: (M3WAN - Core)**
A-	A singlE-stranded linear molecule
B-	A doublE-stranded molecule forming a double helix
C-	A triplE-stranded helix
D-	A circular single strand only

3-	**The two strands of the DNA double helix are described as antiparallel, meaning: (M3WAN - Core)**
A-	One strand runs 3-⟶5- while the other runs 5-⟶3-
B-	Both strands run in the same 5-⟶3- direction
C-	The strands are unrelated in directionality
D-	Only one strand has a defined polarity

4-	**According to the basE-pairing rule, adenine pairs with thymine through how many hydrogen bonds? (Important M3WAN - Core)**						
A-	One	B-	Two	C-	Three	D-	Four

5-	**Guanine pairs with cytosine through how many hydrogen bonds? (Important M3WAN - Core)**						
A-	One	B-	Two	C-	Three	D-	Four

6-	**Because of the specific basE-pairing rule, the two strands of a DNA double helix are described as: (M3WAN - Core)**
A-	Identical to each other
B-	Complementary but not identical to each other
C-	Unrelated in base sequence
D-	Composed of the same nitrogenous bases only

7-	**A nucleotide is composed of a nitrogenous base, a pentose sugar, and: (M3WAN - Core)**		
A-	A phosphate group	B-	An amino group
C-	A carboxyl group	D-	A sulfate group

8-	**The "backbone" of the DNA molecule is formed by which two components? (M3WAN - Core)**		
A-	Nitrogenous bases and hydrogen bonds	B-	Phosphate and sugar
C-	Histones and phosphate	D-	Purines and pyrimidines alone

9-	**The pyrimidine bases that occur in DNA are cytosine and: (M3WAN - Core)**						
A-	Uracil	B-	Thymine	C-	Adenine	D-	Guanine

10-	In RNA, the pyrimidine bases present are cytosine and: (M3WAN - Core)						
A-	Thymine	B-	Uracil	C-	Guanine	D-	Adenine

11-	The principal purine bases, found in both DNA and RNA, are adenine and: (M3WAN - Core)						
A-	Guanine	B-	Thymine	C-	Uracil	D-	Cytosine

12-	A nucleoside is defined as a structure formed by combination of a nitrogenous base with: (M3WAN - Core)		
A-	A phosphate group	B-	A sugar (ribose or deoxyribose)
C-	Another nitrogenous base	D-	A histone protein

13-	The nucleoside formed by combination of thymine with deoxyribose is called: (M3WAN - Core)						
A-	Thymidine	B-	Cytidine	C-	Uridine	D-	Guanosine

14-	The nucleoside formed by combination of uracil with ribose is called: (M3WAN - Core)						
A-	Thymidine	B-	Adenosine	C-	Uridine	D-	Cytidine

15-	Chromosomes are best described as: (M3WAN - Core)
A-	Nucleoproteins formed mainly of DNA and basic proteins
B-	Pure DNA molecules without associated protein
C-	RNA-protein complexes exclusively
D-	LipiD-protein complexes

16-	The normal chromosome number in a human somatic cell is: (M3WAN - Core)						
A-	23	B-	44	C-	46	D-	48

17-	Chromatin is defined as: (M3WAN - Core)
A-	The condensed DNA-protein complex forming chromosomal material
B-	Free nucleotides circulating in the nucleus
C-	A type of RNA polymerase
D-	The nuclear membrane

18-	Condensation of the very long DNA molecule into chromatin is necessary mainly because: (M3WAN - Core)
A-	The total DNA length in a human cell (about 1 meter) must fit into a cell only about 20 micrometers long
B-	DNA is otherwise chemically unstable
C-	Uncondensed DNA cannot replicate
D-	Condensation increases DNA solubility only

19-	The major protein components of chromatin, besides DNA and a small quantity of RNA, are histones and: (M3WAN - Core)						
A-	Protamines	B-	Immunoglobulins	C-	Albumins	D-	Collagen

20-	Histones are small, positively charged proteins at physiological pH mainly due to their high content of histidine, lysine, and: (Important M3WAN - Core)						
A-	Arginine	B-	Glutamate	C-	Aspartate	D-	Tryptophan

21-	The association between histones and DNA within chromatin is maintained mainly by: (Important M3WAN - Core)
A-	Covalent phosphodiester bonds
B-	Ionic bonds between positively charged histones and negatively charged DNA
C-	Hydrophobic interactions exclusively
D-	Disulfide bonds

22-	The genome of an organism is defined as: (M3WAN - Core)
A-	A single chromosome
B-	The total number of genes within one mature cell
C-	The RNA content of a cell
D-	The protein content of a cell

23-	A gene is best defined as: (M3WAN - Core)
A-	A DNA sequence at a specific locus coding for the synthesis of one polypeptide
B-	Any random segment of chromatin
C-	A segment of RNA only
D-	A protein that binds DNA

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
A	B	A	B	C	B	A	B	B	B
11	12	13	14	15	16	17	18	19	20
A	B	A	C	A	C	A	A	A	A
21	22	23	24	25	26	27	28	29	30
B	B	A	-	-	-	-	-	-	-

قال رجل للحسن البصري: يا أباسعيد، أشكو إليك قسوة قلبي، قال: أذبه بالذِّكر.

Lecture (21): Purine and pyrimidine metabolism

Pre Exams , Formatives

1-	Primary metabolic gout Is a disease characterized by increased a of? (2ND TRIAL 35, FINAL 31)				
A-	HGPRt	B-	PRPP	C-	xanthine oxidase

2-	Gout is characterized by increased plasma levels of ? (MID 30)						
A-	Urea	B-	Uric acid	C-	Creatine	D-	Creatinine

3-	Primary metabolic gout can be caused by: (FINAL 36)
A-	Von Gierke disease

4-	What-s the purine base of the following: (FINAL 36)						
A-	Caffeine	B-	cytosine	C-	thymine	D-	uracil

5-	gout is a disease occurs due to.. (Formative 37, Quiz SNU-M1, SNU Bank)		
A-	decrease serum blood uric acid level	B-	increase serum blood uric acid level
C-	decrease serum blood glucose level	D-	increase serum blood glucose level

6-	purine base is (Formative 37, Quiz SNU-M1)						
A-	adenine	B-	thymine	C-	cytosine	D-	uracil

7-	Allopurinol is considered competitive inhibitor of? (FINAL 31)		
A-	xanthine oxidase	B-	pyurvate kinase
C-	glukokinase	D-	Phenyl alanine hydroxylase

8-	gout is a disease of ? (FINAL 32)		
A-	increase serum uric acid	B-	decrease serum uric acid
C-	Increase serum Urea in blood	D-	non of the above

Answers - Pre Exams , Formatives									
31	32	33	34	35	36	37	38	39	40
B	B	A	A	B	A	A	A	-	-

Department Qs

1-	Normal Range Of Serum Uric Acid Is..... (Department Qs33)						
A-	100-200 Mg/Dl	B-	3-7 Mg/Dl	C-	0-40 Mg/Dl	D-	11-15 Mg/Dl

2-	Secondry Metabolic Gout is Due To..... (Department Qs33, Department Qs37)		
A-	Over Activity Of PRPP	B-	Leukemia
C-	HGPRT Defeniciy	D-	Renal Faliure

3-	The end product of purine catabolism in man is: (Department Qs37)						
A-	Inosine	B-	Hypoxanthine	C-	Xanthine	D-	Uric acid

4-	Allopurinol is considered competitive inhibitor of.. (Department Qs37)		
A-	xanthine oxidase	B-	pyruvate kinase
C-	glucokinase	D-	Phenyl alanine hydroxylase

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	B	D	A	-	-	-	-	-	-

Lecture Qs

1-	All Of The Following Are Purine Bases Except : (Lecture Qs33, Lecture Qs37)						
A-	adenine	B-	Guanine	C-	Cytosine	D-	All Of The Above

2-	Gout is a disease characterized by increase in serum.. (Lecture Qs33, Lecture Qs37)						
A-	Bile Acids	B-	Uric Acid	C-	Urea	D-	Creatinine

3-	the Key Regulatory Enzyme In Catabolism Of Purine Is (Lecture Qs33)		
A-	Glucokinase	B-	Xanthine Oxidase
C-	Hexokinase	D-	Pyruvate Dehydrogenase

4-	lynch-nyhan syndrome is caused by deficiency of ..... enzyme (Lecture Qs33, Lecture Qs37)						
A-	HGPRT	B-	PRPP	C-	Xanthine oxidase	D-	Uricase

5-	In purine nucleus nitrogen atom at (1)position is derived from: (Lecture Qs33)								
A-	B- Glutamate	B-		C-	Glycine	D-	Aspartate	E-	D-Alanine

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	B	B	A	C	-	-	-	-	-

M3WAN Bank

1-	**Purine and pyrimidine are the parent compounds of two types of nitrogen-containing: (M3WAN - Core)**		
A-	Heterocyclic aromatic compounds	B-	Aliphatic saturated compounds
C-	Steroid derivatives	D-	Simple monosaccharides

2-	**Caffeine and theobromine, naturally occurring compounds found in coffee and tea, are chemically classified as: (M3WAN - Core)**						
A-	Pyrimidines	B-	Purines	C-	Amino sugars	D-	Flavonoids

3-	**The final end product of purine catabolism in humans is: (Important M3WAN - Core)**						
A-	Urea	B-	Uric acid	C-	Creatinine	D-	Ammonia

4-	**At physiological pH, uric acid exists mainly in a more soluble ionized form known as: (M3WAN - Core)**						
A-	Sodium urate	B-	Xanthine	C-	Hypoxanthine	D-	Allantoin

5-	**The normal serum uric acid concentration is approximately: (M3WAN - Core)**						
A-	0.5–1 mg/dL	B-	3–7 mg/dL	C-	10–15 mg/dL	D-	20–25 mg/dL

6-	**Gout is a metabolic disorder characteristically associated with: (M3WAN - Core)**
A-	Hypouricemia
B-	Overproduction or under-excretion of uric acid leading to hyperuricemia
C-	Excess purine excretion with normal serum levels
D-	Deficiency of xanthine oxidase only

7-	**In severe hyperuricemia, deposits of sodium urate crystals in soft tissues and joints are commonly known as: (Important M3WAN - Core)**						
A-	Tophi	B-	Xanthomas	C-	Ochronosis	D-	Amyloid plaques

8-	**Deposition of urate crystals in joints, producing joint inflammation, results in: (Important M3WAN - Core)**		
A-	Gouty arthritis	B-	Osteoarthritis
C-	Rheumatic fever	D-	Ochronotic arthropathy

9-	**Gout most commonly affects which population group, although postmenopausal women are similarly susceptible? (M3WAN - Core)**		
A-	Males	B-	Premenopausal women
C-	Prepubertal children	D-	Infants

10-	Historically, gout has been particularly associated with which lifestyle factors? (M3WAN - Hard)
A-	High living, overeating, and alcohol consumption
B-	Strict vegetarianism
C-	Chronic starvation
D-	High-fiber, low-fat diets

11-	Lead poisoning is a recognized cause of gout because it: (M3WAN - Hard)		
A-	Increases uric acid excretion	B-	Decreases uric acid excretion
C-	Directly inhibits xanthine oxidase	D-	Directly increases purine synthesis

12-	Primary metabolic gout, an inborn error of purine metabolism, may result from all of the following EXCEPT: (Important M3WAN - Hard)
A-	Increased activity of PRPP synthetase
B-	Overactivity of PRPP amidotransferase
C-	HGPRT deficiency
D-	Deficiency of glucosE-6-phosphatase decreasing purine synthesis

13-	Primary renal gout results from: (M3WAN - Core)		
A-	Overproduction of uric acid in the liver	B-	Failure of uric acid excretion from the body
C-	Excess dietary purine intake only	D-	Increased HGPRT activity

14-	Secondary renal gout occurs as a consequence of: (M3WAN - Core)
A-	Defective glomerular filtration of urate secondary to generalized renal failure
B-	A primary inborn error of purine synthesis
C-	Excess dietary alcohol alone
D-	Increased xanthine oxidase activity

15-	A recognized complication of chronic hyperuricemia, in which uric acid precipitates in the renal tract, is: (M3WAN - Core)		
A-	Kidney stone formation	B-	Osteoporosis
C-	Cardiomyopathy	D-	Cataract formation

16-	To remove urate deposits from an acutely inflamed joint in gout, the drug of choice is: (Important M3WAN - Core)							
A-	Colchicine	B-	Allopurinol	C-	Insulin	D-	Cimetidine

17-	**Uricosuric drugs such as probenecid and sulfinpyrazole treat gout mainly by: (M3WAN - Core)**
A-	Increasing renal excretion of urate from the body
B-	Inhibiting xanthine oxidase
C-	Directly dissolving joint tophi
D-	Increasing purine synthesis

18-	**Allopurinol lowers uric acid production by inhibiting: (Important M3WAN - Core)**		
A-	HGPRT	B-	Xanthine oxidase
C-	PRPP synthetase	D-	Glutathione reductase

19-	**Lesch-Nyhan syndrome results from a deficiency of which enzyme, whose structural gene is located on the X chromosome? (Important M3WAN - Core)**
A-	Xanthine oxidase
B-	HypoxanthinE-guanine phosphoribosyltransferase (HGPRT)
C-	PRPP synthetase
D-	Adenosine deaminase

20-	**Because the HGPRT gene is X-linked, Lesch-Nyhan syndrome characteristically affects: (Important M3WAN - Core)**		
A-	Only males	B-	Only females
C-	Males and females equally	D-	Neither sex specifically

21-	**Clinical features of Lesch-Nyhan syndrome include excessive uric acid production together with: (Important M3WAN - Core)**
A-	Mental retardation, aggressive behavior, and self-mutilation
B-	Hypopigmentation and photophobia
C-	Black urine on standing
D-	Hemolytic anemia

22-	**In Lesch-Nyhan syndrome, allopurinol treatment is effective for controlling hyperuricemia but has: (Important M3WAN - Core)**
A-	No effect on the neurological/behavioral manifestations
B-	Complete curative effect on self-mutilation behavior
C-	No effect on serum uric acid either
D-	A worsening effect on neurological symptoms

23-	**Management of the neuro-behavioral features of Lesch-Nyhan syndrome (such as self-mutilation) is generally limited to: (M3WAN - Hard)**
A-	Behavioral therapy and protective physical devices
B-	High-dose allopurinol alone
C-	Colchicine therapy
D-	Dietary purine restriction alone

24-	**Which of the following is classified as a disorder of pyrimidine, rather than purine, metabolism? (Important M3WAN - Core)**		
A-	Gout	B-	Lesch-Nyhan syndrome
C-	Orotic aciduria	D-	Primary renal gout

Answers - M3WAN Bank									
11	12	13	14	15	16	17	18	19	20
A	B	B	A	B	B	A	A	A	A
21	22	23	24	25	26	27	28	29	30
B	D	B	A	A	A	A	B	B	A
31	32	33	34	35	36	37	38	39	40
A	A	A	C	-	-	-	-	-	-

يا أخي، لله درّ أقوام نعمّهم مولاهم بقربه، فحجبهم عن خطرات الوسواس . حمي إقليم قلوبهم من غبار الشهوات من حمايته بحراس، قبلوا أمره بالقبول، وقاموا به على العينين والرأس، قدّموا زاد الأعمال لسفر الموت وظلمة الأرماس،

يا بطّال، أبطال ميدان الدّجى لله درّهم من أبطال وأفراس، خلع عليهم خلعة الرضا، وناداهم مرحبا بالأحباب الأكياس، ﴿كُنتُمْ خَيْرَ أُمَّةٍ أُخْرِجَتْ لِلنَّاسِ﴾ آل عمران ١١٠ .

ابن الجوزي | بحر الدّموع.

Lecture (22): DNA Replication

Pre Exams , Formatives

1-	Which enzyme is not required for replication (FINAL 34, FINAL 31, FINAL 32)		
A-	DNA ligase	B-	Topoisomerase
C-	Helicase	D-	Reverse transcriptase

2-	Which enzyme is required for separation of double stranded DNA (FINAL 34, FINAL 31, FINAL 32)		
A-	DNA ligase	B-	Topoisomerase
C-	Helicase	D-	Reverse transcriptase

3-	For DNA replication, unwinding of DNA Is done by? (2ND TRIAL 35)						
A-	Helicase	B-	Primase	C-	Ligase	D-	Topoisomerase

4-	Responsible for synthesis of lagging strand in prokaryotes? (2ND TRIAL 35)		
A-	DNA polymerase I	B-	DNA polymerase II
C-	DNA polymerase III	D-	DNA polymerase Y

5-	Okasaki fragments are small bits of ? (MID 30, FINAL 31)		
A-	RNA	B-	DNA
C-	DNA with RNA heads	D-	RNA with DNA heads

6-	Xeroderma pigmentosum results from a defect in ? (MID 30)		
A-	uvr ABC excinuclease	B-	DNA polymerase I
C-	DNA ligase	D-	All of these

7-	DNA replication requires all, EXCEPT? (FINAL 31, FINAL 32)		
A-	Elongation factors	B-	Deoxynucleotide phosphates
C-	DNA polymerase	D-	RNA primer

8-	Which enzyme is required for prevention of positive supercoiling of DNA? (FINAL 31)		
A-	DNA ligase	B-	Topoisomerase
C-	Helicase	D-	Reverse transcriptase

تم إعادة صياغة الاجابات

9-	lagging strand? (FINAL 31)		
A-	Formed in discontinuous pieces (Okazaki fragments), away from the replication fork	B-	Formed towards replication fork
C-	Non of the above	D-	Requires no RNA primers at all

10-	Telomerase activity is highest in the following EXCEPT? (FINAL 31)						
A-	Gamete cell	B-	Cancer cell	C-	Somatic adult cell	D-	all cells

11-	DNA replication is? (FINAL 31)		
A-	Conservative Non-conservative	B-	Non-conservative
C-	Semi-conservative	D-	None

12-	Which of the following is true about DNA polymerase? (FINAL 31, Lecture Qs33)
A-	It can synthesize DNA in the 5- to 3- direction
B-	It can synthesize DNA in the 3- to 5- direction
C-	It can synthesize mRNA in the 3- to 5- direction
D-	It can synthesize mRNA in the 5- to 3- direction

13-	The correct statement concerning RNA and DNA polymerases is? (FINAL 31)		
A-	RNA polymerase use nucleoside diphosphates	B-	RNA polymerase require primers and add
C-	DNA polymerases can add nucleotides at both	D-	All RNA and DNA polymerases can add

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
D	C	A	C	B	A	A	B	AD	C
11	12	13	14	15	16	17	18	19	20
C	A	D	-	-	-	-	-	-	-

Department Qs

1-	All these enzymes involved in double stand separation except: (Department Qs 35)				
A-	DNA helicas	B-	Topoisomerases	C-	Primase

2-	DNA polymeraseIII (Department Qs 35)		
A-	Form Replication of leading strand.	B-	Replication of all lagging strand.
C-	Non of the above.		

3-	lagging strand: (Department Qs 35)		
A-	Formed in pieces.	B-	Formed towards replication fork.
C-	Non of the above.		

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	A	A	-	-	-	-	-	-	-

Lecture Qs

1-	DNA replication is (Lecture Qs33)		
A-	Conservative	B-	Non-conservative
C-	Semi-conservative	D-	None

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
C	-	-	-	-	-	-	-	-	-

M3WAN Bank

1-	DNA replication is best described as (M3WAN - Core)				
A-	Conservative process	B-	Semi-conservative process	C-	Dispersive process only
D-	Non-templated process				

2-	The semi-conservative nature of DNA replication means that each daughter DNA molecule contains: (Important M3WAN - Core)
A-	Two entirely new strands
B-	Two entirely parental (old) strands
C-	One parental strand and one newly synthesized strand
D-	A random mixture of fragments from both parental strands

3-	During DNA replication, each parental strand serves as a: (M3WAN - Core)
A-	Primer only
B-	Template upon which a new complementary strand is synthesized
C-	Site of RNA translation
D-	Source of free nucleotides only

4-	In prokaryotes, excision of RNA primers and filling of the resulting gaps is the function of: (Important M3WAN - Core)		
A-	DNA polymerase I	B-	DNA polymerase II
C-	DNA polymerase III	D-	DNA ligase

5-	Proofreading and repair of newly synthesized DNA strands, by hydrolytic removal of misplaced nucleotides, is mainly the function of: (Important M3WAN - Core)		
A-	DNA polymerase I	B-	DNA polymerase II
C-	DNA polymerase III	D-	Primase

6-	The main prokaryotic enzyme responsible for synthesizing the new DNA strand at the replication fork (leading and lagging strands) is: (Important M3WAN - Core)		
A-	DNA polymerase I	B-	DNA polymerase II
C-	DNA polymerase III	D-	DNA polymerase IV

7-	In eukaryotic cells, mitochondrial DNA replication is carried out by: (M3WAN - Core)		
A-	DNA polymerase α	B-	DNA polymerase δ
C-	DNA polymerase γ	D-	DNA polymerase β

8-	In eukaryotic cells, synthesis of the lagging strand is mainly attributed to: (M3WAN - Hard)		
A-	DNA polymerase α	B-	DNA polymerase δ
C-	DNA polymerase ε	D-	DNA polymerase γ

9-	The enzyme responsible for unwinding the parental DNA double helix by breaking hydrogen bonds between bases, using ATP, is: (Important M3WAN - Core)						
A-	DNA ligase	B-	DNA helicase	C-	Topoisomerase	D-	Primase

10-	The V-shaped region created by the unwinding of the parental DNA helix during replication is called the: (M3WAN - Core)		
A-	Origin of replication	B-	Replication fork
C-	Okazaki fragment	D-	Spliceosome

11-	Proteins that bind to singlE-stranded DNA created by helicase, preventing reannealing and protecting it from nuclease attack, are called: (Important M3WAN - Core)		
A-	SinglE-stranded DNA binding proteins	B-	Histones
C-	Topoisomerases	D-	Primase complexes

12-	The site at which DNA replication begins is called: (M3WAN - Core)		
A-	The termination unit	B-	The origin of replication ("Oric")
C-	The Okazaki site	D-	The promoter

13-	Enzymes that prevent positive supercoiling ahead of the replication fork, facilitating further strand separation, are the: (Important M3WAN - Core)						
A-	Topoisomerases	B-	Ligases	C-	Helicases	D-	Primases

14-	Since DNA polymerase III cannot initiate synthesis of a new strand without a free 3--OH, a short RNA fragment must first be synthesized by: (Important M3WAN - Core)						
A-	DNA ligase	B-	Primase	C-	Topoisomerase	D-	Helicase

15-	The RNA primer synthesized during DNA replication is typically how many nucleotides in length? (M3WAN - Hard)						
A-	2 nucleotides	B-	10 nucleotides	C-	50 nucleotides	D-	100 nucleotides

16-	In eukaryotic DNA replication, the RNA primer is synthesized by: (M3WAN - Hard)		
A-	DNA polymerase α	B-	DNA polymerase δ
C-	DNA polymerase γ	D-	DNA polymerase β

17-	The enzyme that joins the free 3--OH end of one DNA fragment to the 5--phosphate end of another, forming a phosphodiester bond, is: (Important M3WAN - Core)		
A-	DNA ligase	B-	DNA polymerase I
C-	Primase	D-	Topoisomerase

18-	The action of DNA ligase during replication requires: (M3WAN - Core)						
A-	GTP	B-	ATP	C-	NADH	D-	No energy input

19-	As free deoxynucleoside triphosphates (dATP, dGTP, dTTP, dCTP) are incorporated into the growing DNA strand, they lose two phosphate groups to become: (M3WAN - Hard)
A-	Deoxynucleoside monophosphates (dAMP, dGMP, dTMP, dCMP)
B-	Deoxynucleoside diphosphates
C-	Free nitrogenous bases
D-	Deoxyribose only

20-	The leading strand of DNA is synthesized: (Important M3WAN - Core)
A-	Discontinuously, away from the replication fork
B-	Continuously, in the 5-⟶3- direction toward the replication fork
C-	In the 3-⟶5- direction using multiple RNA primers
D-	Without any RNA primer at all

21-	The lagging strand of DNA is synthesized: (Important M3WAN - Core)
A-	Continuously in the 5-⟶3- direction
B-	Discontinuously, forming Okazaki fragments, away from the replication fork
C-	Using only a single RNA primer for the entire strand
D-	In the same continuous manner as the leading strand

22-	**Compared with the leading strand, the lagging strand characteristically requires: (M3WAN - Core)**
A-	Fewer RNA primers
B-	Many RNA primers, one for each Okazaki fragment
C-	No RNA primers at all
D-	Only DNA ligase and no polymerase

23-	**Okazaki fragments are best described as: (M3WAN - Core)**
A-	Small, discontinuous fragments of newly synthesized DNA on the lagging strand
B-	Continuous segments of the leading strand
C-	RNA fragments left permanently in mature DNA
D-	Fragments produced only during mismatch repair

24-	**Which of the following correctly pairs a replication enzyme with its principal function? (Important M3WAN - Core)**
A-	Helicase — joins Okazaki fragments
B-	Topoisomerase — synthesizes RNA primers
C-	Primase — synthesizes short RNA primers required to initiate DNA synthesis
D-	Ligase — unwinds the DNA double helix

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	C	B	A	B	C	C	B	B	B
11	12	13	14	15	16	17	18	19	20
A	B	A	B	B	A	A	B	A	B
21	22	23	24	25	26	27	28	29	30
B	B	A	C	-	-	-	-	-	-

أصول السعادة – أصول التعاسة:

- التوحيد
- الشرك
- السنة
- البدعة
- الطاعة
- المعصية

ابن القيم | الفوائد صـ(٧٦)

Lecture (23): Ribonucleic acids (RNAs)

Pre Exams , Formatives

1-	Which of the following ensure stable binding of RNA polymerase at the promoter site (FINAL 34, FINAL 32)						
A-	DNA photolyase	B-	Sigma factor	C-	DNA glycosylase	D-	Primase

2-	The enzyme involved in amino acid activation is? (2ND TRIAL 35)		
A-	ATP synthetase	B-	aminoacyl tRNA synthetase
C-	aminoacyl MRNA synthetase	D-	aminoacyl rRNA synthetase

3-	The -rho- (p) factor is involved? (MID 30)
A-	To increase the rate of RNA synthesis
B-	In binding catabolite repressor to the promoter region
C-	In proper termination of transcription
D-	To allow proper initiation of transcripton

4-	transacting factors include ? (MID 30)						
A-	Promoters	B-	Repressors	C-	Enhancers	D-	Silencers

5-	Addition of 7-methyl-guanylate capping to heterogeneous nuclear RNA protects from? (MID 30)		
A-	3--5- exonucleases	B-	5--3- exonucleases
C-	Endonuclease	D-	lyase

6-	mRNA of prokaryotes can code for? (MID 30)		
A-	More than one polypeptide	B-	Only one polypeptide
C-	Many exons and introns	D-	Introns only

7-	All tRNA Molecules contain a base paired stem that terminates in the sequence CCA at? (MID 30)						
A-	3- Termini	B-	5- Termini	C-	Anticodon arm	D-	3- and 5- Termini

8-	Capping of the 5- end of immature mRNA is carried out by ..... enzyme (FINAL 36)
A-	guaninE-7-methyl transferase

9-	the tRNA molecule for methionine has the anticodon: (FINAL 37,Lecture Qs37)						
A-	CCU	B-	AUG	C-	UUC	D-	UAC

10-	**Post transcription modification is.... (FINAL 37 SNU)**		
A-	DNA duplication	B-	RNA transcription
C-	DNA transcription	D-	Caping, slicing, tailing

11-	**RNA differs from DNA in? (Formative 35, Department Qs 35)**		
A-	It is double stranded	B-	Only 2 types present
C-	Contain uracil instead of thymine		

12-	**how many types of RNA are made (Formative 37, Quiz SNU-M1, SNU Bank)**						
A-	two types	B-	one type	C-	DNA	D-	all types of RNA

13-	**RNA transcription means (Formative 37, Quiz SNU-M1, SNU Bank)**		
A-	formation of proteins	B-	formation of RNA using DNA
C-	formation of DNA from RNA	D-	formation of nitrogenous bases

14-	**The structure of tRNA appears like a? (FINAL 31, 2ND TRIAL 32)**						
A-	Helix	B-	Hair pin	C-	Clover leaf	D-	Bundle

15-	**Which of the following sequences in mRNA is complementary to the DNA sequence 5--AGCCAATT-3-? (FINAL 31)**						
A-	5--AGCCAAUU-3-	B-	5--UCGGUUAA-3-	C-	5--UUAACCGA-3-	D-	5--AAUUGGCU-3-

16-	**mRNA is complementary to the nucleotide sequence of? (FINAL 31)**						
A-	Coding strand	B-	Ribosomal RNA	C-	tRNA	D-	Template strand

17-	**(5-TATAAT 3) is? (FINAL 31)**						
A-	Hogness box	B-	Pribrow box	C-	Non of the above	D-	In replication

18-	**Name the pyrimidine base which is found in RNA but not in DNA? (FINAL 31)**						
A-	Thymine	B-	Uracil	C-	Adenine	D-	Guanine

19-	**Termination of the synthesis of the RNA molecule is signalled by a sequence in the template strand of the DNA molecule, a signal that is recognized by a termination protein, the? (FINAL 31)**						
A-	Rho (p) factor	B-	a factor	C-	factor	D-	factor

20-	**In the process of transcription in bacterial cells? (FINAL 31)**		
A-	Initiation requires rho protein	B-	RNA polymerase incorporates methylated
C-	Both the sigma unit and core enzymes of RNA	D-	Primase is necessary for initiation

21-	The process of formation of RNA is known as ? (FINAL 32)						
A-	Replication	B-	DNA repair	C-	Translation	D-	Transcription

22-	The nucleic acid base found in mRNA but not in DNA is? (2ND TRIAL 32)						
A-	Adenine	B-	Cytosine	C-	Guanine	D-	Uracil

23-	In contrast to eukaryotic mRNA prokaryotic mRNA (2ND TRIAL 32)		
A-	Can be polycistronic	B-	is synthesized with introns
C-	Can only be monocistronic	D-	Has a poly A tail

24-	The nitrogenous base present in the RNA molecule is? (2ND TRIAL 32)						
A-	Thymine	B-	Uracil	C-	Xanthine	D-	Hypoxanthine

25-	RNA does not contain ? (2ND TRIAL 32)						
A-	Uracil	B-	Adenine	C-	Thymine	D-	Ribose

26-	The sugar moiety present in RNA is ? (2ND TRIAL 32)						
A-	Ribulose	B-	Arabinose	C-	Ribose	D-	Deoxyribose

27-	In every cell, the number of tRNA molecule is at least ? (2ND TRIAL 32)						
A-	10	B-	20	C-	30	D-	40

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
B	B	C	B	B	A	A	A	D	D
11	12	13	14	15	16	17	18	19	20
C	D	B	C	B	D	B	B	A	C
21	22	23	24	25	26	27	28	29	30
D	D	A	B	C	C	B	-	-	-

Department Qs

1-	after RNA transcription the products (SNU Bank)		
A-	do its function	B-	released from the cell
C-	mmust undergo post modification translations	D-	no need post modificational translation

2-	steps of RNA transcription (SNU Bank)						
A-	initiation	B-	elongation	C-	termination	D-	all of the above

3-	before initation step the cell..... (SNU Bank)		
A-	needs basic requirements	B-	doesn-t need basic requirements
C-	needs post modificational translation	D-	will start elongation step

4-	Suger in RNA is (Department Qs33)						
A-	Ribose	B-	Deoxy ribose	C-	Fructose	D-	None of above

5-	Uracile present in (Department Qs33)				
A-	RNA	B-	DNA	C-	Both
D-	, b	E-	None of above		

6-	Smallest RNA (Department Qs33)						
A-	Mrna	B-	Trna	C-	Rrna	D-	None of above

7-	Holo enzyme of RNA polymerase composed of (Department Qs33)						
A-	5 subunits	B-	2 subunits	C-	3 subunits	D-	1 subunits

8-	TATA box present in (Department Qs33)				
A-	Euokaryoyic promoter	B-	Prokaryotic promoter	C-	Both
D-	, b	E-	None of above		

9-	RNAs contain uracil Instead of :- (Department Qs33, Lecture Qs33)						
A-	Adenine	B-	Cytosine	C-	Thymine	D-	Guanine

10-	Ribosomal RNA: (Department Qs 35)
A-	Can only form ribosome without protein.
B-	In eukaryotes they are 55, 5, 85, 185, and 285
C-	Non of the above

11-	Gene promotor: (Department Qs 35)		
A-	Present upstream of gene proper.	B-	Recognized by RNA polymerase.
C-	Non of the above		

12-	RNA polymerase II in Eukaryotes form: (Department Qs 35)				
A-	t-RNA	B-	r-RNA	C-	m-RNA

13-	(5--TATAAT-3-) is: (Department Qs37)		
A-	Hogness box	B-	Pribnow box

C-	None of the above	D-	In replication

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
C	D	A	A	A	B	A	A	C	C
11	12	13	14	15	16	17	18	19	20
A	C	B	-	-	-	-	-	-	-

Lecture Qs

1-	RNAs contain pentose which is:- (Lecture Qs33)						
A-	Ribose	B-	Deoxy Ribose	C-	Ribulose	D-	Xylulose

Answers - Lecture Qs									
1	2	3	4	5	6	7	8	9	10
A	-	-	-	-	-	-	-	-	-

M3WAN Bank

1-	Compared with DNA, RNA is characteristically: (M3WAN - Core)
A-	DoublE-stranded and much less abundant
B-	SinglE-stranded and much more abundant in the cell
C-	Composed of deoxyribose sugar
D-	Devoid of any nitrogenous bases

2-	The pentose sugar present in RNA, distinguishing it from DNA, is: (M3WAN - Core)						
A-	Deoxyribose	B-	Ribose	C-	Ribulose	D-	Xylulose

3-	In RNA, which base replaces thymine, pairing with adenine? (M3WAN - Core)						
A-	Cytosine	B-	Guanine	C-	Uracil	D-	Hypoxanthine

4-	The backbone of an RNA polynucleotide chain is composed of alternating: (M3WAN - Core)
A-	Sugar and phosphate, linked by 3--5- phosphodiester bonds
B-	Nitrogenous bases only
C-	Amino acids and phosphate
D-	Two sugar molecules

5-	Compared with DNA nucleotides, every ribonucleotide in RNA has an additional hydroxyl group located at which position of the ribose? (M3WAN - Hard)

A-	1- position	B-	2- position	C-	4- position	D-	5- position

6-	**Messenger RNA (mRNA) constitutes approximately what proportion of total cellular RNA? (M3WAN - Hard)**						
A-	5%	B-	25%	C-	50%	D-	80%

7-	**The 5- terminal end of eukaryotic mRNA is characteristically capped with: (Important M3WAN - Core)**		
A-	A poly-A tail	B-	7-methylguanosine triphosphate
C-	A hairpin loop structure	D-	An intron sequence

8-	**The primary function of the 5- cap of mRNA is to: (Important M3WAN - Core)**
A-	Facilitate ribosome recognition and protect mRNA from 5- exonuclease digestion
B-	Serve as the start codon for translation
C-	Terminate transcription
D-	Bind directly to tRNA

9-	**The 3- end of most mature mRNAs contains a polymer of adenylate residues known as the: (Important M3WAN - Core)**
A-	Poly-A tail, which protects mRNA from 3- exonuclease attack
B-	TATA box
C-	ShinE-Dalgarno sequence
D-	Anticodon loop

10-	**The region of mRNA located between the non-coding sequences at the 5- and 3- ends, which is actually translated into protein, is called the: (M3WAN - Core)**						
A-	Intron	B-	Coding region	C-	Promoter	D-	Enhancer

11-	**Which of the following is classified as a protein-coding RNA, in contrast to the other non-coding RNA classes? (M3WAN - Core)**		
A-	Messenger RNA (mRNA)	B-	Ribosomal RNA (rRNA)
C-	Transfer RNA (tRNA)	D-	Small nuclear RNA (snRNA)

12-	**Ribosomal RNA (rRNA), the most abundant class of cellular RNA, functions mainly as: (Important M3WAN - Core)**		
A-	The catalytic component of ribosomes	B-	The template for protein synthesis
C-	The adapter molecule carrying amino acids	D-	A regulator of gene silencing only

13-	**The prokaryotic ribosome, sedimenting at 70S, is composed of a small subunit and large subunit of which sizes? (Important M3WAN - Core)**

A-	30S and 50S	B-	40S and 60S	C-	18S and 28S	D-	20S and 40S

14-	**The prokaryotic 30S ribosomal subunit contains one molecule of 16S rRNA together with approximately: (M3WAN - Hard)**						
A-	10 proteins	B-	21 proteins	C-	31 proteins	D-	50 proteins

15-	**The prokaryotic 50S ribosomal subunit contains two rRNA species, 5S and 23S, together with approximately: (M3WAN - Hard)**						
A-	21 proteins	B-	31 proteins	C-	33 proteins	D-	50 proteins

16-	**The eukaryotic ribosome, sedimenting at 80S, is composed of which two subunits? (Important M3WAN - Core)**		
A-	30S and 50S	B-	40S and 60S
C-	18S and 28S alone	D-	45S and 5S

17-	**The eukaryotic 40S ribosomal subunit contains one molecule of 18S rRNA together with approximately: (M3WAN - Hard)**						
A-	21 proteins	B-	33 proteins	C-	50 proteins	D-	80 proteins

18-	**The eukaryotic 60S ribosomal subunit contains three rRNA species (5S, 28S, and 5.8S) together with approximately: (M3WAN - Hard)**						
A-	21 proteins	B-	33 proteins	C-	50 proteins	D-	80 proteins

19-	**During protein synthesis, accurate codon-anticodon recognition between mRNA and tRNA is mainly the role of: (Important M3WAN - Core)**		
A-	The small ribosomal subunit	B-	The large ribosomal subunit
C-	DNA polymerase	D-	RNA polymerase II

20-	**Formation of the peptide bond during elongation of the growing polypeptide chain is mainly attributed to: (Important M3WAN - Core)**		
A-	The small ribosomal subunit	B-	The large ribosomal subunit
C-	Aminoacyl-tRNA synthetase	D-	DNA ligase

21-	**Among the three major species of cellular RNA, transfer RNA (tRNA) is distinguished by being: (M3WAN - Core)**
A-	The largest
B-	The smallest, containing 74–95 nucleotide residues
C-	DoublE-stranded

D-	Devoid of secondary structure

22-	Because of its ready solubility, transfer RNA is also historically referred to as: (M3WAN - Hard)		
A-	Soluble RNA (sRNA)	B-	Heterogeneous RNA
C-	Antisense RNA	D-	Ribozyme RNA

23-	Transfer RNA molecules are also called "adapter molecules" because they: (Important M3WAN - Core)
A-	Act as adapters translating the mRNA nucleotide sequence into a specific amino acid sequence
B-	Adapt the ribosome to different mRNA lengths
C-	Adapt DNA to different replication origins
D-	Serve no functional translation role

24-	The number of distinct tRNA species is generally described as: (M3WAN - Core)
A-	Exactly 4, one per nitrogenous base
B-	At least 20, one corresponding to each amino acid
C-	Exactly 64, one per codon
D-	Only 1, shared by all amino acids

25-	The characteristic secondary structure of tRNA, containing several arms and loops, is known as the: (Important M3WAN - Core)		
A-	Double helix structure	B-	Cloverleaf structure
C-	β-pleated sheet structure	D-	Hairpin-only structure

26-	Which of the following is NOT one of the five main arms of the tRNA cloverleaf structure? (M3WAN - Hard)						
A-	Acceptor arm	B-	Anticodon arm	C-	TψC arm	D-	Poly-A arm

27-	Transcription is defined as the synthesis of RNA using singlE-stranded DNA as a template, catalyzed by: (M3WAN - Core)		
A-	DNA-dependent DNA polymerase	B-	DNA-dependent RNA polymerase
C-	RNA-dependent DNA polymerase	D-	Reverse transcriptase exclusively

28-	Unlike DNA replication, transcription characteristically: (Important M3WAN - Core)
A-	Requires a primer and copies the entire genome
B-	Does not require a primer and transcribes only a short DNA segment
C-	Uses deoxyribonucleotides as substrates
D-	Is semi-conservative

29-	During transcription, RNA synthesis proceeds in which direction, while the DNA template is read in the opposite direction? (M3WAN - Core)		
A-	RNA synthesized 5-⟶3-; template read 3-⟶5-	B-	RNA synthesized 3-⟶5-; template read 5-⟶3-
C-	Both RNA and template read 5-⟶3-	D-	Direction is random

30-	Transcription follows Watson-Crick basE-pairing rules in which adenine on the DNA template pairs with which base in the growing RNA? (M3WAN - Core)						
A-	Thymine	B-	Uracil	C-	Cytosine	D-	Guanine

31-	In eukaryotic cells, synthesis of ribosomal RNA (the most numerous RNA species) is carried out by: (Important M3WAN - Core)		
A-	RNA polymerase I	B-	RNA polymerase II
C-	RNA polymerase III	D-	DNA polymerase I

32-	In eukaryotic cells, synthesis of messenger RNA (the largest RNA species) is carried out by: (Important M3WAN - Core)		
A-	RNA polymerase I	B-	RNA polymerase II
C-	RNA polymerase III	D-	RNA polymerase IV

33-	In eukaryotic cells, synthesis of 5S rRNA and tRNA (the smallest RNA species) is carried out by: (Important M3WAN - Core)		
A-	RNA polymerase I	B-	RNA polymerase II
C-	RNA polymerase III	D-	RNA polymerase γ

34-	The prokaryotic RNA polymerase core enzyme is composed of subunits α2ββ-, while the addition of which subunit converts it into the fully functional holoenzyme? (Important M3WAN - Core)		
A-	σ (sigma) factor	B-	Rho factor
C-	TATA-binding protein	D-	Primase

35-	Within prokaryotic RNA polymerase, the subunit responsible for recognizing and binding the promoter region to initiate transcription is: (M3WAN - Hard)						
A-	α	B-	β	C-	β-	D-	σ

36-	Compared with DNA polymerase, RNA polymerase characteristically: (Important M3WAN - Core)
A-	Requires a primer and has proofreading exonuclease activity
B-	Does not require a primer and lacks proofreading exonuclease activity, giving a higher error rate
C-	Has an identical error rate to DNA polymerase

D-	Cannot initiate new RNA chains

37-	**The prokaryotic promoter sequence located approximately 10 bases upstream of the transcription start site, with consensus sequence TATAAT, is known as the: (Important M3WAN - Core)**						
A-	Pribnow box	B-	TATA box	C-	CAAT box	D-	Enhancer element

38-	**The eukaryotic promoter element with consensus sequence TATAAA, typically located close to the transcription start site, is the: (Important M3WAN - Core)**		
A-	Pribnow box	B-	35 sequence
C-	TATA box	D-	Terminator sequence

39-	**The three sequential steps of transcription are initiation, elongation, and: (M3WAN - Core)**						
A-	Termination	B-	Translation	C-	Splicing	D-	Replication

40-	**Rho-dependent termination of transcription involves a hexameric protein, Rho factor, that functions as: (Important M3WAN - Core)**
A-	An ATP-dependent RNA-DNA helicase that disrupts the nascent RNA/DNA complex
B-	A DNA ligase
C-	A topoisomerase
D-	An RNA-degrading exonuclease only

41-	**Rho-independent termination of transcription characteristically involves formation of a hairpin (stem-loop) structure in the nascent RNA, followed by: (Important M3WAN - Core)**		
A-	A stretch of guanine residues	B-	A stretch of uracil residues, whose weak
C-	: U pairing dissociates from the DNA	D-	Immediate cleavage by RNase H
E-	Binding of DNA polymerase III		

42-	**Eukaryotic transcription requires enzymes, transcription factors, enhancers, and: (M3WAN - Core)**		
A-	Silencers	B-	Ribosomes
C-	tRNA synthetases	D-	Okazaki fragments

43-	**Eukaryotic general transcription factors (such as TFIID, TFIIB, and TFIIH) assemble on the promoter together with RNA polymerase II to form the: (Important M3WAN - Core)**		
A-	Preinitiation (basal transcription) complex	B-	Spliceosome
C-	Replisome	D-	Nucleosome core particle

44-	**Enhancers increase the rate of gene transcription, sometimes by as much as 100-fold, despite: (Important M3WAN - Core)**

A-	Having no promoter activity of their own
B-	Being located exclusively within the coding region
C-	Requiring direct contact with ribosomes
D-	Functioning only in prokaryotes

45-	**Silencers are DNA sequences that regulate transcription by: (M3WAN - Core)**
A-	Binding proteins that inhibit the rate of transcription
B-	Directly synthesizing RNA
C-	Serving as alternative promoters
D-	Enhancing ribosome binding

46-	**The initial, unprocessed RNA transcript synthesized from a eukaryotic protein-coding gene is called: (M3WAN - Core)**
A-	Mature mRNA
B-	Heterogeneous nuclear RNA (hnRNA), or the primary transcript
C-	rRNA precursor
D-	tRNA precursor

47-	**Addition of the 5- cap to eukaryotic mRNA involves guanylyltransferasE-catalyzed addition of GTP, followed by methylation of the terminal guanine using which methyl donor? (Important M3WAN - Core)**		
A-	S-adenosylmethionine (SAM)	B-	Tetrahydrofolate
C-	Biotin	D-	Vitamin B12

48-	**Splicing, the process of removing introns and joining exons to form mature mRNA, is carried out by a complex called the: (Important M3WAN - Core)**		
A-	Spliceosome, composed of multiple snRNPs	B-	Ribosome
C-	Replisome	D-	Nucleosome

49-	**A clinically important genetic disease resulting from defective mRNA splicing is: (Important M3WAN - Core)**		
A-	β-Thalassemia	B-	Xeroderma pigmentosum
C-	Gout	D-	Wilson-s disease

50-	**mRNA editing, an enzymE-mediated alteration of RNA base sequence distinct from splicing, is exemplified by conversion of a CAA codon to a UAA stop codon in the apolipoprotein B gene, catalyzed by: (Important M3WAN - Hard)**		
A-	Cytidine deaminase	B-	Guanylyltransferase
C-	DNA glycosylase	D-	Reverse transcriptase

51-	As a result of mRNA editing of the apolipoprotein B transcript, the liver synthesizes the full-length apoB100 protein, whereas intestinal cells synthesize a shorter protein known as: (Important M3WAN - Hard)						
A-	ApoB48	B-	ApoB100 variant	C-	ApoA1	D-	ApoE

52-	Post-transcriptional modification of tRNA characteristically includes cleavage of the 5- leader sequence, intron splicing, and replacement of the 3- terminal nucleotides with: (M3WAN - Hard)		
A-	A poly-A sequence	B-	CCA
C-	A 7-methylguanosine cap	D-	A hairpin loop only

53-	The mature ribosomal RNAs 28S, 18S, and 5.8S are initially synthesized together as a single long precursor molecule known as: (Important M3WAN - Core)		
A-	The 45S preribosomal RNA	B-	hnRNA
C-	The Okazaki precursor	D-	PrE-tRNA

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	B	C	A	B	A	B	A	A	B
11	12	13	14	15	16	17	18	19	20
A	A	A	B	B	B	B	C	A	B
21	22	23	24	25	26	27	28	29	30
B	A	A	B	B	D	B	B	A	B
31	32	33	34	35	36	37	38	39	40
A	B	C	A	D	B	A	C	A	A
41	42	43	44	45	46	47	48	49	50
B	A	A	A	A	B	A	A	A	A
51	52	53	54	55	56	57	58	59	60
A	B	A	-	-	-	-	-	-	-

أرضُ الفطرةِ رحبةٌ قابلةٌ لما يُغرسُ فيها؛ فإن غرستَ شجرةَ الإيمانِ والتقوى أورثتَ حلاوةَ الأبد، وإن غرستَ شجرةَ الجهلِ والهوى فكلُّ الثمرِ مرٌّ!

• ابنُ القيّم . .

Lecture (24): Protein Synthesis , genetic codons

Pre Exams , Formatives

1-	site of protein synthesis (FINAL 34)
A-	Ribosome

2-	The genetic code operates via ? (MID 30)		
A-	The protein moiety of DNA	B-	The base sequences of DNA
C-	The nucleotide sequence of mRNA	D-	The base sequence of tRNA

3-	preparatory step in translation (FINAL 36)
A-	Amino acyl synthetase

4-	Which of the following is a stop codon? (FINAL 36)						
A-	AUG	B-	UAG	C-	GUG	D-	AAA

5-	In translation, this is not an essential component? (FINAL 37, Lecture Qs33, Lecture Qs37)						
A-	amino acid	B-	ligase	C-	mRNA	D-	anticodon

6-	Folding of synthesized proteins is accelerated by? (MID 32)		
A-	Protein disulphide isomerase	B-	Prolyl cis-trans isomerase
C-	Chaperonins	D-	All of these

Answers - Pre Exams , Formatives

1	2	3	4	5	6	7	8	9	10
A	C	A	B	B	D	-	-	-	-

Department Qs

1-	the first codon to be translated on mRNA is (Department Qs33)						
A-	AUG	B-	GGU	C-	GGA	D-	AAA

2-	AUG, the only identified codon for methionine is important as (Department Qs33)		
A-	A releasing factor for peptide chains	B-	A chain terminating codon
C-	Recognition site on tRNA	D-	A chain initiating codon

3-	In biosynthesis of proteins the chain terminating codons are (Department Qs33)		
A-	UAA, UAG and UGA	B-	UGG, UGU and AGU
C-	AAU, AAG and GAU	D-	GCG, GCA and GCU

4-	The formation of initiation complex during protein synthesis requires a factor: (Department Qs33)						
A-	IF-III	B-	EF-I	C-	EF-II	D-	IF-I

5-	Initiation of protein synthesis begins with binding of (Department Qs33)		
A-	40S ribosomal unit on mRNA	B-	60S ribosomal unit
C-	Charging of tRNA with specific amino acid	D-	Attachment of aminoacyl tRNA on mRNA

6-	Initiation of protein synthesis requires (Department Qs33)						
A-	ATP	B-	AMP	C-	GDP	D-	GTP

7-	The enzyme amino acyl tRNA synthetase is involved in (Department Qs33)
A-	Dissociation of discharged tRNA from 80S ribosome
B-	Charging of tRNA with specific amino acids
C-	Termination of protein synthesis
D-	Nucleophilic attack on esterified carboxyl group of peptidyl tRNA

8-	In the process of activation of amino acids for protein synthesis, the number of high energy phosphate bond equivalent utilised is (Department Qs33)						
A-	0	B-	1	C-	2	D-	4

9-	Translation results in a product known as (Department Qs33)						
A-	Protein	B-	tRNA	C-	mRNA	D-	rRNA

10-	Tetracylin prevents synthesis of polypeptide by (Department Qs33)
A-	Blocking mRNA formation from DNA
B-	Releasing peptides from mRNA-tRNA complex
C-	Competing with mRNA for ribosomal binding sites
D-	Preventing binding of aminoacyl tRNA

11-	In prokaryotes, chloramphenicol (Department Qs33)
A-	Causes premature release of the polypeptide chain
B-	Causes misreading of the mRNA
C-	Depolymerises DNA
D-	Inhibits peptidyl transferase activity

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
A	D	A	A	A	D	B	C	A	D

11	12	13	14	15	16	17	18	19	20
D	-	-	-	-	-	-	-	-	-

Lecture Qs

1-	Protein synthesis corresponds to the process of? (Lecture Qs33)
A-	duplicating required DNA for synthesis of proteins
B-	formation of amino acids from mRNA
C-	formation of mRNA from DNA template
D-	formation of amino acids from DNA template directly

2-	This is considered to be the start codon? (Lecture Qs33, Lecture Qs37)						
A-	AGG	B-	UAG	C-	GUG	D-	AUG

3-	This elongation factor is known as translocase? (Lecture Qs33, Lecture Qs37)						
A-	EFtu	B-	EF2	C-	EFts	D-	EF 5

4-	This is incorrect about the nature of genetic code? (Lecture Qs33, Lecture Qs37)						
A-	universal	B-	overlapping	C-	commaless	D-	Triplet

5-	In protein synthesis, translocation is initiated with the movement of? (Lecture Qs33)		
A-	tRNA from P-site to the A-site	B-	dipeptidyl tRNA from A-site to P-site
C-	tRNA from A-site to P-site	D-	tRNA from P-site to E-site

Answers - Lecture Qs

1	2	3	4	5	6	7	8	9	10
B	D	B	B	B	-	-	-	-	-

M3WAN Bank

1-	A codon is defined as: (M3WAN - Core)
A-	A single nucleotide base coding for an amino acid
B-	A series of three adjacent bases in mRNA that codes for a specific amino acid
C-	A sequence of five bases in tRNA
D-	A stretch of DNA coding for an entire protein

2-	Given that mRNA contains four possible nucleotide bases arranged in threE-base codons, the total number of possible codons is: (M3WAN - Core)						
A-	16	B-	20	C-	64	D-	128

3-	The first codon in the reading frame of virtually every protein, coding for methionine, is: (Important M3WAN - Core)						
A-	UAA	B-	AUG	C-	UGA	D-	GUG

4-	Which of the following is a stop (termination) codon? (Important M3WAN - Core)						
A-	AUG	B-	UUU	C-	UAA	D-	CAU

5-	The property of the genetic code whereby each codon specifies only one particular amino acid is called: (M3WAN - Core)						
A-	Degeneracy	B-	Specificity	C-	Universality	D-	Wobble

6-	The property of the genetic code whereby a single amino acid may be specified by more than one codon (e-g-, arginine has 6 codons) is called: (Important M3WAN - Core)		
A-	Specificity	B-	Degeneracy
C-	Non-overlapping	D-	CommA-less reading

7-	The property whereby a given codon specifies the same amino acid in essentially all living organisms, from bacteria to humans, is called: (M3WAN - Core)						
A-	Universality	B-	Degeneracy	C-	Specificity	D-	Wobble

8-	The property of the genetic code meaning that no single base can be shared as a member of two consecutive codons is termed: (M3WAN - Hard)						
A-	Degeneracy	B-	Non-overlapping	C-	Universality	D-	CommA-less

9-	The property of the genetic code meaning that codons are read as a continuous sequence without gaps or punctuation between them is termed: (M3WAN - Hard)						
A-	CommA-less	B-	Non-overlapping	C-	Degeneracy	D-	Wobble

10-	Protein synthesis as a whole comprises two major sequential processes: (M3WAN - Core)		
A-	Replication and translation	B-	Transcription and translation
C-	Transcription and replication	D-	Splicing and replication

11-	DNA remains confined to the nucleus during protein synthesis mainly because it is: (M3WAN - Core)
A-	Too large and doublE-stranded to cross the nuclear membrane
B-	Chemically unstable outside the nucleus
C-	Unable to be transcribed
D-	Composed of ribose sugar

12-	Translation, the process of assembling a polypeptide from an mRNA template, occurs at the: (M3WAN - Core)						
A-	Nucleolus	B-	Ribosome	C-	Golgi apparatus	D-	Lysosome

13-	Activation of amino acids prior to protein synthesis, a preparatory step requiring ATP, is catalyzed by: (Important M3WAN - Core)		
A-	Aminoacyl-tRNA synthetase	B-	RNA polymerase
C-	Peptidyl transferase	D-	Helicase

14-	The three sequential steps of the translation process are initiation, elongation, and: (M3WAN - Core)						
A-	Termination	B-	Splicing	C-	Capping	D-	Replication

15-	Formation of the 43S preinitiation complex during translation initiation requires the 40S ribosomal subunit, GTP, initiation factors, and: (Important M3WAN - Core)		
A-	Methionyl-tRNA	B-	Aminoacyl-tRNA for the second amino acid
C-	Peptidyl transferase	D-	Rho factor

16-	Formation of the 48S preinitiation complex occurs when the 43S complex additionally associates with: (Important M3WAN - Core)		
A-	mRNA (via eIF4)	B-	The large ribosomal subunit
C-	Poly-A binding protein alone	D-	DNA polymerase

17-	The fully assembled 80S initiation complex forms after joining of the 48S complex with: (M3WAN - Core)		
A-	The large (60S) ribosomal subunit	B-	A second molecule of mRNA
C-	DNA ligase	D-	RNA polymerase III

18-	During translation, the anticodon of methionyl-tRNA that pairs with the start codon AUG is: (Important M3WAN - Core)						
A-	UAC	B-	AUG	C-	UGA	D-	CAU

19-	During the elongation phase, delivery of a new aminoacyl-tRNA to the ribosomal A site, coupled with hydrolysis of GTP, is mediated by: (Important M3WAN - Core)		
A-	eEF1α	B-	eIF4
C-	eRF	D-	EF-2 (translocase)

20-	Formation of the peptide bond during elongation involves transfer of the growing peptide chain from the tRNA in the P site to the: (Important M3WAN - Core)						
A-	A site	B-	E site	C-	5- cap	D-	Poly-A tail

21-	After peptide bond formation, translocation of the ribosome three bases along the mRNA (toward the 3- end) is catalyzed by translocase together with: (Important M3WAN - Core)		
A-	eEF1α	B-	EF-2
C-	eIF2	D-	Peptidyl transferase alone

22-	During translocation, the uncharged (deacylated) tRNA is released from which ribosomal site? (M3WAN - Core)						
A-	A site	B-	P site	C-	E site	D-	Acceptor site

23-	Translation terminates when the ribosome encounters which type of codon? (M3WAN - Core)		
A-	A start codon	B-	One of the stop codons (UAA, UGA, UAG)
C-	A silent codon	D-	A wobble codon

24-	Release of the completed polypeptide chain at termination is mediated by eukaryotic releasing factors (eRF) that activate: (Important M3WAN - Core)		
A-	Peptidyl transferase	B-	Aminoacyl-tRNA synthetase
C-	RNA polymerase II	D-	DNA ligase

25-	Completion of translation termination, including release of the polypeptide and mRNA, requires energy in the form of: (M3WAN - Hard)						
A-	ATP only	B-	GTP	C-	NADH	D-	No energy input

26-	Reduced specificity at the third position of a codon, such that a change in this position often does not alter the amino acid encoded, is known as: (Important M3WAN - Core)		
A-	The wobble (Crick-s wobble) hypothesis	B-	Degeneracy of the first base
C-	FramE-shifting	D-	Codon capping

27-	According to the wobble phenomenon, both UUU and UUC code for phenylalanine because the change occurs at the: (Important M3WAN - Core)		
A-	First base of the codon	B-	Second base of the codon
C-	Third base of the codon	D-	Anticodon loop of the ribosome

28-	**In contrast, a change in the first or second base of a codon (e-g-, UUU to CUU) typically: (M3WAN - Core)**
A-	Has no effect on the amino acid encoded
B-	Changes the amino acid encoded (e-g-, phenylalanine to leucine)
C-	Converts the codon into a stop codon
D-	Always restores the same amino acid

29-	**One major biological significance of the wobble hypothesis is that it: (Important M3WAN - Core)**
A-	Allows fewer than 61 tRNA species to recognize all 61 sense codons
B-	Requires exactly 61 distinct tRNA species, one per codon
C-	Eliminates the need for stop codons
D-	Prevents any amino acid from having more than one codon

30-	**The wobble hypothesis also helps explain why: (M3WAN - Core)**
A-	Certain amino acids are specified by more than one codon
B-	All amino acids have exactly one codon
C-	Start codons vary between organisms
D-	mRNA cannot be translated in bacteria

Answers - M3WAN Bank									
1	2	3	4	5	6	7	8	9	10
B	C	B	C	B	B	A	B	A	B
11	12	13	14	15	16	17	18	19	20
A	B	A	A	A	A	A	A	A	A
21	22	23	24	25	26	27	28	29	30
B	C	B	A	B	A	C	B	A	A

عن وهيب بن الورد رحمه الله قال: بلغني أن موسى نبي الله عليه السلام قال يا رب أخبرني عن آية رضاك عن عبدك فأوحى الله تعالى إليه إذا رأيتني أهيء له طاعتي وأصرفه عن معصيتي فذاك آية رضائي عنه

حلية الأولياء .

Lecture (25):) DNA alteration (Mutations)

Pre Exams , Formatives

1-	protein is used in Mismatch repair in Pyrimidine dimer? (2ND TRIAL 35)
A-	mut L

2-	Xeroderma Pigmentosum? (2ND TRIAL 35)
A-	it is an example of a defective mechanism for the repair of pyrimidine dimers in DNA

3-	P 53 gene? (MID 30)		
A-	A proto-oncogene	B-	An oncogene
C-	A tumour suppressor gene	D-	None of these

4-	DNA repair enzymes are example of ? (MID 30)		
A-	Primary antioxidant	B-	Secondary antioxidant
C-	Tertiary antioxidant	D-	Non of the above

5-	In the following partial sequence of mRNA, a mutation of the template DNA results in a change of 1 to UAA. The type of mutation is (88 89 90 91 92 93 94) GUC GAC CAG UAG GGC UAA CCG (MID 30)						
A-	Missene	B-	Silent	C-	Nonsense	D-	Frame shift

6-	Mut proteins are used for repair of ? (MID 30)		
A-	ThyminE-thymine dimer	B-	double DNA strand breaks
C-	Single DNA strand breaks	D-	Miss match mutation

7-	Mut L repair: (FINAL 37)		
A-	Recombinational double strand breaks	B-	Base excision repair
C-	Mismatch Repair	D-	thyminE-thymine dimer

8-	Which type of anemia represent a point mutation? (FINAL 31, FINAL 32)		
A-	Iron deficiency anemia	B-	sickle cell anemia
C-	Thalassemia	D-	Megaloblstic anemia

9-	Mut L enzyme involved in ..... Repair? (FINAL 31)		
A-	Mismatch	B-	thymine - thymine dimmer
C-	non of the above	D-	a and b

10-	Point mutation may produce? (FINAL 32)		
A-	Non functioning protein	B-	premature stopping of protein chain
C-	Both a , b	D-	non of the above

Answers - Pre Exams , Formatives									
1	2	3	4	5	6	7	8	9	10
A	A	C	C	C	D	C	B	A	C

Department Qs

1-	A mutation that convert an amino acid codon to stop codon is a : (Department Qs37)		
A-	Frame shift mutation	B-	Non sense mutation
C-	Silent mutation	D-	Transversion

Answers - Department Qs									
1	2	3	4	5	6	7	8	9	10
B	-	-	-	-	-	-	-	-	-

M3WAN Bank

1-	A mutation is defined as: (M3WAN - Core)
A-	A change in the sequence of nucleotide bases of the genetic code
B-	Any change in protein amino acid sequence only
C-	A normal variant of chromatin packaging
D-	A change restricted to RNA only

2-	Which of the following is NOT a recognized cause of mutation? (Important M3WAN - Hard)		
A-	Faulty DNA replication	B-	Physical agents such as X-rays
C-	Chemical agents such as free radicals	D-	Normal Watson-Crick base pairing

3-	A transition mutation is defined as: (Important M3WAN - Core)
A-	Change of one purine to the other purine, or one pyrimidine to the other pyrimidine
B-	Change of a purine to a pyrimidine, or vice versa
C-	Insertion of an extra nucleotide
D-	Deletion of a codon

4-	A transversion mutation is defined as: (Important M3WAN - Core)
A-	Change of one purine to the other purine
B-	Change of a purine to either of the two pyrimidines, or a pyrimidine to either of the two purines

C-	Change of one pyrimidine to the other pyrimidine only
D-	A silent, non-consequential base change

5-	**The two broad categories of mutation, based on mechanism, are point mutations and: (M3WAN - Core)**		
A-	Frameshift mutations	B-	Splicing mutations
C-	Rho-dependent mutations	D-	Wobble mutations

6-	**A point mutation is best described as: (M3WAN - Core)**
A-	Insertion or deletion of multiple nucleotides
B-	A single base substitution (replacement of one base for another)
C-	Loss of an entire chromosome
D-	A change confined exclusively to introns

7-	**Frameshift mutations result from: (Important M3WAN - Core)**
A-	Deletion or addition of one or more nucleotides, altering the reading frame
B-	Substitution of a single base without altering the reading frame
C-	Methylation of cytosine residues
D-	Loss of the poly-A tail only

8-	**If exactly three nucleotides are inserted or deleted in a frameshift mutation, the resulting protein characteristically shows: (M3WAN - Hard)**
A-	No change in the reading frame, with one amino acid added or missing
B-	A completely garbled reading frame from that point onward
C-	Immediate loss of the start codon
D-	No effect on the protein at all

9-	**A frameshift mutation involving a single nucleotide deletion (not a multiple of three) typically causes: (Important M3WAN - Core)**
A-	Garbled translation of the mRNA distal to the deletion, often with premature termination
B-	A silent mutation with no effect
C-	Exact restoration of the original reading frame after one codon
D-	Enhanced protein stability

10-	**A point mutation in which the altered base still codes for the same amino acid (typically due to a change at the third codon position) is called a: (Important M3WAN - Core)**		
A-	Missense mutation	B-	Nonsense mutation
C-	Silent mutation	D-	Frameshift mutation

11-	A point mutation in which the altered base changes the codon to specify a different amino acid is called a: (Important M3WAN - Core)		
A-	Silent mutation	B-	Missense mutation
C-	Nonsense mutation	D-	Neutral mutation

12-	A point mutation in which the altered base converts a sense codon into a premature stop codon, producing a truncated protein, is called a: (Important M3WAN - Core)		
A-	Silent mutation	B-	Missense mutation
C-	Nonsense mutation	D-	Wobble mutation

13-	DNA repair mechanisms that correct damage affecting only a single strand are collectively known as: (M3WAN - Core)		
A-	Excision repair	B-	Recombinational repair
C-	Semi-conservative repair	D-	Translesion synthesis only

14-	Nucleotide excision repair is specifically used to correct: (Important M3WAN - Core)		
A-	ThyminE-thymine (pyrimidine) dimers	B-	DoublE-strand breaks
C-	Mismatched base pairs from replication errors	D-	Deaminated cytosine only

15-	In nucleotide excision repair, the enzyme that recognizes a thymine dimer and makes an initial nick in the affected strand is: (Important M3WAN - Core)		
A-	DNA ligase	B-	UV-specific endonuclease
C-	DNA polymerase III	D-	MutS

16-	After the initial nick by UV-specific endonuclease, the gap is filled with new nucleotides, using the healthy strand as template, by: (M3WAN - Core)		
A-	DNA polymerase I, via nick translation	B-	DNA polymerase III
C-	Primase	D-	RNA polymerase II

17-	Completion of nucleotide excision repair, joining the new and original DNA strands, requires: (M3WAN - Core)		
A-	DNA ligase	B-	Topoisomerase
C-	Helicase	D-	Reverse transcriptase

18-	Mismatch repair corrects copying errors such as G-T mispairing, with the mismatched base recognized by a group of proteins that identify the parental strand by its: (Important M3WAN - Core)		
A-	Methylation status	B-	Acetylation status
C-	Phosphorylation status	D-	Glycosylation status

19-	The bacterial proteins responsible for recognizing mismatches and identifying the parental DNA strand in mismatch repair are: (Important M3WAN - Core)		
A-	MutS, MutL, and MutH	B-	Ku and DNA-PK
C-	TFIID and TFIIB	D-	eIF1 and eIF2

20-	Base excision repair corrects damage such as spontaneous deamination of cytosine to uracil, adenine to hypoxanthine, and guanine to: (Important M3WAN - Core)						
A-	Xanthine	B-	Thymine	C-	Cytosine	D-	Adenine

21-	In base excision repair, the abnormal base (e-g-, uracil) is first recognized and removed by: (Important M3WAN - Core)
A-	DNA glycosylase, creating an apurinic/apyrimidinic (AP) site
B-	UV-specific endonuclease
C-	MutH
D-	Primase

22-	Following creation of an AP site in base excision repair, the deoxyribosE-phosphate backbone is cleaved by: (M3WAN - Hard)		
A-	AP-endonuclease	B-	DNA ligase
C-	Topoisomerase	D-	Aminoacyl-tRNA synthetase

23-	Repair of doublE-strand DNA breaks caused by ionizing radiation, chemotherapy, or oxidative free radicals principally involves two proteins: Ku and: (Important M3WAN - Core)		
A-	DNA-dependent protein kinase	B-	MutS
C-	eRF	D-	Rho factor

24-	Xeroderma pigmentosum, an autosomal recessive disease characterized by extreme sunlight sensitivity and high skin cancer risk, results from a defect in: (Important M3WAN - Core)
A-	Mismatch repair
B-	Nucleotide excision repair (absence of UV-specific endonuclease)
C-	Base excision repair of deaminated cytosine
D-	Recombinational doublE-strand break repair

25-	Ataxia telangiectasia, an autosomal recessive disease with increased sensitivity to X-ray-induced damage and malignancy of the reticuloendothelial system, results from impaired repair of: (Important M3WAN - Core)		
A-	Thymine dimers	B-	DNA strand breaks
C-	Deaminated bases	D-	Mismatched bases only

26-	**Fanconi-s anemia, an autosomal recessive disorder with increased malignancy risk, results from failure to correct: (Important M3WAN - Core)**
A-	Cross-linkage DNA damage (e-g-, thyminE-thymine dimers)
B-	Simple point mutations only
C-	Poly-A tail defects
D-	Ribosomal RNA processing defects

27-	**Hereditary nonpolyposis colorectal cancer (Lynch syndrome), associated with cancers of the colon, stomach, and uterus, results from defective mismatch repair due to mutations in: (Important M3WAN - Core)**		
A-	hMSH2 and hMLH1	B-	BRCA1 and BRCA2
C-	p53 and RB1	D-	HGPRT and PRPP synthetase

Answers - M3WAN Bank									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	**D**	**A**	**B**	**A**	**B**	**A**	**A**	**A**	**C**
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	**C**	**A**	**A**	**B**	**A**	**A**	**A**	**A**	**A**
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	**A**	**A**	**B**	**B**	**A**	**A**	**-**	**-**	**-**

تم بفضل الله ، هذا وما كان من توفيق فمن الله وحده وما كان من خطأ أو زلل فمن أنفسنا ومن الشيطان ، وفقنا الله وإياكم لخيري الدنيا والآخرة وأعاننا على طاعته .

نسألكم الدعاء لكل من سبقنا من إخواننا إلى الله أن يتغمدهم بواسع رحمته وأن يرفع قدرهم ويتجاوز عن سيئاتهم وأن يجمعنا بهم في جناته .